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- Nash equilibrium; Cournot; Bertrand (block 2): Ch. 5, §5.1–5.2 (p. 79–95)
- Mixed strategies and mixed Nash equilibrium (block 2): Ch. 6, §6.1–6.2 (p. 101–114)
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4

Rationality and Common Knowledge

In this chapter we study the implications of imposing the assumptions of rationality as well as common knowledge of rationality. We derive and explore some solution concepts that result from these two assumptions and seek to understand the restrictions that each of the two assumptions imposes on the way in which players will play games.

4.1 Dominance in Pure Strategies

It will be useful to begin by introducing some new notation. We denoted the payoff of a player i from a profile of strategies $s = (s_1, s_2, \ldots, s_{i-1}, s_i, s_{i+1}, \ldots, s_n)$ as $v_i(s)$. It will soon be very useful to refer specifically to the strategies of a player's opponents in a game. For example, the actions chosen by the players who are *not* player i are denoted by the profile

$$(s_1, s_2, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n) \in S_1 \times S_2 \times \cdots \times S_{i-1} \times S_{i+1} \times \cdots \times S_n.$$

To simplify we will hereafter use a common shorthand notation as follows: We define $S_{-i} \equiv S_1 \times S_2 \times \cdots \times S_{i-1} \times S_{i+1} \times \cdots \times S_n$ as the set of all the strategy sets of *all players who are not* player i. We then define $s_{-i} \in S_{-i}$ as a particular possible profile of strategies for all players who are not i. Hence we can rewrite the payoff of player i from strategy s as $v_i(s_i, s_{-i})$, where $s = (s_i, s_{-i})$.

4.1.1 Dominated Strategies

The Prisoner's Dilemma was easy to analyze: each of the two players has an action that is best *regardless* of what his opponent chooses. Suggesting that each player will choose this action seems natural because it is consistent with the basic concept of rationality. If we assume that the players are rational, then we should expect them to choose whatever they deem to be best for them. If it turns out that a player's best strategy does not depend on the strategies of his opponents then we should be all the more confident that he will choose it.

It is not too often that we will find ourselves in situations in which we have a best action that does not depend on the actions of our opponents. We begin, therefore, with a less demanding concept that follows from rationality. In particular consider the strategy mum in the Prisoner's Dilemma:

		Player 2	
		M	F
Player 1	M	$-2, -2$	$-5, -1$
	F	$-1, -5$	$-4, -4$

As we argued earlier, playing M is worse than playing F for each player *regardless* of what the player's opponent does. What makes it unappealing is that there is another strategy, F, that is better than M regardless of what one's opponent chooses. We say that such a strategy is *dominated*. Formally we have

Definition 4.1 Let $s_i \in S_i$ and $s_i' \in S_i$ be possible strategies for player i. We say that s_i' is **strictly dominated** by s_i if for any possible combination of the other players' strategies, $s_{-i} \in S_{-i}$, player i's payoff from s_i' is strictly less than that from s_i. That is,

$$v_i(s_i, s_{-i}) > v_i(s_i', s_{-i}) \quad \text{for all } s_{-i} \in S_{-i}.$$

We will write $s_i \succ_i s_i'$ to denote that s_i' is strictly dominated by s_i.

Now that we have a precise definition for a dominated strategy, it is straightforward to draw an obvious conclusion:

Claim 4.1 *A rational player will never play a strictly dominated strategy.*

This claim is obvious. If a player plays a dominated strategy then he cannot be playing optimally because, by the definition of a dominated strategy, the player has another strategy that will yield him a higher payoff *regardless of the strategies of his opponents.* Hence knowledge of the game implies that a player should recognize dominated strategies, and rationality implies that these strategies will be avoided.

When we apply the notion of a dominated strategy to the Prisoner's Dilemma we argue that each of the two players has one dominated strategy that he should never use, and hence each player is left with one strategy that is not dominated. Therefore, for the Prisoner's Dilemma, rationality alone is enough to offer a prediction about which outcome will prevail: (F, F) is this outcome.

Many games, however, will not be as special as the Prisoner's Dilemma, and rationality alone will not suggest a clear-cut, unique prediction. As an example, consider the following advertising game. Two competing brands can choose one of three marketing campaigns—low (L), medium (M), and high (H)—with payoffs given by the following matrix:

		Player 2		
		L	M	H
Player 1	L	6, 6	2, 8	0, 4
	M	8, 2	4, 4	1, 3
	H	4, 0	3, 1	2, 2

It is easy to see that each player has one dominated strategy, which is L. However, neither M nor H is dominated. For example, if player 2 plays M then player 1 should also play M, while if player 2 plays H then player 1 should also play H. Hence rationality alone does not offer a unique prediction. It is nonetheless worth spending some time on the extreme cases in which it does.

4.1.2 Dominant Strategy Equilibrium

Because a strictly dominated strategy is one to avoid at all costs,[1] there is a counterpart strategy, represented by F in the Prisoner's Dilemma, that would be desirable. This is a strategy that is *always* the best thing you can do, regardless of what your opponents choose. Formally we have

Definition 4.2 $s_i \in S_i$ is a **strictly dominant strategy** for i if every other strategy of i is strictly dominated by it, that is,

$$v_i(s_i, s_{-i}) > v_i(s_i', s_{-i}) \quad \text{for all } s_i' \in S_i, \;\; s_i' \neq s_i, \;\; \text{and all } s_{-i} \in S_{-i}.$$

If, as in the Prisoner's Dilemma, every player had such a wonderful dominant strategy, then it would be a very sensible predictor of behavior because it follows from rationality alone. We can introduce this simple idea as our first solution concept:

Definition 4.3 The strategy profile $s^D \in S$ is a **strict dominant strategy equilibrium** if $s_i^D \in S_i$ is a strict dominant strategy for all $i \in N$.

This gives a formal definition for the outcome "both players fink," or (F, F), in the Prisoner's Dilemma: it is a dominant strategy equilibrium. In this equilibrium the payoffs are $(-4, -4)$ for players 1 and 2, respectively.

Caveat Be careful not to make a common error by referring to the pair of payoffs $(-4, -4)$ as the solution. The solution should always be described as the strategies that the players will choose. Strategies are a set of actions by the players, and payoffs are a result of the outcome. When we talk about predictions, or equilibria, we will always refer to *what players do* as the equilibrium, not their payoffs.

Using this solution concept for any game is not that difficult. It basically requires that we identify a strict dominant strategy for each player and then use this profile of strategies to predict or prescribe behavior. If, as in the Prisoner's Dilemma, we are lucky enough to find a dominant strategy equilibrium for other games, then this solution concept has a very appealing property:

Proposition 4.1 *If the game* $\Gamma = \langle N, \{S_i\}_{i=1}^n, \{v_i\}_{i=1}^n \rangle$ *has a strictly dominant strategy equilibrium* s^D*, then* s^D *is the unique dominant strategy equilibrium.*

This proposition is rather easy to prove, and that proof is left as exercise 4.1 at the end of the chapter.

1. This is a good point at which to stop and reflect on a very simple yet powerful lesson. In any situation, look first for your dominated strategies and avoid them!

4.1.3 Evaluating Dominant Strategy Equilibrium

Proposition 4.1 is very useful in addressing one of our proposed criteria for evaluating a solution concept: when it exists, the strict-dominance solution concept guarantees uniqueness. However, what do we know about existence? A quick observation will easily convince you that this is a problem.

Consider the Battle of the Sexes game introduced in Section 3.3 and described again in the following matrix:

		Chris: *O*	Chris: *F*
Alex	*O*	2, 1	0, 0
	F	0, 0	1, 2

Neither player has a dominated strategy, implying that neither has a dominant strategy either. The best strategy for Chris depends on what Alex chooses and vice versa. Thus if we stick to the solution concept of strict dominance we will encounter games, in fact many of them, for which there will be no equilibrium. This unfortunate conclusion implies that the strict-dominance solution concept will often fail to predict the choices that players ought to, or will, choose in many games.

Regarding the invariance criterion, the strict-dominance solution concept does comply. From definition 4.2, s^D is a strictly dominant strategy equilibrium if and only if $v_i(s_i^D, s_{-i}) > v_i(s_i', s_{-i})$ for all $s_i' \in S_i$ and all $s_{-i} \in S_{-i}$. Because the inequality is strict, we can find a small enough value $\varepsilon > 0$ such that if we either add or subtract ε from any payoff $v_i(s_i, s_{-i})$ the inequality will still hold.

We now turn to the Pareto criterion of equilibrium outcomes when a strict-dominance solution exists. The Prisoner's Dilemma has an equilibrium prediction using the strict-dominance solution concept, so we can evaluate the efficiency properties of the unique strictly dominant strategy equilibrium for that game. It is easy to see that the outcome prescribed by this solution is *not Pareto optimal:* both players would be better off if they could each commit to play *M*, yet left to their own devices they will not do this. Of course, in other games the solution may be Pareto optimal (see, for example, the "altruistic" Prisoner's Dilemma in Section 3.3).

The failure of Pareto optimality is *not* a failure of the solution concept. The assumption that players are rational causes them to fink in the Prisoner's Dilemma if we restrict attention to self-enforcing behavior. The failure of Pareto optimality implies that the players would benefit from modifying the environment in which they find themselves to create other enforcement mechanisms—for example, creating a "mafia" with norms of conduct that enforce implicit agreements so as to punish those who fink.

To explicitly see how this can work, imagine that a mafia member who finks on another member is very seriously reprimanded, which will change the payoff structure of the Prisoner's Dilemma if he is caught. If the pain from mafia punishment is equivalent to z, then we have to subtract z units of payoff for each player who finks. The "mafia-modified" Prisoner's Dilemma is represented by the following matrix:

		Player 2: *M*	Player 2: *F*
Player 1	*M*	$-2, -2$	$-5, -1-z$
	F	$-1-z, -5$	$-4-z, -4-z$

If z is strictly greater than 1 then this punishment will be enough to flip our predicted equilibrium outcome of the game because then M becomes the strict dominant strategy (and (M, M) is Pareto optimal).

This example demonstrates that "institutional design," which changes the game that players play, can be very useful in affecting the well-being of players. By introducing this external enforcer, or institution, we are able to get the players to choose outcomes that make them both better off compared to what they can achieve without the additional institution. Moreover, notice that if the players believe that the mafia will enforce the code of conduct then there is no need to actually enforce it—the players choose not to fink, and the enforcement of punishments need not happen. However, we need to be suspicious of whether such an institution will be *self-enforcing,* that is, whether the mafia will indeed enforce the punishments. And, for it to be self-enforcing, we need to model the behavior of potential punishers and whether they themselves will have the selfish incentives to carry out the enforcement activities. This is something we will explore at length when we consider multistage and repeated games in Chapters Chapters 9 and 10.

Remark A related notion is that of **weak dominance.** We say that s_i' is *weakly dominated* by s_i if, for any possible combination of the other players' strategies, player i's payoff from s_i' is weakly less than that from s_i. That is,

$$v_i(s_i, s_{-i}) \geq v_i(s_i', s_{-i}) \quad \text{for all } s_{-i} \in S_{-i}.$$

This means that for some $s_{-i} \in S_{-i}$ this weak inequality may hold strictly, while for other $s_{-i}' \in S_{-i}$ it will hold with equality. We define a strategy to be **weakly dominant** in a similar way. This is still useful because if we can find a dominant strategy for a player, be it weak or strict, this seems like the most obvious thing to prescribe. An important difference between weak and strict dominance is that if a weakly dominant equilibrium exists, it need not be unique. To show this is left as exercise 4.2.

4.2 Iterated Elimination of Strictly Dominated Pure Strategies

As we saw in the previous chapter, our requirement that players be rational implied two important conclusions:

1. A rational player will never play a dominated strategy.
2. If a rational player has a strictly dominant strategy then he will play it.

We used this second conclusion to define the solution concept of strict dominance, which is very appealing because, when it exists, it requires only rationality as its driving force. A drawback of the dominant strategy solution concept is, however, that it will often fail to exist. Hence if we wish to develop a predictive theory of behavior in games then we must consider alternative approaches that will apply to a wide variety of games.

4.2.1 Iterated Elimination and Common Knowledge of Rationality

We begin with the premise that players are rational, and we build on the first conclusion in the previous section, which claims that a rational player will never play a

dominated strategy. This conclusion is by itself useful in that it rules out what players *will not do*. As a result, we conclude that rationality tells us which strategies will never be played.

Now turn to another important assumption introduced earlier: the structure of the game and the rationality of the players are common knowledge among the players. The introduction of *common knowledge of rationality* allows us to do much more than identify strategies that rational players will avoid. If indeed all the players *know* that each player will never play a strictly dominated strategy, they can effectively ignore those strictly dominated strategies that their opponents will never play, and their opponents can do the same thing. If the original game has some players with some strictly dominated strategies, then all the players know that *effectively* they are facing a "smaller" restricted game with fewer total strategies.

This logic can be taken further. Because it is *common knowledge* that all players are rational, then everyone knows that everyone knows that the game is effectively a smaller game. In this smaller restricted game, everyone knows that players will not play strictly dominated strategies. In fact we may indeed find additional strategies that are dominated in the restricted game that *were not dominated* in the original game. Because it is common knowledge that players will perform this kind of reasoning again, the process can continue until no more strategies can be eliminated in this way.

To see this idea more concretely, consider the following two-player finite game:

		Player 2		
		L	*C*	*R*
Player 1	*U*	4, 3	5, 1	6, 2
	M	2, 1	8, 4	3, 6
	D	3, 0	9, 6	2, 8

(4.1)

A quick observation reveals that there is no strictly dominant strategy, neither for player 1 nor for player 2. Also note that there is no strictly dominated strategy for player 1. There is, however, a strictly dominated strategy for player 2: the strategy C is strictly dominated by R because $2 > 1$ (row U), $6 > 4$ (row M), and $8 > 6$ (row D). Thus, because this is common knowledge, both players know that we can effectively eliminate the strategy C from player 2's strategy set, which results in the following reduced game:

		Player 2	
		L	*R*
Player 1	*U*	4, 3	6, 2
	M	2, 1	3, 6
	D	3, 0	2, 8

In this reduced game, both M and D are strictly dominated by U for player 1, allowing us to perform a second round of eliminating strategies, this time for player 1. Eliminating these two strategies yields the following trivial game:

		Player 2	
		L	*R*
Player 1	*U*	**4, 3**	6, 2

in which player 2 has a strictly dominated strategy, playing R. Thus for this example the iterated process of eliminating dominated strategies yields a unique prediction: the strategy profile we expect these players to play is (U, L), giving the players the payoffs of (4, 3).

As the example demonstrates, this process of **iterated elimination of strictly dominated strategies** (IESDS) builds on the assumption of common knowledge of rationality. The first step of iterated elimination is a consequence of player 2's rationality; the second stage follows because players *know* that players are rational; the third stage follows because players *know that players know* that they are rational, and this ends in a unique prediction.

More generally we can apply this process to games in the following way. Let S_i^k denote the strategy set of player i that survives k rounds of IESDS. We begin the process by defining $S_i^0 = S_i$ for each i, the original strategy set of player i in the game.

Step 1: Define $S_i^0 = S_i$ for each i, the original strategy set of player i in the game, and set $k = 0$.

Step 2: Are there players for whom there are strategies $s_i \in S_i^k$ that are strictly dominated? If yes, go to step 3. If not, go to step 4.

Step 3: For all the players $i \in N$, remove any strategies $s_i \in S_i^k$ that are strictly dominated. Set $k = k + 1$, and define a new game with strategy sets S_i^k that *do not include* the strictly dominated strategies that have been removed. Go back to step 2.

Step 4: The remaining strategies in S_i^k are reasonable predictions for behavior.

In this chapter we refrain from giving a precise mathematical definition of the process because this requires us to consider richer behavior by the players, in particular, allowing them to choose randomly between their different pure strategies. We will revisit this approach briefly when such stochastic play, or **mixed strategies,** is introduced later.[2]

Using the process of IESDS we can define a new solution concept:

Definition 4.4 We will call any strategy profile $s^{ES} = (s_1^{ES}, \ldots, s_n^{ES})$ that survives the process of IESDS an **iterated-elimination equilibrium.**

Like the concept of a strictly dominant strategy equilibrium, the iterated-elimination equilibrium starts with the premise of rationality. However, in addition to rationality, IESDS requires a lot more: common knowledge of rationality. We will discuss the implications of this requirement later in this chapter.

4.2.2 Example: Cournot Duopoly

Recall the Cournot duopoly example we introduced in Section 3.1.2, but consider instead a simpler example of this problem in which the firms have linear rather than quadratic costs: the cost for each firm for producing quantity q_i is given by $c_i(q_i) = 10q_i$ for $i \in \{1, 2\}$. (Using economics jargon, this is a case of constant marginal cost

2. Just to satisfy your curiosity, think of the Battle of the Sexes, and imagine that Chris can pull out a coin and flip between the decision of opera or football. This by itself introduces a "new" strategy, and we will exploit such strategies to develop a formal definition of IESDS.

equal to 10 and no fixed costs.) Let the demand be given by $p(q) = 100 - q$, where $q = q_1 + q_2$.

Consider first the profit (payoff) function of firm 1:

$$\begin{aligned} v_1(q_1, q_2) &= \overbrace{(100 - q_1 - q_2)q_1}^{\text{Revenue}} - \overbrace{10q_1}^{\text{Costs}} \\ &= 90q_1 - q_1^2 - q_1q_2. \end{aligned}$$

What should firm 1 do? If it knew what quantity firm 2 will choose to produce, say some value of q_2, then the profits of firm 1 would be maximized when the first-order condition is satisfied, that is, when $90 - 2q_1 - q_2 = 0$. Thus, for any given value of q_2, firm 1 maximizes its profits when it sets its own quantity according to the function

$$q_1 = \frac{90 - q_2}{2}. \tag{4.2}$$

Though it is true that the choice of firm 1 depends on what it believes firm 2 is choosing, equation (4.2) implies that firm 1 will *never* choose to produce more than $q_1 = 45$. This follows from the simple observation that q_2 is never negative, in which case equation (4.2) implies that $q_1 \leq 45$. In fact, this is equivalent to showing that any quantity $q_1 > 45$ is strictly dominated by $q_1 = 45$. To see this, for any q_2, the profits from setting $q_1 = 45$ are given by

$$v_1(45, q_2) = (100 - 45 - q_2)45 - 450 = 2025 - 45q_2.$$

The profits from choosing any other q_1 are given by

$$v_1(q_1, q_2) = (100 - q_1 - q_2)q_1 - 10q_1 = 90q_1 - q_1q_2 - q_1^2.$$

Thus we can subtract $v_1(q_1, q_2)$ from $v_1(45, q_2)$ and obtain

$$\begin{aligned} v_1(45, q_2) - v_1(q_1, q_2) &= 2025 - 45q_2 - (90q_1 - q_1q_2 - q_1^2) \\ &= 2025 - q_1(90 - q_1) - q_2(45 - q_1). \end{aligned}$$

It is easy to check that for any $q_1 > 45$ this difference is positive regardless of the value of q_2.[3] Hence we conclude that any $q_1 > 45$ is strictly dominated by $q_1 = 45$.

It is easy to see that firm 2 faces exactly the same profit function, which implies that any $q_2 > 45$ is strictly dominated by $q_2 = 45$. This observation leads to our first round of iterated elimination: a rational firm produces no more than 45 units, implying that the effective strategy space that survives one round of elimination is $q_i \in [0, 45]$ for $i \in \{1, 2\}$.

We can now turn to the second round of elimination. Because $q_2 \leq 45$, equation (4.2) implies that firm 1 will choose a quantity no less than 22.5, and a symmetric argument applies to firm 2. Hence the second round of elimination implies that the surviving strategy sets are $q_i \in [22.5, 45]$ for $i \in \{1, 2\}$.

The next step of this process will reduce the strategy set to $q_i \in [22.5, 33\frac{3}{4}]$, and the process will continue on and on. Interestingly the set of strategies that survives

3. This follows because if $q_1 > 45$ then $q_1(90 - q_1) < 2025$ and $q_2(45 - q_1) \leq 0$ for any $q_2 \geq 0$.

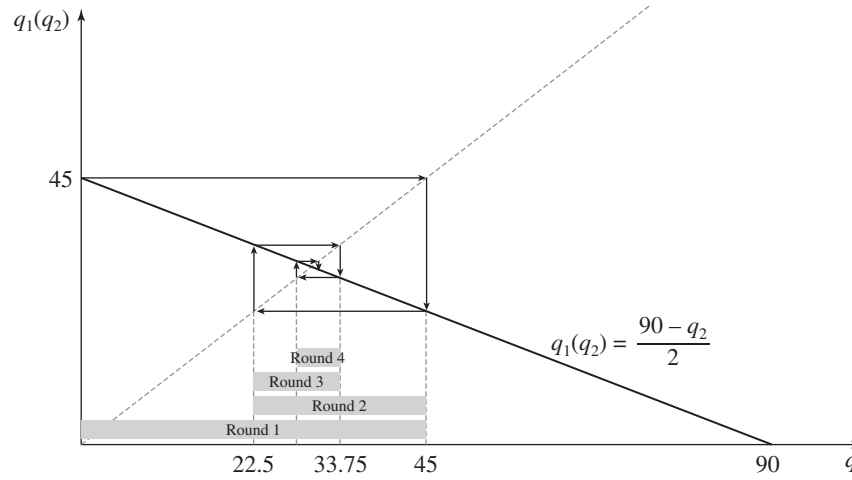


FIGURE 4.1 IESDS convergence in the Cournot game.

this process converges to a single quantity choice of $q_i = 30$. To see this, notice how we moved from one surviving interval to the next. We started by noting that $q_2 \geq 0$, and using equation (4.2) we found that $q_1 \leq 45$, creating the first-round interval of [0, 45]. Then, by symmetry, it follows that $q_2 \leq 45$, and using equation (4.2) again we conclude that $q_1 \geq 22.5$, creating the second-round interval [22.5, 45]. We can see this process graphically in Figure 4.1, where we use the upper (lower) end of the previous interval to determine the lower (upper) end of the next one. If this were to converge to an interval and not to a single point, then by the symmetry between both firms, the resulting interval for each firm would be $[q_{\min}, q_{\max}]$ that simultaneously satisfy two equations with two unknowns: $q_{\min} = \frac{90 - q_{\max}}{2}$ and $q_{\max} = \frac{90 - q_{\min}}{2}$. However, the only solution to these two equations is $q_{\min} = q_{\max} = 30$. Hence using IESDS for the Cournot game results in a unique predictor of behavior where $q_1 = q_2 = 30$, and each firm earns a profit of $v_1 = v_2 = 900$.

4.2.3 Evaluating IESDS

We turn to evaluate the IESDS solution concept using the criteria we introduced earlier. Start with existence and note that, unlike the concept of strict dominance, we can apply IESDS to *any* game by applying the algorithm just described. It does not require the existence of a strictly dominant strategy, nor does it require the existence of strictly dominated strategies. It is the latter characteristic, however, that gives this concept some bite: when strictly dominated strategies exist, the process of IESDS is able to say something about how common knowledge of rationality restricts behavior.

It is worth noting that this existence result is a consequence of assuming common knowledge of rationality. By doing so we are giving the players the ability to reason through the strategic implications of rationality, and to do so over and over again, while correctly anticipating that other players can perform the same kind of reasoning. Rationality alone does not provide this kind of reasoning.

It is indeed attractive that an IESDS solution always exists. This comes, however, at the cost of uniqueness. In the simple 3×3 matrix game described in (4.1) and the

Cournot duopoly game, IESDS implied the survival of a unique strategy. Consider instead the Battle of the Sexes, given by the following matrix:

		Chris	
		O	F
Alex	O	2, 1	0, 0
	F	0, 0	1, 2

IESDS cannot restrict the set of strategies here for the simple reason that neither O nor F is a strictly dominated strategy for each player. As we can see, this solution concept can be applied (it exists) to any game, but it will often fail to provide a unique solution. For the Battle of the Sexes game, IESDS can only conclude that "anything can happen."

After analyzing the efficiency of the outcomes that can be derived from strict dominance in Section 4.1.3, you may have anticipated the possible efficiency of IESDS equilibria. An easy illustration can be provided by the Prisoner's Dilemma. IESDS leaves (F, F) as the unique survivor, or IESDS equilibrium, after only one round of elimination. As we already demonstrated, the outcome from (F, F) is not Pareto optimal. Similarly, both previous examples (the 3×3 matrix game in (4.1) and the Cournot game) provide further evidence that Pareto optimality need not be achieved by IESDS: In the 3×3 matrix example, both strategy profiles (M, C) and (D, C) yield higher payoffs for both players—(8, 4) and (9, 6), respectively—than the unique IESDS equilibrium, which yields (4, 3). For the Cournot game, producing $q_1 = q_2 = 30$ yields profits of 900 for each firm. If instead they would both produce $q_1 = q_2 = 20$ then each would earn profits of 1000. Thus common knowledge of rationality does not mean that players can guarantee the best outcome for themselves when their own incentives dictate their behavior.

On a final note, it is interesting to observe that there is a simple and quite obvious relationship between the IESDS solution concept and the strict-dominance solution concept:

Proposition 4.2 *If for a game* $\Gamma = \langle N, \{S_i\}_{i=1}^n, \{v_i\}_{i=1}^n \rangle$ s^* *is a* strict dominant strategy equilibrium, *then* s^* uniquely survives IESDS.

Proof If $s^* = (s_1^*, \ldots, s_n^*)$ is a strict dominant strategy equilibrium then, by definition, for every player i all other strategies s_i' are strictly dominated by s_i^*. This implies that after one stage of elimination we will be left with a single profile of strategies, which is exactly s^*, and this concludes the proof. ■

This simple proposition is both intuitive and straightforward. Because rationality is the only requirement needed in order to eliminate all strictly dominated strategies in one round, then if all strategies but one are strictly dominated for each and every player, both IESDS and strict dominance will result in the same outcome. This shows us that whenever strict dominance results in a unique outcome, then IESDS will result in the same unique outcome after one round. However, as we saw earlier, IESDS may offer a fine prediction when strict dominance does not apply. This is exactly what the extra assumption of common knowledge of rationality delivers: a more widely applicable solution concept. However, the assumption of common knowledge of rationality is far from innocuous. It requires the players to be, in some way, extremely

intelligent and to possess unusual levels of foresight. For the most part, game theory relies on this strong assumption, and hence it must be applied to the real world with caution. Remember the rule about how assumptions drive conclusions: garbage in, garbage out.

4.3 Beliefs, Best Response, and Rationalizability

Both of the solution concepts we have seen so far, strict dominance and IESDS, are based on eliminating actions that players would never play. An alternative approach is to ask: what possible strategies might players choose to play and under what conditions? When we considered eliminating strategies that no rational player would choose to play, it was by finding some strategy that is *always* better or, as we said, that dominates the eliminated strategies. A strategy that cannot be eliminated, therefore, suggests that *under some conditions* this strategy is the one that the player may like to choose. When we qualify a strategy to be the best one a player can choose *under some conditions,* these conditions must be expressed in terms that are rigorous and are related to the game that is being played.

To set the stage, think about situations in which you were puzzled about the behavior of someone you knew. To consider his choice as irrational, or simply stupid, you would have to consider whether there is a way in which he could defend his action as a good choice. A natural way to determine whether this is the case is to simply ask him, "What were you thinking?" If the response lays out a plausible situation for which his choice was a good one, then you cannot question his rationality. (You may of course question his wisdom, or even his sanity, if his thoughts seem bizarre.)

This is a type of reasoning that we will formalize and discuss in this chapter. If a strategy s_i is not strictly dominated for player i then it must be that there are combinations of strategies of player i's opponents for which the strategy s_i is player i's best choice. This reasoning will allow us to justify or *rationalize* the choice of player i.

4.3.1 The Best Response

As we discussed early in Chapter 3, what makes a game different from a single-player decision problem is that once you understand the actions, outcomes, and preferences of a decision problem, then you can choose your best or optimal action. In a game, however, your optimal decision not only depends on the structure of the game, but it will often depend on what the other players are doing.

Take the Battle of the Sexes as an example:

		Chris	
		O	*F*
Alex	*O*	2, 1	0, 0
	F	0, 0	1, 2

As the matrix demonstrates, the best choice of Alex depends on what Chris will do. If Chris goes to the opera then Alex would rather go to the opera instead of going to the football game. If, however, Chris goes to the football game then Alex's optimal action is switched around.

increases up to one increment (say, \$0.01) above the *second highest* proxy bid. For example, imagine that three people have placed bids on a used laptop of \$55, \$98, and \$112. The current price will be at \$98.01, and if the auction ended the player who bid \$112 would win at a price of \$98.01. If you were to place a bid of \$103.45 then the player who bid \$112 would still win, but at a price of \$103.46, while if your bid was \$123.12 then you would win at a price of \$112.01.

Now consider eBay's historical recommendation that you think hard about the value you impute to the good and that you enter your true value as your bid—no more, no less. Assume that the value of the good for each potential bidder is independent of how much other bidders value it.

a. Argue that bidding more than your valuation is weakly dominated by actually bidding your valuation.
b. Argue that bidding less than your valuation is weakly dominated by actually bidding your valuation.
c. Use your analysis to make sense of eBay's recommendation. Would you follow it?

4.5 **Iterated Elimination:** In the following normal-form game, which strategy profiles survive iterated elimination of strictly dominated strategies?

		Player 2		
		L	*C*	*R*
Player 1	*U*	6, 8	2, 6	8, 2
	M	8, 2	4, 4	9, 5
	D	8, 10	4, 6	6, 7

4.6 **Roommates:** Two roommates each need to choose to clean their apartment, and each can choose an amount of time $t_i \geq 0$ to clean. If their choices are t_i and t_j, then player i's payoff is given by $(10 - t_j)t_i - t_i^2$. (This payoff function implies that the more one roommate cleans, the less valuable is cleaning for the other roommate.)

a. What is the best response correspondence of each player i?
b. Which choices survive one round of IESDS?
c. Which choices survive IESDS?

4.7 **Campaigning:** Two candidates, 1 and 2, are running for office. Each has one of three choices in running his campaign: focus on the positive aspects of one's own platform (call this a positive campaign $[P]$), focus on the positive aspects of one's own platform while attacking one's opponent's campaign (call this a balanced campaign $[B]$), or finally focus only on attacking one's opponent (call this a negative campaign $[N]$). All a candidate cares about is the probability of winning, so assume that if a candidate expects to win with probability $\pi \in [0, 1]$, then his payoff is π. The probability that a candidate wins depends on his choice of campaign and his opponent's choice. The probabilities of winning are given as follows:

If both choose the same campaign then each wins with probability 0.5.
If candidate i uses a positive campaign while $j \neq i$ uses a balanced one then i loses for sure.

If candidate i uses a positive campaign while $j \neq i$ uses a negative one then i wins with probability 0.3.

If candidate i uses a negative campaign while $j \neq i$ uses a balanced one then i wins with probability 0.6.

a. Model this story as a normal-form game. (It suffices to be specific about the payoff function of one player and to explain how the other player's payoff function is different and why.)
b. Write down the game in matrix form.
c. What happens at each stage of elimination of strictly dominated strategies? Will this procedure lead to a clear prediction?

4.8 **Beauty Contest Best Responses:** Consider the p-beauty contest presented in Section 4.3.5.

a. Show that if player i believes that everyone else is choosing 20 then 19 is not the only best response for any number of players n.
b. Show that the set of best-response strategies to everyone else choosing the number 20 depends on the number of players n.

4.9 **Beauty Contest Rationalizability:** Consider the p-beauty contest presented in Section 4.3.5. Show that if the number of players $n > 2$ then the choices $\{0, 1\}$ for each player are both rationalizable, while if $n = 2$ then only the choice of $\{0\}$ by each player is rationalizable.

4.10 **Popsicle Stands:** Five lifeguard towers are lined up along a beach; the leftmost tower is number 1 and the rightmost tower is number 5. Two vendors, players 1 and 2, each have a popsicle stand that can be located next to one of five towers. There are 25 people located next to each tower, and each person will purchase a popsicle from the stand that is closest to him or her. That is, if player 1 locates his stand at tower 2 and player 2 at tower 3, then 50 people (at towers 1 and 2) will purchase from player 1, while 75 (from towers 3, 4, and 5) will purchase from vendor 2. Each purchase yields a profit of \$1.

a. Specify the strategy set for each player. Are there any strictly dominated strategies?
b. Find the set of strategies that survive rationalizability.

5

Pinning Down Beliefs: Nash Equilibrium

We have seen three solution concepts that offer some insights into predicting the behavior of rational players in strategic (normal-form) games. The first, strict dominance, relied only on rationality, and in some cases, like the Prisoner's Dilemma, it predicted a unique outcome, as it would in any game for which a dominant strategy equilibrium exists. However, it often fails to exist. The two sister concepts of IESDS and rationalizability relied on more than rationality by requiring common knowledge of rationality. In return a solution existed for every game, and for some games there was a unique prediction. Moreover, whenever there is a strict dominant equilibrium, it also uniquely survives IESDS and rationalizability. Even for some games for which the strict-dominance solution did not apply, like the Cournot duopoly, we obtained a unique prediction from IESDS and rationalizability.

However, when we consider a game like the Battle of the Sexes, none of these concepts had any bite. Dominant strategy equilibrium did not apply, and both IESDS and rationalizability could not restrict the set of reasonable behavior:

		Chris	
		O	*F*
Alex	*O*	2, 1	0, 0
	F	0, 0	1, 2

For example, we cannot rule out the possibility that Alex goes to the opera while Chris goes to the football game, because Alex may behave optimally given his belief that Chris is going to the opera, and Chris may behave optimally given his belief that Alex is going to the football game. Yet there is something troubling about this outcome. If we think of this pair of actions not only as actions, but as a system of actions and beliefs, then there is something of a dissonance: indeed the players are playing best responses to their beliefs, but their beliefs are wrong!

In this chapter we make a rather heroic leap that ties together beliefs and actions and results in the most central and best-known solution concept in game theory. As already mentioned, for dominant strategy equilibrium we required only that players be rational, while for IESDS and rationalizability we required common knowledge

of rationality. Now we introduce a much more demanding concept, that of the Nash equilibrium, first put forth by John Nash (1950a), who received the Nobel Prize in Economics for this achievement.[1]

5.1 Nash Equilibrium in Pure Strategies

To cut to the chase, a Nash equilibrium is a system of beliefs and a profile of actions for which each player is playing a best response to his beliefs and, moreover, players have *correct beliefs.* Another common way of defining a Nash equilibrium, which does not refer to beliefs, is as a profile of strategies for which *each* player is choosing a best response to the strategies of *all other* players. Formally we have:

Definition 5.1 The pure-strategy profile $s^* = (s_1^*, s_2^*, \ldots, s_n^*) \in S$ is a **Nash equilibrium** if s_i^* is a best response to s_{-i}^*, for all $i \in N$, that is,

$$v_i(s_i^*, s_{-i}^*) \geq v_i(s_i', s_{-i}^*) \quad \text{for all } s_i' \in S_i \text{ and all } i \in N.$$

Consider as an example the following two-player discrete game, which we used to demonstrate IESDS:

		Player 2		
		L	C	R
Player 1	U	$\boxed{4, 3}$	5, 1	6, 2
	M	2, 1	8, 4	3, 6
	D	3, 0	9, 6	2, 8

In this game, the only pair of pure strategies that survived IESDS is the pair (U, L). As it turns out, this is also the only pair of strategies that constitutes a Nash equilibrium. If player 2 is playing the column L, then player 1's best response is $BR_1(L) = \{U\}$; at the same time, if player 1 is playing the row U, then player 2's best response is $BR_2(U) = \{L\}$.

What about the other games we saw? In the Prisoner's Dilemma, the unique Nash equilibrium is (F, F). This should be easy to see: if each player is playing a dominant strategy then he is by definition playing a best response to *anything* his opponent is choosing, and hence it must be a Nash equilibrium. As we will soon see in Section 5.2.3, and as you may have already anticipated, the unique Nash equilibrium in the Cournot duopoly game that we discussed earlier is $(q_1, q_2) = (30, 30)$.

The relationship between strict-dominance, IESDS, rationalizability, and Nash equilibrium outcomes in many of the examples we have analyzed is no coincidence. There is a simple relationship between the concepts we previously explored and that of Nash equilibrium, as the following proposition clearly states:

Proposition 5.1 *Consider a strategy profile $s^* = (s_1^*, s_2^*, \ldots, s_n^*)$. If s^* is either*

1. *a strict dominant strategy equilibrium,*
2. *the unique survivor of IESDS, or*

1. Nash's life was the subject of a very successful movie, *A Beautiful Mind,* based on the book by Sylvia Nasar (1998).

3. *the unique rationalizable strategy profile,*

then s^* is the unique Nash equilibrium.

This proposition is not difficult to prove, and the proof is left as exercise 5.1 at the end of the chapter. The intuition is of course quite straightforward: we know that if there is a strict dominant strategy equilibrium then it uniquely survives IESDS and rationalizability, and this in turn must mean that each player is playing a best response to the other players' strategies.

At the risk of being repetitive, let me emphasize the requirements for a Nash equilibrium:

1. Each player is playing a *best response* to his *beliefs.*
2. The *beliefs* of the players about their opponents *are correct.*

The first requirement is a direct consequence of rationality. It is the second requirement that is very demanding and is a tremendous leap beyond the requirements we have considered so far. It is one thing to ask people to behave rationally given their beliefs (play a best response), but it is a totally different thing to ask players to predict the behavior of their opponents correctly.

Then again it may be possible to accept such a strong requirement if we allow for some reasoning that is beyond the physical structure of the game. For example, imagine, in the Battle of the Sexes game, that Alex is an influential person—people just seem to follow Alex, and this is something that Alex knows well. In this case Chris should believe, knowing that Alex is so influential, that Alex would expect Chris to go to the opera. Knowing this, Alex should believe that Chris will indeed believe that Alex is going to the opera, and so Chris will go to the opera too.

It is important to note that this argument *is not* that Chris likes to please Alex—such an argument would change the payoff of the game and increase Chris's payoff from pleasing Alex. Instead this argument is only about beliefs that are "self-fulfilling." That is, if these beliefs have some weight to them, which may be based on past experience or on some kind of deductive reasoning, then they will be self-fulfilling in that they support the behavior that players believe will occur.

Indeed (O, O) is a Nash equilibrium. However, notice that we can make the symmetric argument about Chris being an influential person: (F, F) is also a Nash equilibrium. As the external game theorist, however, we should not say more than "one of these two outcomes is what we predict." (You should be able to convince yourself that no other pair of pure strategies in the Battle of the Sexes game is a Nash equilibrium.)

5.1.1 Pure-Strategy Nash Equilibrium in a Matrix

This short section presents a simple method to find all the pure-strategy Nash equilibria in a matrix game if at least one exists. Consider the following two-person finite game in matrix form:

		Player 2		
		L	*C*	*R*
Player 1	*U*	7, 7	4, 2	1, 8
	M	2, 4	5, 5	2, 3
	D	8, 1	3, 2	0, 0

It is easy to see that no strategy is dominated and thus that strict dominance cannot be applied to this game. For the same reason, IESDS and rationalizability will conclude that anything can happen. However, a pure-strategy Nash equilibrium does exist, and in fact it is unique. To find it we use a simple method that builds on the fact that any Nash equilibrium must call for a pair of strategies in which each of the two players is playing a best response to his opponent's strategy. The procedure is best explained in three steps:

Step 1: For every *column,* which is a strategy of player 2, find the highest payoff entry for player 1. By definition this entry must be in the row that is a best response for the particular column being considered. Underline the pair of payoffs in this row under this column:

		Player 2		
		L	*C*	*R*
Player 1	*U*	7, 7	4, 2	1, 8
	M	2, 4	$\underline{5, 5}$	$\underline{2, 3}$
	D	$\underline{8, 1}$	3, 2	0, 0

Step 1 identifies the best response of player 1 *for each of the pure strategies* (columns) of player 2. For instance, if player 2 is playing L, then player 1's best response is D, and we underline the payoffs associated with this row in column 1. After performing this step we see that there are three pairs of pure strategies at which player 1 is playing a best response: (D, L), (M, C), and (M, R).

Step 2: For every *row,* which is a strategy of player 1, find the highest payoff entry for player 2. By definition this entry must be in the column that is a best response for the particular row being considered. Overline the pair of payoffs in this entry:

		Player 2		
		L	*C*	*R*
Player 1	*U*	7, 7	4, 2	$\overline{1, 8}$
	M	2, 4	$\overline{\underline{5, 5}}$	$\underline{2, 3}$
	D	$\underline{8, 1}$	$\overline{3, 2}$	0, 0

Step 2 similarly identifies the pairs of strategies at which player 2 is playing a best response. For instance, if player 1 is playing D, then player 2's best response is C, and we overline the payoffs associated with this column in row 3. We can continue to conclude that player 2 is playing a best response at three strategy pairs: (D, C), (M, C), and (U, R).

Step 3: If any matrix entry has both an under- and an overline, it is the outcome of a Nash equilibrium in pure strategies.

This follows immediately from the fact that both players are playing a best response at any such pair of strategies. In this example we find that (M, C) is the unique pure-strategy Nash equilibrium—it is the only pair of pure strategies for which both players are playing a best response. If you apply

this approach to the Battle of the Sexes, for example, you will find both pure-strategy Nash equilibria, (O, O) and (F, F). For the Prisoner's Dilemma only (F, F) will be identified.

5.1.2 Evaluating the Nash Equilibria Solution

Considering our criteria for evaluating solution concepts, we can see from the Battle of the Sexes example that we may not have a unique Nash equilibrium. However, as alluded to in our earlier discussion of the Battle of the Sexes game, there is no reason to expect that we should. Indeed we may need to entertain other aspects of an environment in which players interact, such as social norms and historical beliefs, to make precise predictions about which of the possible Nash equilibria may result as the more likely outcome.

In Section 5.2.4 we will analyze a price competition game in which a Nash equilibrium may fail to exist. It turns out, however, that for quite general conditions games will have at least one Nash equilibrium. For the interested reader, Section 6.4 discusses some conditions that guarantee the existence of a Nash equilibrium, which was a central part of Nash's Ph.D. dissertation. This fact gives the Nash solution concept its power—like IESDS and rationalizability, the solution concept of Nash is widely applicable. It will, however, usually lead to more refined predictions than those of IESDS and rationalizability, as implied by proposition 5.1.

As with the previous solution concepts, we can easily see that Nash equilibrium does not guarantee Pareto optimality. The theme should be obvious by now: left to their own devices, people in many situations will do what is best for them, at the expense of social efficiency. This point was made quite convincingly and intuitively in Hardin's (1968) "tragedy of the commons" argument, which we explore in Section 5.2.2. This is where our focus on *self-enforcing outcomes* has its bite: our solution concepts took the game as given, and they imposed rationality and common knowledge of rationality to try to see what players would choose to do. If they each seek to maximize their individual well-being then the players may hinder their ability to achieve socially optimal outcomes.

5.2 Nash Equilibrium: Some Classic Applications

The previous section introduced the central pillar of modern noncooperative game theory, the Nash equilibrium solution concept. It has been applied widely in economics, political science, legal studies, and even biology. In what follows we demonstrate some of the best-known applications of the concept.

5.2.1 Two Kinds of Societies

The French philosopher Jean-Jacques Rousseau presented the following situation that describes a trade-off between playing it safe and relying on others to achieve a larger gain. Two hunters, players 1 and 2, can each choose to hunt a stag (S), which provides a rather large and tasty meal, or hunt a hare (H)—also tasty, but much less filling. Hunting stags is challenging and requires mutual cooperation. If either hunts a stag alone, the chance of success is negligible, while hunting hares is an individualistic enterprise that is not done in pairs. Hence hunting stags is most beneficial for society but requires "trust" between the hunters in that each believes that the other is joining

forces with him. The game, often referred to as the Stag Hunt game, can be described by the following matrix:

	S	H
S	5, 5	0, 3
H	3, 0	3, 3

It is easy to see that the game has two pure-strategy equilibria: (S, S) and (H, H). However, the payoff from (S, S) Pareto dominates that from (H, H). Why then would (H, H) ever be a reasonable prediction? This is precisely the strength of the Nash equilibrium concept. If each player anticipates that the other will not join forces, then he knows that going out to hunt the stag alone is not likely to be a successful enterprise and that going after the hare will be better. This belief would result in a society of individualists who do not cooperate to achieve a better outcome. In contrast, if the players expect each other to be cooperative in going after the stag, then this anticipation is self-fulfilling and results in what can be considered a cooperative society. In the real world, societies that may look very similar in their endowments, access to technology, and physical environments have very different achievements, all because of self-fulfilling beliefs or, as they are often called, *norms* of behavior.[2]

5.2.2 The Tragedy of the Commons

The *tragedy of the commons* refers to the conflict over scarce resources that results from the tension between individual selfish interests and the common good; the concept was popularized by Hardin (1968). The central idea has proven useful for understanding how we have come to be on the brink of several environmental catastrophes.

Hardin introduces the hypothetical example of a pasture shared by local herders. Each herder wants to maximize his yield, increasing his herd size whenever possible. Each additional animal has a positive effect for its herder, but the cost of that extra animal, namely degradation of the overall quality of the pasture, is shared by all the other herders. As a consequence the individual incentive for each herder is to grow his own herd, and in the end this scenario causes tremendous losses for everyone. To those trained in economics, it is yet another example of the distortion that results from the "free-rider" problem. It should also remind you of the Prisoner's Dilemma, in which individuals driven by selfish incentives cause pain to the group.

In the course of his essay, Hardin develops the theme, drawing on examples of such latter-day commons as the atmosphere, oceans, rivers, fish stocks, national parks, advertising, and even parking meters. A major theme running throughout the essay is the growth of human populations, with the earth's resources being a global commons. (Given that this example concerns the addition of extra "animals" to the population, it is the closest to his original analogy.)

2. For an excellent exposition of the role that beliefs play in societies, see Greif (2006). The idea that coordinated changes are needed for developing countries to move out of poverty and into industrial growth dates back to Paul Rosenstein-Rodan's theory of the "big push," which is explored further in Murphy et al. (1989).

Let's put some game theoretic analysis behind this story. Imagine that there are n players, say firms, in the world, each choosing how much to produce. Their production activity in turn consumes some of the clean air that surrounds our planet. There is a total amount of clean air equal to K, and any consumption of clean air comes out of this common resource. Each player i chooses his own consumption of clean air for production, $k_i \geq 0$, and the amount of clean air left is therefore $K - \sum_{i=1}^{n} k_i$. The benefit of consuming an amount $k_i \geq 0$ gives player i a benefit equal to $\ln(k_i)$, and no other player benefits from i's choice. Each player also enjoys consuming the remainder of the clean air, giving each a benefit $\ln(K - \sum_{i=1}^{n} k_i)$. Hence the payoff for player i from the choice $k = (k_1, k_2, \ldots, k_n)$ is equal to

$$v_i(k_i, k_{-i}) = \ln(k_i) + \ln\left(K - \sum_{j=1}^{n} k_j\right). \tag{5.1}$$

To solve for a Nash equilibrium we can compute the best-response correspondences for each player and then find a strategy profile for which all the best-response functions are satisfied together. This is an important point that warrants further emphasis. We know that given k_{-i}, player i will want to choose an element in $BR_i(k_{-i})$. Hence if we find some profile of choices $(k_1^*, k_2^*, \ldots, k_n^*)$ for which $k_i^* = BR_i(k_{-i}^*)$ for all $i \in N$ then this must be a Nash equilibrium.

This means that if we derive all n best-response correspondences, and it turns out that they are functions (unique best responses), then we have a system of n equations, one for each player's best-response function, with n unknowns, the choices of each player. Solving this system will yield a Nash equilibrium. To get player i's best-response function (and we will verify that it is a function), we write down the first-order condition of his payoff function:

$$\frac{\partial v_i(k_i, k_{-i})}{\partial k_i} = \frac{1}{k_i} - \frac{1}{K - \sum_{j=1}^{n} k_j} = 0,$$

and this gives us player i's best response function,[3]

$$BR_i(k_{-i}) = \frac{K - \sum_{j \neq i} k_j}{2}.$$

We therefore have n such equations, one for each player, and if we substitute the choice k_i instead of $BR_i(k_{-i})$ we get the n equations with n unknowns that need to be solved.

We proceed to solve the equilibrium for two players and leave the n-player case as exercise 5.7 at the end of the chapter. Letting $k_i(k_j)$ be the best response of player i, we have two best-response equations:

$$k_1(k_2) = \frac{K - k_2}{2} \qquad \text{and} \qquad k_2(k_1) = \frac{K - k_1}{2}.$$

These two equations are plotted in Figure 5.1. As the figure illustrates, the more player j consumes, the less player i wants to consume. In particular if player 2 consumes nothing (effectively not existing), then player 1 will consume $k_1 = \frac{K}{2}$, and as player

3. Of course, we are implicitly assuming that $\sum_{j=1}^{n} k_j \leq K$.

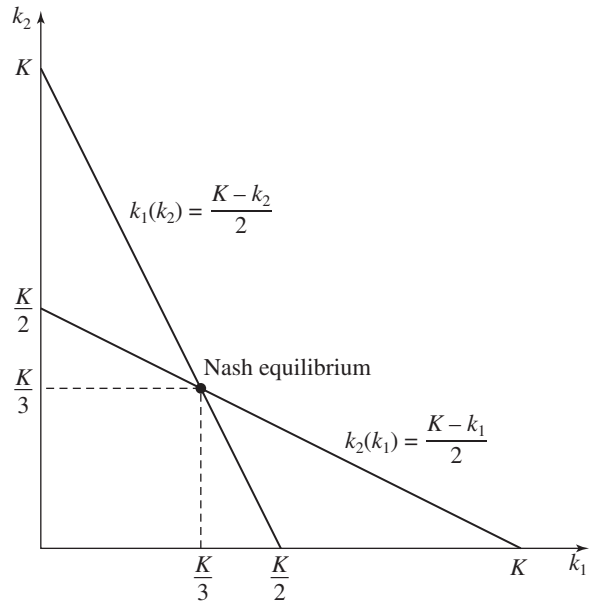


FIGURE 5.1 Best-response functions: two-player tragedy of the commons.

2's consumption increases toward K, player 1's consumption decreases toward zero. If we solve the two best-response functions simultaneously, we find the unique Nash equilibrium, which has both players playing $k_1 = k_2 = \frac{K}{3}$, as shown in Figure 5.1.

Now we can ask whether this two-player society could do better. Is consuming $\frac{K}{3}$ for each player too much or too little? The right way to answer these questions is using the Pareto criterion: can we find another consumption profile that will make everyone better off? If we can, we can compare that with the Nash equilibrium to answer this question. To find such a profile we'll use a little trick: we will maximize the sum of all the payoff functions, which we can think of as the "world's payoff function," $w(k_1, k_2)$. I won't go into the moral justification for using this approach, but it will turn out to be a useful tool.[4] The function we are maximizing is, therefore,

$$\max_{k_1,k_2} \; w(k_1, k_2) = \sum_{i=1}^{2} v_i(k_1, k_2) = \sum_{i=1}^{2} \ln(k_i) + 2 \ln \left(K - \sum_{i=1}^{2} k_i \right).$$

The first-order conditions for this problem are

$$\frac{\partial w(k_1, k_2)}{\partial k_1} = \frac{1}{k_1} - \frac{2}{K - k_1 - k_2} = 0$$

and

$$\frac{\partial w(k_1, k_2)}{\partial k_2} = \frac{1}{k_2} - \frac{2}{K - k_1 - k_2} = 0.$$

4. In general, maximizing the sum of utility functions, or maximizing total welfare, will result in a Pareto-optimal outcome, but it need not be the only one. In this example, this maximization gives us the symmetric Pareto-optimal consumption profile because the payoff function of each player is concave in his own consumption with $\frac{\partial v_i}{\partial k_i} > 0$, $\frac{\partial^2 v_i}{\partial k_i^2} < 0$, and $\lim_{k_i \to 0} \frac{\partial v_i}{\partial k_i} = \infty$.

Solving these two equations simultaneously will result in Pareto-optimal choices for k_1 and k_2. The unique solution to these two equations yields $k_1 = k_2 = \frac{K}{4}$, which means that from a social point of view the Nash equilibrium has the two players each consuming too much clean air. Indeed they would both be better off if each consumed $k_i = \frac{K}{4}$ instead of $k_i = \frac{K}{3}$.[5] In exercise 5.7 at the end of the chapter you are asked to show the consequences of having more than two players.

Thus, as Hardin puts it, giving people the freedom to make choices may make them all worse off than if those choices were somehow regulated. Of course the counterargument is whether we can trust a regulator to keep things under control; if not, the question remains which is the better of the two evils—an answer that game theory cannot offer!

5.2.3 Cournot Duopoly

Let's revisit the Cournot game with demand $P = a - bq$ and cost functions $c_i(q_i) = c_i q_i$ for firms $i \in \{1, 2\}$. The maximization problem that firm i faces when it believes that its opponent chooses quantity q_j is

$$\max_{q_i} \; v_i(q_i, q_j) = (a - bq_i - bq_j)q_i - c_i q_i.$$

Recall that the best-response function for each firm is given by the first-order condition, so that

$$BR_i(q_j) = \frac{a - bq_j - c_i}{2b}.$$

This means that each firm chooses quantities as follows:

$$q_1 = \frac{a - bq_2 - c_1}{2b} \qquad \text{and} \qquad q_2 = \frac{a - bq_1 - c_2}{2b}. \tag{5.2}$$

A pair of quantities (q_1, q_2) that are *mutual best responses* will be a Cournot-Nash equilibrium, which occurs when we solve both best-response functions (5.2) simultaneously. The best-response functions shown in Figure 5.2 depict the special case we solved earlier, in which $a = 100$, $b = 1$, and $c_1 = c_2 = 10$, in which case the unique Nash equilibrium is $q_1 = q_2 = 30$.

Notice that the Nash equilibrium coincides with the unique strategies that survive IESDS and rationalizability, which is the conclusion of proposition 5.1. An exercise that is left for you (exercise 5.8) is to explore the Cournot model with many firms.

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5. To see this we can calculate Δv_i, the difference between a player's payoff when we maximize total surplus $\left(\text{which we solved as } k_i = \frac{K}{4}\right)$ and his Nash equilibrium payoff:

$$\begin{aligned}
\Delta v_i &= \ln\left(\frac{K}{4}\right) + \ln\left(\frac{K}{2}\right) - \ln\left(\frac{K}{3}\right) - \ln\left(\frac{K}{3}\right) \\
&= \ln(K) - \ln(4) + \ln(K) - \ln(2) - \ln(K) + \ln(3) - \ln(K) + \ln(3) \\
&= 2\ln(3) - \ln(2) - \ln(4) \\
&= 0.051 > 0.
\end{aligned}$$

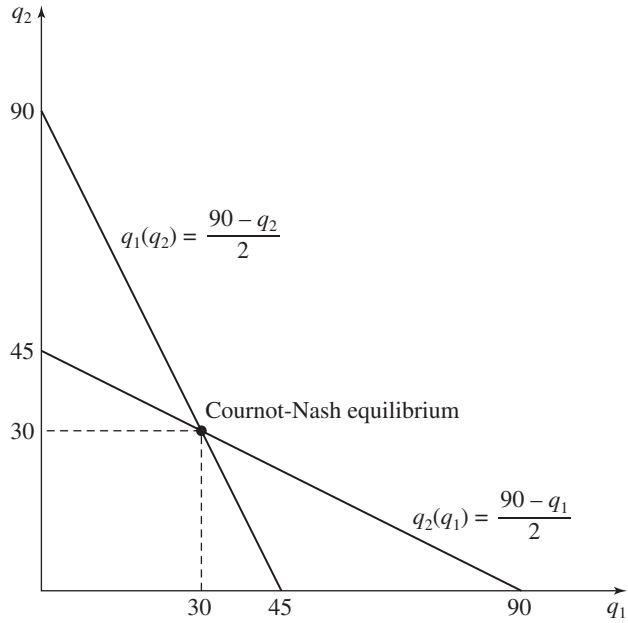


FIGURE 5.2 Cournot duopoly game: best-response functions and Nash equilibrium.

5.2.4 Bertrand Duopoly

The Cournot model assumed that the firms choose quantities and the market price adjusts to clear the demand. However, one can argue that firms often set prices and let consumers choose from where they will purchase, rather than setting quantities and waiting for the market price to equilibrate demand. We now consider the game in which each firm posts a price for otherwise identical goods. This was the situation modeled and analyzed by Joseph Bertrand (1883).

As before, assume that demand is given by $p = 100 - q$ and cost functions $c_i(q_i) = 10q_i$ for firms $i \in \{1, 2\}$. Clearly we would expect all buyers to buy from the firm whose price is the lowest. What happens if there is a tie? Let's assume that the market splits equally between the two firms. This gives us the following normal-form game:

Players: $N = \{1, 2\}$.

Strategy sets: $S_i = [0, \infty]$ for $i \in \{1, 2\}$, and firms choose prices $p_i \in S_i$.

Payoffs: To calculate payoffs we need to know what the quantities will be for each firm. Given our assumption on ties, the quantities are given by

$$q_i(p_i, p_j) = \begin{cases} 100 - p_i & \text{if } p_i < p_j \\ 0 & \text{if } p_i > p_j \\ \frac{100 - p_i}{2} & \text{if } p_i = p_j, \end{cases}$$

which in turn means that the payoff function is given by

$$v_i(p_i, p_j) = \begin{cases} (100 - p_i)(p_i - 10) & \text{if } p_i < p_j \\ 0 & \text{if } p_i > p_j \\ \frac{100 - p_i}{2}(p_i - 10) & \text{if } p_i = p_j. \end{cases}$$

Now that the description of the game is complete, we can try to calculate the best-response functions of both firms. To do this we will start with a slight modification that is motivated by reality: assume that prices cannot be *any* real number $p \geq 0$, but instead are limited to increments of some small fixed value, say $\varepsilon > 0$, which implies that the strategy (price) sets are $S_i = \{0, \varepsilon, 2\varepsilon, 3\varepsilon, \ldots\}$. For example, if we are considering cents as the price increment, so that $\varepsilon = 0.01$, then the strategy set will be $S_i = \{0, 0.01, 0.02, 0.03, \ldots\}$. We will very soon see what happens when this small denomination ε becomes very small and approaches zero.

We derive the best response of a firm by exhausting the relevant situations that it can face. It is useful to start with the situation in which only one monopolistic firm is in the market. We can calculate the monopoly price, which is the price that would maximize a single firm's profits if there were no competitors. This would be obtained by maximizing $v_i(p) = pq - 10q = (100 - p)(p - 10)$ and the first-order condition is $110 - 2p = 0$, resulting in an optimal price of $p = 55$, in a quantity of $q = 45$, and in profits equal to \$2025.

Let us now turn back to the duopoly with two firms and consider the case in which $p_j > 55$. It is easy to see that firm i can act as if there was no competition: just set the monopoly price of 55 and get the whole market. Hence we conclude that if $p_j > 55$ then the best response of firm i is to set $p_i = 55$.

It is also easy to see that in the case in which $p_j < 10$ then the best response of firm i is to set a price that is higher than that set by firm j. If it charges a price $p_i \leq p_j$ then it will sell a positive quantity at a price that is lower than its costs, causing firm i to lose money. If it charges a price $p_i > p_j$ then it sells nothing and loses nothing.

Now consider the case $55 \geq p_j \geq 10.02$. Firm i can choose one of three options: either set $p_i > p_j$ and get nothing, set $p_i = p_j$ and split the market, or set $p_i < p_j$ and get the whole market. It is not too hard to establish that firm i wants to just undercut firm j and capture the whole market, a goal that can be accomplished by setting a price of $p_i = p_j - 0.01$. To see this, observe that if $p_j > 10.01$ then by setting $p_i = p_j$ firm i gets $v_i = \frac{100-p_j}{2}(p_j - 10)$, while if it sets $p_i = p_j - 0.01$ it will get $v_i' = (100 - (p_j - 0.01))((p_j - 0.01) - 10)$. We can calculate the difference between the two as follows:

$$\Delta v_i = v_i' - v_i = (100 - (p_j - 0.01))((p_j - 0.01) - 10) - \frac{100 - p_j}{2}(p_j - 10)$$

$$= 55.02p_j - 0.5p_j^2 - 501.1.$$

It is easy to check that Δv_i is positive at $p_j = 10.02$ (it is equal to 0.0002). To see that it is positive for all values of $p \in [10.02, 55]$, we show that Δv_i has a positive derivative for any $p_j \in [10.02, 55]$, which implies that this difference grows even more positive as p_j increases in this domain. That is,

$$\frac{d\Delta v_i}{dp_j} = 55.02 - p_j > 0 \quad \text{for all } p < 55.02.$$

Thus we conclude that when $p_j \in [10.02, 55]$ the best response of firm i is to charge \$0.01 less, that is, $p_i = p_j - 0.01$.

To complete the analysis, we have to explore two final cases: $p_j = 10.01$ and $p_j = 10$. The three options to consider are setting $p_i = p_j$, $p_i > p_j$, or $p_i < p_j$. When $p_j = 10.01$ then undercutting j's price means setting $p_i = 10$, which gives i zero profits and is the same as setting $p_i > p_j$. Thus the best response is setting $p_i = p_j = 10.01$ and splitting the market with very low profits. Finally, if $p_j = 10$

then any choice of price $p_i \geq p_j$ will yield firm i zero profits, whereas setting $p_i < p_j$ causes losses. Therefore any price $p_i \geq p_j$ is a best response when $p_j = 10$.

In summary we calculated:

$$BR_i(p_j) = \begin{cases} 55 & \text{if } p_j > 55 \\ p_j - 0.01 & \text{if } 55 \geq p_j \geq 10.02 \\ 10.01 & \text{if } p_j = 10.01 \\ p_i \in \{10, 10.01, 10.02, 10.03, \ldots\} & \text{if } p_j = 10. \end{cases}$$

Now given that firm j's best response is exactly symmetric, it should not be hard to see that there are two Nash equilibria that follow immediately from the form of the best-response functions: *The best response* to $p_j = 10.01$ is $BR_i(10.01) = 10.01$, and *a best response* to $p_j = 10$ is $p_i = 10$ or $10 \in BR_i(10)$. Thus the two Nash equilibria are

$$(p_1, p_2) \in \{(10, 10), (10.01, 10.01)\}.$$

It is worth pausing here for a moment to address a common point of confusion, which often arises when a player has more than one best response to a certain action of his opponents. In this example, when $p_2 = 10$, player 1 is indifferent regarding *any price at or above 10 that he chooses:* if he splits the market with $p_1 = 10$ he gets half the market with no profits, and if he sets $p_1 > 10$ he gets no customers and has no profits. One may be tempted to jump to the following conclusion: if player 2 is choosing $p_2 = 10$ then any choice of $p_1 \geq 10$ together with $p_2 = 10$ will be a Nash equilibrium. This is incorrect! It is true that player 1 is playing a best response with any one of his choices, but if $p_1 > 10$ then $p_2 = 10$ is *not* a best response of player 2 to p_1, as we can observe from the foregoing analysis.

Comparing the outcome of the Bertrand game to that of the Cournot game is an interesting exercise. Notice that when firms choose quantities (Cournot), the unique Nash equilibrium is $q_1 = q_2 = 30$. A quick calculation shows that for the aggregate quantity of $q = q_1 + q_2 = 60$ we get a demand price of $p = 40$ and each firm makes a profit of \$900. When instead these firms compete on prices, the two possible equilibria have either zero profits when both choose $p_1 = p_2 = \$10$ or negligible profits (about \$0.45) when they each choose $p_1 = p_2 = \$10.01$. Interestingly, for both the Cournot and Bertrand games, if we had only one player then he would maximize profits by choosing the monopoly price (or quantity) of \$55 (or 45 units) and earn a profit of \$2025.

The message of this analysis is quite striking: one firm may have monopoly power, but when we let one more firm compete, and they compete on prices, then the market will behave competitively—if both choose a price of \$10 then price will equal marginal costs! Notice that if we add a third and a fourth firm this will not change the outcome; prices will have to be \$10 (or practically the same at \$10.01) for all firms in the Nash (Bertrand) equilibrium. This is not the case for Cournot competition, in which firms manage to obtain some market power as long as the number of firms is not too large.

A quick observation should lead you to realize that if we let ε be smaller than \$0.01, the conclusions we reached earlier will be sustained, and we will have two Nash equilibria, one with $p_1 = p_2 = 10$ and one with $p_1 = p_2 = 10 + \varepsilon$. Clearly these two equilibria become "closer" in profits as ε becomes smaller, and they converge to each other as ε approaches zero.

It turns out that if we assume that prices can be chosen as *any* real number, we get a very "clean" result: the unique Bertrand-Nash equilibrium will have prices equal to marginal costs, implying a competitive outcome. We prove this for the more general symmetric case in which $c_i(q_i) = cq_i$ for both firms and demand is equal to $p = a - bq$ with $a > c$.[6]

Proposition 5.2 *For $\varepsilon = 0$ (prices can be any real number) there is a unique Nash equilibrium: $p_1 = p_2 = c$.*

Proof First note that in any equilibrium $p_i \geq c$ for both firms—otherwise at least one firm offering a price lower than c will lose money (pay the consumers to take its goods!). We therefore need to show that $p_i > c$ cannot be part of any equilibrium. We can see this in two steps:

1. If $p_1 = p_2 = \hat{p} > c$ then each would benefit from lowering its price to some price $\hat{p} - \varepsilon$ (ε very small) and get the whole market for almost the same price.
2. If $p_1 > p_2 \geq c$ then player 2 would want to deviate to $p_1 - \varepsilon$ (ε very small) and earn higher profits.

It is easy to see that $p_1 = p_2 = c$ is an equilibrium: Firm i's best response to $p_j = c$ is $p_i(c) \geq c$. That is, any price at or above marginal costs c is a best response to the other player charging c. Hence $p_1 = p_2 = c$ is the unique equilibrium because each is playing a best response to the other's choice and neither wants to deviate to a different price. ■

We will now see an interesting variation of the Bertrand game. Assume that $c_i(q_i) = c_i q_i$ represents the cost of each firm as before. Now, however, let $c_1 = 1$ and $c_2 = 2$ so that the two firms are not identical: firm 1 has a cost advantage. Let the demand still be $p = 100 - q$.

Now consider the case with discrete price jumps with $\varepsilon = 0.01$. The firms are not symmetric in that firm 1 has a lower marginal cost than firm 2, and unlike the example in which both had the same costs we cannot have a Nash equilibrium in which both charge the same price. To see this, imagine that $p_1 = p_2 = 2.00$. Firm 2 has no incentive to deviate, but this is no longer true for firm 1, which will be happy to cut its price by 0.01. We know that firm 2 will not be willing to sell at a price below $p = 2$, so one possible Nash equilibrium (you are asked to find more in exercise 5.12 at the end of this chapter) is

$$(p_1^*, p_2^*) = (1.99, 2.00).$$

Now we can ask ourselves what happens if $\varepsilon = 0$. If we would think of using a "limit" approach to answer this question then we may expect a result similar to the one we saw before: if we focus on the equilibrium pair $(p_1^*, p_2^*) = (2 - \varepsilon, 2)$ then as $\varepsilon \to 0$ we must get the Nash equilibrium $p_1 = p_2 = 2$. But is this really an equilibrium? Interestingly, the answer is no! To see this, consider the best response of firm 1. Its payoff function is not continuous when firm 2 offers a price of 2 (or any other positive price). The profit function of firm 1, as a function of p_1 when $p_2 = 2$, is depicted in Figure 5.3. The figure first draws the profits of firm 1 as if it were a monopolist with

6. The condition $a > c$ is necessary for firms to be able to produce positive quantities and not lose money.

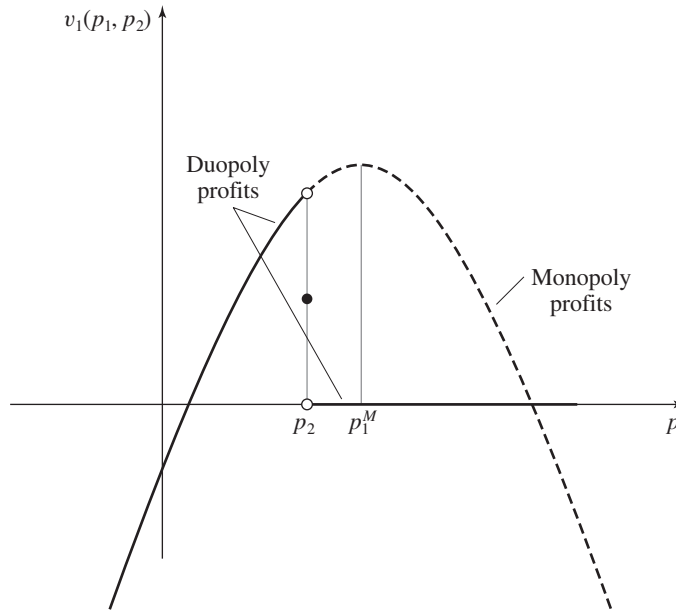


FIGURE 5.3 The profit function in the Bertrand duopoly game.

no competition (the hump-shaped curve), and if this were the case it would charge its monopoly price $p_1^M = 50.5$.[7] If firm 2 charges more than the monopoly price, this will have no impact on the choice of firm 1—it will still charge the monopoly price. If, however, firm 2 charges a price p_2 that is less than the monopoly price then there is a **discontinuity in the payoff function** of firm 1: as its price p_1 approaches p_2 from below, its profits rise. However, when it hits p_2 exactly then it will split the market and see its profits drop by half. Algebraically the profit function of firm 1 when $p_2 = 2$ is given by

$$v_i(p_1, 2) = \begin{cases} (100 - p_1)(p_1 - 1) & \text{if } p_1 < 2 \\ \frac{(100-p_1)(p_1-1)}{2} & \text{if } p_1 = 2 \\ 0 & \text{if } p_1 > 2. \end{cases}$$

This discontinuity causes firm 1 to *not* have a well-defined best response correspondence when $p_2 < 50.5$. Firm 1 wants to set a price *as close to* p_2 *as it can,* but it does not want to reach p_2 because then it splits the market and experiences a sizable decrease in profits. Once its price goes above p_2 then firm 1's profits drop further to zero. Indeed the consequence must be that a Nash equilibrium does not exist precisely because firm 1 does not have a "well-behaved" payoff function.

To see this directly, first observe that there cannot be a Nash equilibrium with $p_i \geq p_j > 2$: firm i would want to deviate to some $p \in (2, p_j)$. Second, observe that there cannot be a Nash equilibrium with $p_i \leq p_j < 1$: firm i would want to deviate to any $p > p_j$. Hence if there is a Nash equilibrium, it must have the prices between 1 and 2. Similar to the first observation, within this range we cannot have $p_1 \geq p_2$ (firm

7. The maximization here is for the profit function $v_1 = (100 - p)(p - c_1)$, where $c_1 = 1$.

1 would want to deviate just slightly below p_2), and similar to the second observation, we cannot have $p_2 \leq p_1$ (firm 2 would want to deviate to any price above p_1).

This problem of payoff discontinuity is one that we will avoid in the remainder of this book precisely because it will often lead to problems of nonexistence of equilibrium. We need to remember that it is a problem that disappears when we have some discreteness in the actions of players, and if we think that our choice of continuous strategies is one of convenience (to use calculus for optimization), then we may feel comfortable enough ignoring such anomalies. We discuss the existence of Nash equilibria further in Section 6.4.

Remark It is worth pointing out an interesting difference between the Cournot game and the Bertrand game. In the Cournot game the best-response function of each player is downward sloping. That is, the more player j produces, the lower is the best-response quantity of player i. In the Bertrand game, however, for prices between marginal costs (equal to 10 in the leading example) and the monopoly price (equal to 45), the higher the price set by player j, the higher is the best-response price of player i. These differences have received some attention in the literature. Games for which the best response of one player decreases in the choice of the other, like the Cournot game, are called games with **strategic substitutes.** Another example of a game with strategic substitutes is the tragedy of the commons. In contrast, games for which the best response of one player increases in the choice of the other, like the Bertrand game, are called games with **strategic complements.** Another example of a game with strategic complements appears in exercise 5.10 at the end of the chapter. There are several interesting insights to be derived from distinguishing between strategic substitutes and strategic complements. For a nice example see Fudenberg and Tirole (1984).

5.2.5 Political Ideology and Electoral Competition

Given a population of citizens who vote for political candidates, how should candidates position themselves along the political spectrum? One view of the world is that a politician cares only about representing his true beliefs, and that drives the campaign. Another more cynical view is that politicians care only about getting elected and hence will choose a platform that maximizes their chances. This is precisely the view taken in the seminal model introduced by Hotelling (1929).[8]

To consider a simple variant of Hotelling's original model, imagine that there are two politicians, each caring only about being elected. There are 101 citizens, each labeled by an integer $-50, -49, \ldots, 0, \ldots, +49, +50$. Each citizen has political preferences: for simplicity let's call the "-50" citizen the most "left"-leaning citizen and the "$+50$" citizen the most "right"-leaning citizen.

Each candidate i chooses his platform as a policy $a_i \in \{-50, -49, \ldots, 0, \ldots, +49, +50\}$ so that each policy is associated with the citizen for whom this policy is ideal. Each citizen chooses the candidate whose platform is closest to his political preferences. For example, if candidate 1 chooses platform $a_1 = -15$ while candidate 2 chooses platform $a_2 = +22$, then all the citizens at or above $+22$ will surely vote

8. Hotelling's main object of analysis was competition between firms. However, he did also discuss the example of electoral competition, yielding important insights into rational choice–based political science.

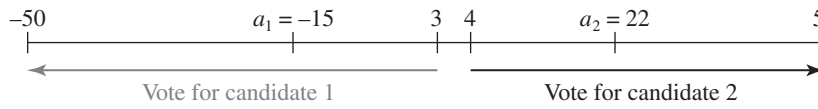


FIGURE 5.4 The Hotelling model of voting behavior.

for candidate 2, all the citizens at or below -15 will surely vote for candidate 1, and those in between will split between the candidates. In particular those citizens at or below 3 will vote for candidate 1, while those at or above 4 will vote for candidate 2. The reason is that -15 is at a distance of 18 away from citizen 3, while $+22$ is at a distance of 19 away. This is shown in Figure 5.4.

The outcome is determined by majority rule: if a majority of citizens vote for candidate i then he wins. Since there is an odd number of voters, unless someone is indifferent, one candidate will always win. In the event that a citizen is indifferent between the candidates then the citizen tosses a coin to determine for whom to vote. Assume that our candidates want to win, so that they prefer winning to a tie, and they prefer a tie to losing. Now consider the best response of player i. If player j chooses a policy $a_j > 0$ then by choosing $a_i = a_j$ or $a_i = -a_j$ there will be a tie.[9] By choosing $a_i > a_j$ or $a_i < -a_j$, player i will surely lose, while by choosing $a_i \in [-a_j + 1, a_j - 1]$, player i will win, so any platform in this interval is a best response to $a_j > 0$. Similarly a symmetric argument implies that any platform in the interval $a_i \in [a_j + 1, -a_j - 1]$ is a best response to $a_j < 0$. Observe that the best response to $a_j = 0$ is zero.[10] Thus we can write the best-response correspondence of each player as

$$BR_i(a_j) = \begin{cases} [a_j + 1, -a_j - 1] & \text{if } a_j < 0 \\ 0 & \text{if } a_j = 0 \\ [-a_j + 1, a_j - 1] & \text{if } a_j > 0. \end{cases}$$

From here it is easy to see that there is a unique Nash equilibrium, $a_1 = a_2 = 0$, implying that both candidates position their platforms smack in the middle of the political spectrum! Indeed, as Hotelling (1929, p. 54) wrote, "The competition for votes between the Republican and Democratic parties does not lead to a clear drawing of issues, and adoption of two strongly contrasted positions between which the voter may choose. Instead, each party strives to make its platform as much like the other's as possible." A fine insight in 1929, and one that is echoed frequently today.

This simple example is related to a powerful result known as the **median voter theorem.** It states that if voters are different from one another along a single-dimensional "preference" line, as in Hotelling's model, and if each prefers his own political location, with other platforms being less and less attractive the farther away

9. For example, if candidate 1 chooses $a_1 = -10$ and candidate 2 chooses $a_2 = 10$ then all the "positive" citizens will vote for candidate 2, all the "negative" ones will vote for candidate 1, and citizen 0 is indifferent so he may choose not to vote. In this event there will be a tie. If we force every citizen to vote one way or another then there will be no ties because we assumed an odd number of citizens.

10. Notice that the best response to $a_j = -1$ or $a_j = +1$ is also unique and equal to $a_i = 0$, because in these cases the interval previously identified collapses to 0.

they fall to either side of that location,[11] then the political platform located at the median voter will defeat *any other platform* in a simple majority vote. The theorem was first articulated by Black (1948), and it received prominence in Downs's famous 1957 book. Nevertheless one can see how the seed of the idea had been planted as far back as Hotelling's formalization of spatial competition.

Remark In the more common representation of the Hotelling competition model, the citizens are a continuum of voters, say, given by the interval $A = [\underline{a}, \overline{a}]$, with distribution $F(a)$ to determine the distribution of each political preference. We define the **median voter** as that voter a^m for which $\Pr\{a \leq a^m\} = \frac{1}{2}$. $\Big($For example, if $F(\cdot)$ is uniform then $a^m = \frac{\underline{a}+\overline{a}}{2}$.$\Big)$ The best-response correspondence is similar to that given earlier, but for all choices of player j that are not equal to a^m player i's best response is an open interval. For example, if $a_j > a^m$, and if $a' < a^m$ is an "opposite" policy that ties with a_j, then the best response of player i is to choose any platform that lies in the open interval (a', a_j). If, however, $a_j = a^m$ then player i's unique best response is to choose $a_i = a^m$, implying that both candidates choosing a^m is the unique Nash equilibrium. You are asked to prove this in exercise 5.15.

5.3 Summary

- Any strategy profile for which players are playing mutual best responses is a Nash equilibrium, making this equilibrium concept self-enforcing.
- If a profile of strategies is the unique survivor of IESDS or is the unique rationalizable profile of strategies then it is a Nash equilibrium.
- If a profile of strategies is a Nash equilibrium then it must survive IESDS and it must be rationalizable, but not every strategy that survives IESDS or that is rationalizable is a Nash equilibrium.
- Nash equilibrium analysis can shed light on phenomena such as the tragedy of the commons and the nature of competition in markets and in politics.

5.4 Exercises

5.1 **Prove Proposition 5.1.**

5.2 **Weak Dominance:** A strategy $s^W \in S$ is a weakly dominant strategy equilibrium if $s_i^W \in S_i$ is a weakly dominant strategy for all $i \in N$, that is, if $v_i(s_i^W, s_{-i}) \geq v_i(s_i', s_{-i})$ for all $s_i' \in S_i$ and for all $s_{-i} \in S_{-i}$. Provide an example of a game for which there is a weakly dominant strategy equilibrium as well as another Nash equilibrium.

5.3 **Nash and IESDS:** Consider a two-player game with m pure strategies for each player that can be represented by an $m \times m$ matrix.

a. Show that if $m = 2$ and the game has a unique pure-strategy Nash equilibrium then this is the unique strategy profile that survives IESDS.

11. This condition on preferences is called that of "single-peaked" preferences: if we draw the utility function of some voter with his political "bliss point" being $a \in A$, the utility is highest at a and declines in both directions, hence the single peak at a.

b. Show that if $m = 3$ and the game has a unique pure-strategy equilibrium then it may not be the only strategy profile that survives IESDS.

5.4 **Splitting Pizza:** You and a friend are in an Italian restaurant, and the owner offers both of you a free eight-slice pizza under the following condition. Each of you must simultaneously announce how many slices you would like; that is, each player $i \in \{1, 2\}$ names his desired amount of pizza, $0 \leq s_i \leq 8$. If $s_1 + s_2 \leq 8$ then the players get their demands (and the owner eats any leftover slices). If $s_1 + s_2 > 8$, then the players get nothing. Assume that you each care only about how much pizza you individually consume, and the more the better.

a. Write out or graph each player's best-response correspondence.
b. What outcomes can be supported as pure-strategy Nash equilibria?

5.5 **Public Good Contribution:** Three players live in a town, and each can choose to contribute to fund a streetlamp. The value of having the streetlamp is 3 for each player, and the value of not having it is 0. The mayor asks each player to contribute either 1 or nothing. If at least two players contribute then the lamp will be erected. If one player or no players contribute then the lamp will not be erected, in which case any person who contributed will not get his money back.

a. Write out or graph each player's best-response correspondence.
b. What outcomes can be supported as pure-strategy Nash equilibria?

5.6 **Hawk-Dove:** The following game has been widely used in evolutionary biology to understand how fighting and display strategies by animals could coexist in a population. For a typical Hawk-Dove game there are resources to be gained (e.g., food, mates, territories), denoted as v. Each of two players can choose to be aggressive, as Hawk (H), or compromising, as Dove (D). If both players choose H then they split the resources but lose some payoff from injuries, denoted as k. Assume that $k > \frac{v}{2}$. If both choose D then they split the resources but engage in some display of power that carries a display cost d, with $d < \frac{v}{2}$. Finally, if player i chooses H while j chooses D then i gets all the resources while j leaves with no benefits and no costs.

a. Describe this game in a matrix.
b. Assume that $v = 10$, $k = 6$, and $d = 4$. What outcomes can be supported as pure-strategy Nash equilibria?[12]

5.7 **The *n*-Player Tragedy of the Commons:** Suppose there are n players in the tragedy of the commons example in Section 5.2.2.

a. Find the Nash equilibrium of this game. How does n affect the Nash outcome?
b. Find the socially optimal outcome with n players. How does n affect this outcome?
c. How does the Nash equilibrium outcome compare to the socially efficient outcome as n approaches infinity?

5.8 **The *n*-Firm Cournot Model:** Suppose there are n firms in the Cournot oligopoly model. Let q_i denote the quantity produced by firm i, and let

12. In the evolutionary biology literature, the analysis performed is of a very different nature. Instead of considering the Nash equilibrium analysis of a static game, the analysis is a dynamic one in which successful strategies "replicate" in a large population. This analysis is part of a methodology called evolutionary game theory. For more on the subject see Gintis (2000).

$Q = q_i + \cdots + q_n$ denote the aggregate production. Let $P(Q)$ denote the market clearing price (when demand equals Q) and assume that the inverse demand function is given by $P(Q) = a - Q$, where $Q \leq a$. Assume that firms have no fixed cost and that the cost of producing quantity q_i is cq_i (all firms have the same marginal cost, and assume that $c < a$).

a. Model this as a normal-form game.
b. What is the Nash (Cournot) equilibrium of the game in which firms choose their quantities simultaneously?
c. What happens to the equilibrium price as n approaches infinity? Is this familiar?

5.9 **Tragedy of the Roommates:** You and your $n - 1$ roommates each have five hours of free time you could spend cleaning your apartment. You all dislike cleaning, but you all like having a clean apartment: each person's payoff is the total hours spent (by everyone) cleaning, minus a number c times the hours spent (individually) cleaning. That is,

$$v_i(s_1, s_2, \ldots, s_n) = -c \cdot s_i + \sum_{j=1}^{n} s_j.$$

Assume everyone chooses simultaneously how much time to spend cleaning.

a. Find the Nash equilibrium if $c < 1$.
b. Find the Nash equilibrium if $c > 1$.
c. Set $n = 5$ and $c = 2$. Is the Nash equilibrium Pareto efficient? If not, can you find an outcome in which everyone is better off than in the Nash equilibrium outcome?

5.10 **Synergies:** Two division managers can invest time and effort in creating a better working relationship. Each invests $e_i \geq 0$, and if both invest more then both are better off, but it is costly for each manager to invest. In particular the payoff function for player i from effort levels (e_i, e_j) is $v_i(e_i, e_j) = (a + e_j)e_i - e_i^2$.

a. What is the best-response correspondence of each player?
b. In what way are the best-response correspondences different from those in the Cournot game? Why?
c. Find the Nash equilibrium of this game and argue that it is unique.

5.11 **Wasteful Shipping Costs:** Consider two countries, A and B, each with a monopolist that owns the only active coal mine in the country. Let firm 1 be the firm located in country A and firm 2 the one in country B. Let q_i^j, $i \in \{1, 2\}$ and $j \in \{A, B\}$ denote the quantity that firm i sells in country j. Consequently let $q_i = q_i^A + q_i^B$ be the total quantity produced by firm $i \in \{1, 2\}$ and let $q^j = q_1^j + q_2^j$ be the total quantity sold in country $j \in \{A, B\}$. The demand for coal in countries A and B is then given respectively by

$$p^j = 90 - q^j, \quad j \in \{A, B\},$$

and the cost of production for each firm is given by

$$c_i(q_i) = 10q_i, \quad i \in \{1, 2\}.$$

a. Assume that the countries do not have a trade agreement and, in fact, that the importation of coal into either country is prohibited.

6

Mixed Strategies

In the previous chapters we restricted players to using pure strategies and we postponed discussing the option that a player may choose to randomize between several of his pure strategies. You may wonder why anyone would wish to randomize between actions. This turns out to be an important type of behavior to consider, with interesting implications and interpretations. In fact, as we will now see, there are many games for which there will be no equilibrium predictions if we do not consider the players' ability to choose stochastic strategies.

Consider the following classic zero-sum game called Matching Pennies.[1] Players 1 and 2 each put a penny on a table simultaneously. If the two pennies come up the same side (heads or tails) then player 1 gets both; otherwise player 2 does. We can represent this in the following matrix:

		Player 2	
		H	T
Player 1	H	$\underline{1, -1}$	$\overline{-1, 1}$
	T	$\overline{-1, 1}$	$\underline{1, -1}$

The matrix also includes the best-response choices of each player using the method we introduced in Section 5.1.1 to find pure-strategy Nash equilibria. As you can see, this method does not work: Given a belief that player 1 has about player 2's choice, he always wants to match it. In contrast, given a belief that player 2 has about player 1's choice, he would like to choose the opposite orientation for his penny. Does this mean that a Nash equilibrium fails to exist? We will soon see that a Nash equilibrium will indeed exist *if we allow players to choose random strategies,* and there will be an intuitive appeal to the proposed equilibrium.

Matching Pennies is not the only simple game that fails to have a pure-strategy Nash equilibrium. Recall the child's game rock-paper-scissors, in which rock beats

1. A *zero-sum game* is one in which the gains of one player are the losses of another, hence their payoffs always sum to zero. The class of zero-sum games was the main subject of analysis before Nash introduced his solution concept in the 1950s. These games have some very nice mathematical properties and were a central object of analysis in von Neumann and Morgenstern's (1944) seminal book.

scissors, scissors beats paper, and paper beats rock. If winning gives the player a payoff of 1 and the loser a payoff of -1, and if we assume that a tie is worth 0, then we can describe this game by the following matrix:

		Player 2		
		R	P	S
Player 1	R	0, 0	-1, 1	1, -1
	P	1, -1	0, 0	-1, 1
	S	-1, 1	1, -1	0, 0

It is rather straightforward to write down the best-response correspondence for player 1 when he believes that player 2 will play one of his pure strategies as follows:

$$s_1(s_2) = \begin{cases} P & \text{when } s_2 = R \\ S & \text{when } s_2 = P \\ R & \text{when } s_2 = S, \end{cases}$$

and a similar (symmetric) list would be the best-response correspondence of player 2. Examining the two best-response correspondences immediately implies that there is no pure-strategy equilibrium, just like in the Matching Pennies game. The reason is that, starting with any pair of pure strategies, at least one player is not playing a best response and will want to change his strategy in response.

6.1 Strategies, Beliefs, and Expected Payoffs

We now introduce the possibility that players choose stochastic strategies, such as flipping a coin or rolling a die to determine what they will choose to do. This approach will turn out to offer us several important advances over that followed so far. Aside from giving the players a richer set of actions from which to choose, it will more importantly give them a richer set of possible beliefs that capture an uncertain world. If player i can believe that his opponents are choosing stochastic strategies, then this puts player i in the same kind of situation as a decision maker who faces a decision problem with probabilistic uncertainty. If you are not familiar with such settings, you are encouraged to review Chapter 2, which lays out the simple decision problem with random events.

6.1.1 Finite Strategy Sets

We start with the basic definition of random play when players have finite strategy sets S_i:

Definition 6.1 Let $S_i = \{s_{i1}, s_{i2}, \ldots, s_{im}\}$ be player i's finite set of pure strategies. Define $\triangle S_i$ as the **simplex** of S_i, which is the set of all probability distributions over S_i. A **mixed strategy** for player i is an element $\sigma_i \in \triangle S_i$, so that $\sigma_i = \{\sigma_i(s_{i1}), \sigma_i(s_{i2}), \ldots, \sigma_i(s_{im}))$ is a probability distribution over S_i, where $\sigma_i(s_i)$ is the probability that player i plays s_i.

That is, a mixed strategy for player i is just a probability distribution over his pure strategies. Recall that any probability distribution $\sigma_i(\cdot)$ over a finite set of elements (a finite state space), in our case S_i, must satisfy two conditions:

1. $\sigma_i(s_i) \geq 0$ for all $s_i \in S_i$, and
2. $\sum_{s_i \in S_i} \sigma_i(s_i) = 1$.

That is, the probability of any event happening must be nonnegative, and the sum of the probabilities of all the possible events must add up to one.[2] Notice that every pure strategy is a mixed strategy with a *degenerate* distribution that picks a single pure strategy with probability one and all other pure strategies with probability zero.

As an example, consider the Matching Pennies game described earlier, with the matrix

		Player 2	
		H	T
Player 1	H	1, −1	−1, 1
	T	−1, 1	1, −1

For each player i, $S_i = \{H, T\}$, and the simplex, which is the set of mixed strategies, can be written as

$$\triangle S_i = \{(\sigma_i(H), \sigma_i(T)) : \sigma_i(H) \geq 0, \sigma_i(T) \geq 0, \sigma_i(H) + \sigma_i(T) = 1\}.$$

We read this as follows: the set of mixed strategies is the set of all pairs $(\sigma_i(H), \sigma_i(T))$ such that both are nonnegative numbers, and they both sum to one.[3] We use the notation $\sigma_i(H)$ to represent the probability that player i plays H and $\sigma_i(T)$ to represent the probability that player i plays T.

Now consider the example of the rock-paper-scissors game, in which $S_i = \{R, P, S\}$ (for rock, paper, and scissors, respectively). We can define the simplex as

$$\triangle S_i = \{(\sigma_i(R), \sigma_i(P), \sigma_i(S)) : \sigma_i(R), \sigma_i(P), \sigma_i(S) \geq 0, \sigma_i(R) + \sigma_i(P) + \sigma_i(S) = 1\},$$

which is now three numbers, each defining the probability that the player plays one of his pure strategies. As mentioned earlier, a pure strategy is just a special case of a mixed strategy. For example, in this game we can represent the pure strategy of playing R with the degenerate mixed strategy: $\sigma(R) = 1$, $\sigma(P) = \sigma(S) = 0$.

From our definition it is clear that when a player uses a mixed strategy, he may choose not to use all of his pure strategies in the mix; that is, he may have some pure strategies that are not selected with positive probability. Given a player's

2. The notation $\sum_{s_i \in S_i} \sigma(s_i)$ means the sum of $\sigma(s_i)$ over all the $s_i \in S_i$. If S_i has m elements, as in the definition, we could write this as $\sum_{k=1}^{m} \sigma_i(s_{ik})$.

3. The simplex of this two-element strategy set can be represented by a single number $p \in [0, 1]$, where p is the probability that player i plays H and $1 - p$ is the probability that player i plays T. This follows from the definition of a probability distribution over a two-element set. In general the simplex of a strategy set with m pure strategies will be in an $(m - 1)$-dimensional space, where each of the $m - 1$ numbers is in $[0, 1]$, and will represent the probability of the first $m - 1$ pure strategies. All sum to a number equal to or less than one so that the remainder is the probability of the mth pure strategy.

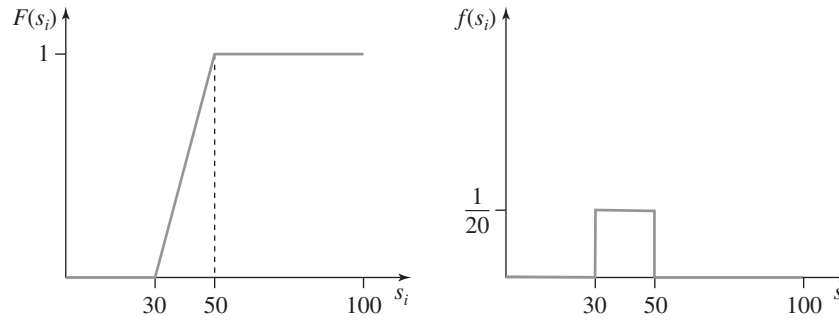


FIGURE 6.1 A continuous mixed strategy in the Cournot game.

mixed strategy $\sigma_i(\cdot)$, it will be useful to distinguish between pure strategies that are chosen with a positive probability and those that are not. We offer the following definition:

Definition 6.2 Given a mixed strategy $\sigma_i(\cdot)$ for player i, we will say that a pure strategy $s_i \in S_i$ is **in the support of** $\sigma_i(\cdot)$ if and only if it occurs with positive probability, that is, $\sigma_i(s_i) > 0$.

For example, in the game of rock-paper-scissors, a player can choose rock or paper, each with equal probability, and not choose scissors. In this case $\sigma_i(R) = \sigma_i(P) = 0.5$ and $\sigma_i(S) = 0$. We will then say that R and P are in the support of $\sigma_i(\cdot)$, but S is not.

6.1.2 Continuous Strategy Sets

As we have seen with the Cournot and Bertrand duopoly examples, or the tragedy of the commons example in Section 5.2.2, the pure-strategy sets that players have need not be finite. In the case in which the pure-strategy sets are well-defined intervals, a mixed strategy will be given by a cumulative distribution function:

Definition 6.3 Let S_i be player i's pure-strategy set and assume that S_i is an interval. A **mixed strategy** for player i is a cumulative distribution function $F_i : S_i \to [0, 1]$, where $F_i(x) = \Pr\{s_i \leq x\}$. If $F_i(\cdot)$ is differentiable with density $f_i(\cdot)$ then we say that $s_i \in S_i$ is in the support of $F_i(\cdot)$ if $f_i(s_i) > 0$.

As an example, consider the Cournot duopoly game with a capacity constraint of 100 units of production, so that $S_i = [0, 100]$ for $i \in \{1, 2\}$. Consider the mixed strategy in which player i chooses a quantity between 30 and 50 using a uniform distribution. That is,

$$F_i(s_i) = \begin{cases} 0 & \text{for } s_i < 30 \\ \frac{s_i - 30}{20} & \text{for } s_i \in [30, 50] \\ 1 & \text{for } s_i > 50 \end{cases} \quad \text{and} \quad f_i(s_i) = \begin{cases} 0 & \text{for } s_i < 30 \\ \frac{1}{20} & \text{for } s_i \in [30, 50] \\ 0 & \text{for } s_i > 50. \end{cases}$$

These two functions are depicted in Figure 6.1.

We will typically focus on games with finite strategy sets to illustrate most of the examples with mixed strategies, but some interesting examples will have infinite strategy sets and will require the use of cumulative distributions and densities to explore behavior in mixed strategies.

6.1.3 Beliefs and Mixed Strategies

As we discussed earlier, introducing probability distributions not only enriches the set of actions from which a player can choose but also allows us to enrich the beliefs that players can have. Consider, for example, player i, who plays against opponents $-i$. It may be that player i is uncertain about the behavior of his opponents for many reasons. For example, he may believe that his opponents are indeed choosing mixed strategies, which immediately implies that their behavior is not fixed but rather random. An alternative interpretation is the situation in which player i is playing a game against an opponent that he does not know, whose background will determine how he will play. This interpretation will be revisited in Section 12.5, and it is a very appealing justification for beliefs that are random and behavior that is consistent with these beliefs.

To introduce beliefs about mixed strategies formally we define them as follows:

Definition 6.4 A **belief** for player i is given by a *probability distribution* $\pi_i \in \triangle S_{-i}$ over the strategies of his opponents. We denote by $\pi_i(s_{-i})$ the probability player i assigns to his opponents playing $s_{-i} \in S_{-i}$.

Thus a *belief* for player i is a probability distribution over the strategies of his opponents. Notice that the belief of player i lies in the same set that represents the profiles of mixed strategies of player i's opponents. For example, in the rock-paper-scissors game, we can represent the beliefs of player 1 as a triplet, $(\pi_1(R), \pi_1(P), \pi_1(S))$, where by definition $\pi_1(R), \pi_1(P), \pi_1(S) \geq 0$ and $\pi_1(R) + \pi_1(P) + \pi_1(S) = 1$. The interpretation of $\pi_1(s_2)$ is the probability that player 1 assigns to player 2 playing some particular $s_2 \in S_2$. Recall that the strategy of player 2 is a triplet $\sigma_2(R), \sigma_2(P), \sigma_2(S) \geq 0$, with $\sigma_2(R) + \sigma_2(P) + \sigma_2(S) = 1$, so we can clearly see the analogy between π and σ.

6.1.4 Expected Payoffs

Consider the Matching Pennies game described previously, and assume for the moment that player 2 chooses the mixed strategy $\sigma_2(H) = \frac{1}{3}$ and $\sigma_2(T) = \frac{2}{3}$. If player 1 plays H then he will win and get 1 with probability $\frac{1}{3}$ while he will lose and get -1 with probability $\frac{2}{3}$. If, however, he plays T then he will win and get 1 with probability $\frac{2}{3}$ while he will lose and get -1 with probability $\frac{1}{3}$. Thus by choosing different actions player 1 will face different lotteries, as described in Chapter 2.

To evaluate these lotteries we will resort to the notion of expected payoff over lotteries as presented in Section 2.2. Thus we define the expected payoff of a player as follows:

Definition 6.5 The **expected payoff** of player i when he chooses the pure strategy $s_i \in S_i$ and his opponents play the mixed strategy $\sigma_{-i} \in \Delta S_{-i}$ is

$$v_i(s_i, \sigma_{-i}) = \sum_{s_{-i} \in S_{-i}} \sigma_{-i}(s_{-i}) v_i(s_i, s_{-i}).$$

Similarly the expected payoff of player i when he chooses the mixed strategy $\sigma_i \in \Delta S_i$ and his opponents play the mixed strategy $\sigma_{-i} \in \Delta S_{-i}$ is

$$v_1(\sigma_i, \sigma_{-i}) = \sum_{s_i \in S_i} \sigma_i(s_i) v_i(s_i, \sigma_{-i}) = \sum_{s_i \in S_i} \left(\sum_{s_{-i} \in S_{-i}} \sigma_i(s_i) \sigma_{-i}(s_{-i}) v_i(s_i, s_{-i}) \right).$$

The idea is a straightforward adaptation of definition 2.3 in Section 2.2.1. The randomness that player i faces if he chooses some $s_i \in S_i$ is created by the random selection of $s_{-i} \in S_{-i}$ that is described by the probability distribution $\sigma_{-i}(\cdot)$. Clearly the definition we just presented is well defined only for finite strategy sets S_i. The analog to interval strategy sets is a straightforward adaptation of the second part of definition 2.3.[4]

As an example, recall the rock-paper-scissors game:

		Player 2		
		R	P	S
Player 1	R	0, 0	−1, 1	1, −1
	P	1, −1	0, 0	−1, 1
	S	−1, 1	1, −1	0, 0

and assume that player 2 plays $\sigma_2(R) = \sigma_2(P) = \frac{1}{2}; \sigma_2(S) = 0$. We can now calculate the expected payoff for player 1 from any of his pure strategies,

$$v_1(R, \sigma_2) = \tfrac{1}{2} \times 0 + \tfrac{1}{2} \times (-1) + 0 \times 1 = -\tfrac{1}{2}$$
$$v_1(P, \sigma_2) = \tfrac{1}{2} \times 1 + \tfrac{1}{2} \times 0 + 0 \times (-1) = \tfrac{1}{2}$$
$$v_1(S, \sigma_2) = \tfrac{1}{2} \times (-1) + \tfrac{1}{2} \times 1 + 0 \times 0 = 0.$$

It is easy to see that player 1 has a unique best response to this mixed strategy of player 2. If he plays P, he wins or ties with equal probability, while his other two pure strategies are worse: with R he either loses or ties and with S he either loses or wins. Clearly if his beliefs about the strategy of his opponent are different then player 1 is likely to have a different best response.

It is useful to consider an example in which the players have strategy sets that are intervals. Consider the following game, known as an **all-pay auction,** in which two players can bid for a dollar. Each can submit a bid that is a real number (we are not restricted to penny increments), so that $S_i = [0, \infty)$, $i \in \{1, 2\}$. The person with the higher bid gets the dollar, but the twist is that both bidders have to pay their bids (hence the name of the game). If there is a tie then both pay and the dollar is awarded to each player with an equal probability of 0.5. Thus if player i bids s_i and player $j \neq i$ bids s_j then player i's payoff is

4. Consider a game in which each player has a strategy set given by the interval $S_i = [\underline{s}_i, \overline{s}_i]$. If player 1 is playing s_1 and his opponents, players $j = 2, 3, \ldots, n$, are using the mixed strategies given by the density function $f_j(\cdot)$ then the expected payoff of player 1 is given by

$$\int_{\underline{s}_2}^{\overline{s}_2} \int_{\underline{s}_3}^{\overline{s}_3} \cdots \int_{\underline{s}_n}^{\overline{s}_n} v_i(s_i, s_{-i}) f_2(s_2) f_3(s_3) \cdots f_n(s_n) ds_2 ds_3 \cdots ds_n.$$

For more on this topic see Section 19.4.4.

$$v_i(s_i, s_{-i}) = \begin{cases} -s_i & \text{if } s_i < s_j \\ \frac{1}{2} - s_i & \text{if } s_i = s_j \\ 1 - s_i & \text{if } s_i > s_j. \end{cases}$$

Now imagine that player 2 is playing a mixed strategy in which he is *uniformly* choosing a bid between 0 and 1. That is, player 2's mixed strategy σ_2 is a uniform distribution over the interval 0 and 1, which is represented by the cumulative distribution function and density

$$F_2(s_2) = \begin{cases} s_2 & \text{for } s_2 \in [0, 1] \\ 1 & \text{for } s_2 > 1 \end{cases} \qquad \text{and} \qquad f_2(s_2) = \begin{cases} 1 & \text{for } s_2 \in [0, 1] \\ 0 & \text{for } s_2 > 1. \end{cases}$$

The expected payoff of player 1 from offering a bid $s_i > 1$ is $1 - s_i < 0$ because he will win for sure, but this would not be wise. The expected payoff from bidding $s_i < 1$ is[5]

$$\begin{aligned} v_1(s_1, \sigma_2) &= \Pr\{s_1 < s_2\}(-s_1) + \Pr\{s_1 = s_2\}\left(\tfrac{1}{2} - s_1\right) + \Pr\{s_1 > s_2\}\left(1 - s_1\right) \\ &= (1 - F_2(s_1))(-s_1) + 0\left(\tfrac{1}{2} - s_1\right) + F_2(s_1)(1 - s_1) \\ &= 0. \end{aligned}$$

Thus when player 2 is using a uniform distribution between 0 and 1 for his bid, then player 1 cannot get any positive *expected* payoff from any bid he offers: any bid less than 1 offers an expected payoff of 0, and any bid above 1 guarantees getting the dollar at an inflated price. This game is one to which we will return later, as it has several interesting features and twists.

6.2 Mixed-Strategy Nash Equilibrium

Now that we are equipped with a richer space for both strategies and beliefs, we are ready to restate the definition of a Nash equilibrium for this more general setup as follows:

Definition 6.6 The mixed-strategy profile $\sigma^* = (\sigma_1^*, \sigma_2^*, \ldots, \sigma_n^*)$ is a *Nash equilibrium* if for each player σ_i^* is a best response to σ_{-i}^*. That is, for all $i \in N$,

$$v_i(\sigma_i^*, \sigma_{-i}^*) \geq v_i(\sigma_i, \sigma_{-i}^*) \;\; \forall\, \sigma_i \in \Delta S_i.$$

This definition is the natural generalization of definition 5.1. We require that each player be choosing a strategy $\sigma_i^* \in \Delta S_i$ that is (one of) the best choice(s) he can make when his opponents are choosing some profile $\sigma_{-i}^* \in \Delta S_{-i}$.

As we discussed previously, there is another interesting interpretation of the definition of a Nash equilibrium. We can think of σ_{-i}^* as the belief of player i about his opponents, π_i, which captures the idea that player i is uncertain of his opponents' behavior. The profile of mixed strategies σ_{-i}^* thus captures this uncertain belief over all of the pure strategies that player i's opponents can play. Clearly rationality requires

5. If player 2 is using a uniform distribution over [0, 1] then $\Pr\{s_1 = s_2\} = 0$ for any $s_1 \in [0, 1]$.

that a player play a best response given his beliefs (and this now extends the notion of rationalizability to allow for uncertain beliefs). A Nash equilibrium requires that these beliefs be correct.

Recall that we defined a pure strategy $s_i \in S_i$ to be *in the support of* σ_i if $\sigma_i(s_i) > 0$, that is, if s_i is played with positive probability (see definition 6.2). Now imagine that in the Nash equilibrium profile σ^* the support of i's mixed strategy σ_i^* contains more than one pure strategy—say s_i and s_i' are both in the support of σ_i^*.

What must we conclude about a rational player i if σ_i^* is indeed part of a Nash equilibrium $(\sigma_i^*, \sigma_{-i}^*)$? By definition σ_i^* is a best response against σ_{-i}^*, which means that given σ_{-i}^* player i cannot do better than to randomize between more than one of his pure strategies, in this case, s_i and s_i'. But when would a player be willing to randomize between two alternative pure strategies? The answer is predictable:

Proposition 6.1 *If σ^* is a Nash equilibrium, and both s_i and s_i' are in the support of σ_i^*, then*

$$v_i(s_i, \sigma_{-i}^*) = v_i(s_i', \sigma_{-i}^*) = v_i(\sigma_i^*, \sigma_{-i}^*).$$

The proof is quite straightforward and follows from the observation that if a player is randomizing between two alternatives then he must be indifferent between them. If this were not the case, say $v_i(s_i, \sigma_{-i}^*) > v_i(s_i', \sigma_{-i}^*)$ with both s_i and s_i' in the support of σ_i^*, then by reducing the probability of playing s_i' from $\sigma_i^*(s_i')$ to zero, and increasing the probability of playing s_i from $\sigma_i^*(s_i)$ to $\sigma_i^*(s_i) + \sigma_i^*(s_i')$, player i's expected payoff must go up, implying that σ_i^* could not have been a best response to σ_{-i}^*.

This simple observation will play an important role in computing mixed-strategy Nash equilibria. In particular we know that if a player is playing a mixed strategy then he must be indifferent between the actions he is choosing with positive probability, that is, the actions that are in the support of his mixed strategy. One player's indifference will impose restrictions on the behavior of other players, and these restrictions will help us find the mixed-strategy Nash equilibrium.

For games with many players, or with two players who have many strategies, finding the set of mixed-strategy Nash equilibria is a tedious task. It is often done with the help of computer algorithms, because it generally takes on the form of a linear programming problem. Nevertheless it will be useful to see how one computes mixed-strategy Nash equilibria for simpler games.

6.2.1 Example: Matching Pennies

Consider the Matching Pennies game,

		Player 2	
		H	T
Player 1	H	$1, -1$	$-1, 1$
	T	$-1, 1$	$1, -1$

and recall that we showed that this game does not have a pure-strategy Nash equilibrium. We now ask, does it have a mixed-strategy Nash equilibrium? To answer this, we have to find mixed strategies for both players that are mutual best responses.

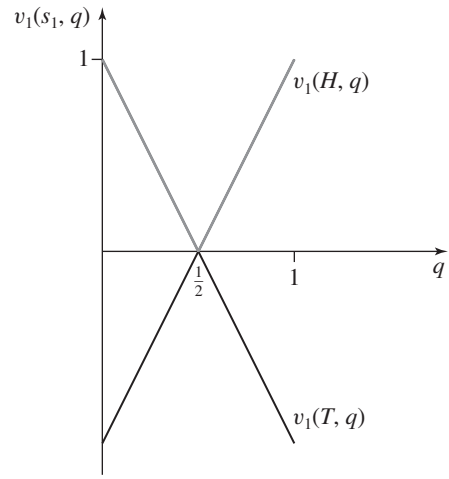


FIGURE 6.2 Expected payoffs for player 1 in the Matching Pennies game.

To simplify the notation, define mixed strategies for players 1 and 2 as follows: Let p be the probability that player 1 plays H and $1 - p$ the probability that he plays T. Similarly let q be the probability that player 2 plays H and $1 - q$ the probability that he plays T.

Using the formulas for expected payoffs in this game, we can write player 1's expected payoff from each of his two pure actions as follows:

$$v_1(H, q) = q \times 1 + (1 - q) \times (-1) = 2q - 1 \tag{6.1}$$

$$v_1(T, q) = q \times (-1) + (1 - q) \times 1 = 1 - 2q. \tag{6.2}$$

With these equations in hand, we can calculate the best response of player 1 for any choice q of player 2. In particular player 1 will prefer to play H over playing T if and only if $v_1(H, q) > v_1(T, q)$. Using (6.1) and (6.2), this will be true if and only if

$$2q - 1 > 1 - 2q,$$

which is equivalent to $q > \frac{1}{2}$. Similarly playing T will be strictly better than playing H for player 1 if and only if $q < \frac{1}{2}$. Finally, when $q = \frac{1}{2}$ player 1 will be indifferent between playing H or T.

It is useful to graph the expected payoff of player 1 from choosing either H or T as a function of player 2's choice of q, as shown in Figure 6.2.

The expected payoff of player 1 from playing H was given by the function $v_1(H, q) = 2q - 1$, as described in (6.1). This is the rising linear function in the figure. Similarly $v_1(T, q) = 1 - 2q$, described in (6.2), is the declining function. Now it is easy to see what determines the best response of player 1. The gray "upper envelope" of the graph will show the highest payoff that player 1 can achieve when player 2 plays any given level of q. When $q < \frac{1}{2}$ this is achieved by playing T; when $q > \frac{1}{2}$ this is achieved by playing H; and when $q = \frac{1}{2}$ both H and T are equally good for player 1, giving him an expected payoff of zero.

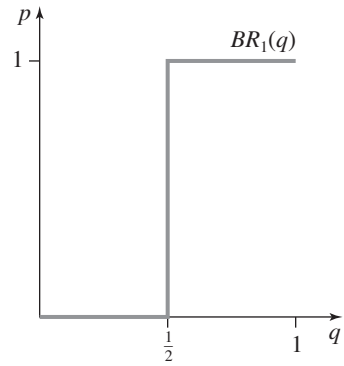


FIGURE 6.3 Player 1's best-response correspondences in the Matching Pennies game.

This simple analysis results in the best-response correspondence of player 1, which is

$$BR_1(q) = \begin{cases} p = 0 & \text{if } q < \frac{1}{2} \\ p \in [0, 1] & \text{if } q = \frac{1}{2} \\ p = 1 & \text{if } q > \frac{1}{2} \end{cases}$$

and is depicted in Figure 6.3. Notice that this is a best-response *correspondence*, and not a function, because at the value of $q = \frac{1}{2}$ any value of $p \in [0, 1]$ is a best response.

In a similar way we can calculate the payoffs of player 2 given a mixed-strategy p of player 1 to be

$$v_2(p, H) = p \times (-1) + (1 - p) \times 1 = 1 - 2p$$
$$v_2(p, T) = p \times 1 + (1 - p) \times (-1) = 2p - 1,$$

and this implies that player 2's best response is

$$BR_2(p) = \begin{cases} q = 1 & \text{if } p < \frac{1}{2} \\ q \in [0, 1] & \text{if } p = \frac{1}{2} \\ q = 0 & \text{if } p > \frac{1}{2}. \end{cases}$$

To find a Nash equilibrium we are looking for a pair of choices (p, q) for which the two best-response correspondences cross. Were we to superimpose the best response of player 2 onto Figure 6.3 then we would see that the two best-response correspondences cross at $p = q = \frac{1}{2}$. Nevertheless it is worth walking through the logic of this solution.

We know from proposition 6.1 that when player 1 is mixing between H and T, both with positive probability, then it must be the case that his payoffs from H and from T are identical. This, it turns out, imposes a restriction on the behavior of player 2, given by the choice of q. Player 1 is willing to mix between H and T if and only if $v_1(H, q) = v_1(T, q)$, which will hold if and only if $q = \frac{1}{2}$. This is the way in which *the indifference of player 1 imposes a restriction on player 2:* only when player 2 is playing $q = \frac{1}{2}$ will player 1 be willing to mix between his actions H and T. Similarly player 2 is willing to mix between H and T only when $v_2(p, H) = v_2(p, T)$, which

is true only when $p = \frac{1}{2}$. We have come to the conclusion of our quest for a Nash equilibrium in this game. We can see that there is indeed a pair of mixed strategies that form a Nash equilibrium, and these are precisely when $(p, q) = \left(\frac{1}{2}, \frac{1}{2}\right)$.

There is a simple logic, which we can derive from the Matching Pennies example, that is behind the general method for finding mixed-strategy equilibria in games. The logic relies on a fact that we have already discussed: if a player is mixing several strategies then he *must be indifferent* between them. What a particular player i is willing to do depends on the strategies of his opponents. Therefore, to find out when player i is willing to mix some of his pure strategies, we must find strategies of his opponents, $-i$, that make him indifferent between some of his pure actions.

For the Matching Pennies game this can be easily illustrated as follows. First, we ask *which strategy of player 2* will make *player 1 indifferent* between playing H and T. The answer to this question (assuming it is unique) *must be player 2's strategy in equilibrium*. The reason is simple: if player 1 is to mix in equilibrium, then player 2 must be playing a strategy for which player 1's best response is mixing, and player 2's strategy must therefore make player 1 indifferent between playing H and T. Similarly we ask which strategy of player 1 will make player 2 indifferent between playing H and T, and this must be player 1's equilibrium strategy.

Remark The game of Matching Pennies is representative of situations in which one player wants to match the actions of the other, while the other wants to avoid that matching. One common example is penalty goals in soccer. The goalie wishes to jump in the direction that the kicker will kick the ball, while the kicker wishes to kick the ball in the opposite direction from the one in which the goalie chooses to jump. When they go in the same direction then the goalie wins and the kicker loses, while if they go in different directions then the opposite happens. As you can see, this is exactly the structure of the Matching Pennies game. Other common examples of such games are bosses monitoring their employees and the employees' decisions about how hard to work, or police monitoring crimes and the criminals who wish to commit them.

6.2.2 Example: Rock-Paper-Scissors

When we have games with more than two strategies for each player, then coming up with quick ways to solve mixed-strategy equilibria is a bit more involved than in 2×2 games, and it will usually involve more tedious algebra that solves several equations with several unknowns. If we consider the game of rock-paper-scissors, for example, there are many mixing combinations for each player, and we can't simply draw graphs the way we did for the Matching Pennies game.

		Player 2		
		R	P	S
Player 1	R	0, 0	−1, 1	1, −1
	P	1, −1	0, 0	−1, 1
	S	−1, 1	1, −1	0, 0

To find the Nash equilibrium of the rock-paper-scissors game we proceed in three steps. First we show that there is no Nash equilibrium in which at least one player

plays a pure strategy. Then we show that there is no Nash equilibrium in which at least one player mixes only between two pure strategies. These steps will imply that in any Nash equilibrium, both players must be mixing with all three pure strategies, and this will lead to the solution.

Claim 6.1 *There can be no Nash equilibrium in which one player plays a pure strategy and the other mixes.*

To see this, suppose that player i plays a pure strategy. It's easy to see from looking at the payoff matrix that player j always receives different payoffs from each of his pure strategies whenever i plays a pure strategy. Therefore player j cannot be indifferent between any of his pure strategies, so j cannot be playing a mixed strategy if i plays a pure strategy. But we know that there are no pure-strategy equilibria, and hence we conclude that there are no Nash equilibria where either player plays a pure strategy.

Claim 6.2 *There can be no Nash equilibrium in which at least one player mixes only between two pure strategies.*

To see this, suppose that i mixes between R and P. Then j always gets a strictly higher payoff from playing P than from playing R, so no strategy requiring j to play R with positive probability can be a best response for j, and j can't play R in any Nash equilibrium. But if j doesn't play R then i gets a strictly higher payoff from S than from P, so no strategy requiring i to play P with positive probability can be a best response to j not playing R. But we assumed that i was mixing between R and P, so we've reached a contradiction. We conclude that in equilibrium i cannot mix between R and P. We can apply similar reasoning to i's other pairs of pure strategies. We conclude that in any Nash equilibrium of this game, no player can play a mixed strategy in which he only plays two pure strategies with positive probability.

If by now you've guessed that the mixed strategies $\sigma_1^* = \sigma_2^* = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$ form a Nash equilibrium then you are right. If player i plays σ_i^* then j will receive an expected payoff of 0 from every pure strategy, so j will be indifferent between all of his pure strategies. Therefore $BR_j(\sigma_i^*)$ includes all of j's mixed strategies and in particular $\sigma_j^* \in BR_j(\sigma_i^*)$. Similarly $\sigma_i^* \in BR_i(\sigma_j^*)$. We conclude that σ_1^* and σ_2^* form a Nash equilibrium. We will prove that (σ_1^*, σ_2^*) is *the unique Nash equilibrium.*

Suppose player i plays R with probability $\sigma_i(R) \in (0, 1)$, P with probability $\sigma_i(P) \in (0, 1)$, and S with probability $1 - \sigma_i(R) - \sigma_i(P)$. Because we proved that both players have to mix with all three pure strategies, it follows that $\sigma_i(R) + \sigma_i(P) < 1$ so that $1 - \sigma_i(R) - \sigma_i(P) \in (0, 1)$. It follows that player j receives the following payoffs from his three pure strategies:

$$v_j(R, \sigma_i) = -\sigma_i(P) + 1 - \sigma_i(R) - \sigma_i(P) = 1 - \sigma_i(R) - 2\sigma_i(P)$$

$$v_j(P, \sigma_i) = \sigma_i(R) - (1 - \sigma_i(R) - \sigma_i(P)) = 2\sigma_i(R) + \sigma_i(P) - 1$$

$$v_j(S, \sigma_i) = -\sigma_i(R) + \sigma_i(P).$$

In any Nash equilibrium in which j plays all three of his pure strategies with positive probability, he must receive the same expected payoff from all strategies. Therefore, in any equilibrium, we must have $v_j(R, \sigma_i) = v_j(P, \sigma_i) = v_j(S, \sigma_i)$. If we set these

payoffs equal to each other and solve for $\sigma_i(R)$ and $\sigma_i(P)$, we get $\sigma_i(R) = \sigma_i(P) = 1 - \sigma_i(R) - \sigma_i(P) = \frac{1}{3}$. We conclude that j is willing to include all three of his pure strategies in his mixed strategy if and only if i plays $\sigma_i^* = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$. Similarly i will be willing to play all his pure strategies with positive probability if and only if j plays $\sigma_j^* = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$. Therefore there is no other Nash equilibrium in which both players play all their pure strategies with positive probability.

6.2.3 Multiple Equilibria: Pure and Mixed

In the Matching Pennies and rock-paper-scissors games, the unique Nash equilibrium was a mixed-strategy Nash equilibrium. It turns out that mixed-strategy equilibria need not be unique when they exist. In fact when a game has multiple pure-strategy Nash equilibria, it will almost always have other Nash equilibria in mixed strategies. Consider the following game:

		Player 2	
		C	R
Player 1	M	0, 0	3, 5
	D	4, 4	0, 3

It is easy to check that (M, R) and (D, C) are both pure-strategy Nash equilibria. It turns out that in 2×2 matrix games like this one, when there are two distinct pure-strategy Nash equilibria then there will almost always be a third one in mixed strategies.[6]

For this game, let player 1's mixed strategy be given by $\sigma_1 = (\sigma_1(M), \sigma_1(D))$, with $\sigma_1(M) = p$ and $\sigma_1(D) = 1 - p$, and let player 2's mixed strategy be given by $\sigma_2 = (\sigma_2(C), \sigma_2(R))$, with $\sigma_2(C) = q$ and $\sigma_2(R) = 1 - q$. Player 1 will mix when $v_1(M, q) = v_1(D, q)$, or when

$$q \times 0 + (1 - q) \times 3 = q \times 4 + (1 - q) \times 0$$
$$\Rightarrow q = \tfrac{3}{7},$$

and player 2 will mix when $v_2(p, C) = v_2(p, R)$, or when

$$p \times 0 + (1 - p) \times 4 = p \times 5 + (1 - p) \times 3$$
$$\Rightarrow p = \tfrac{1}{6}.$$

This yields our third Nash equilibrium: $(\sigma_1^*, \sigma_2^*) = \left(\left(\frac{1}{6}, \frac{5}{6}\right), \left(\frac{3}{7}, \frac{4}{7}\right)\right)$.

6. The statement "almost always" is not defined here, but it effectively means that if we draw numbers at random from some set of distributions to fill a game matrix, and it will result in more than one pure-strategy Nash equilibrium, then with probability 1 it will also have at least one mixed-strategy equilibrium. In fact a game will typically have an odd number of equilibria. This result is known as an *index theorem* and is far beyond the scope of this text.

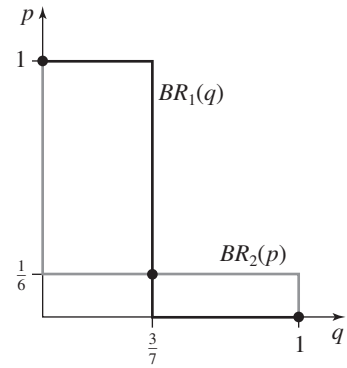


FIGURE 6.4 Best-response correspondences and Nash equilibria.

It is interesting to see that all three equilibria would show up in a careful drawing of the best-response functions. Using the payoff functions $v_1(M, q)$ and $v_1(D, q)$ we have

$$BR_1(q) = \begin{cases} p = 1 & \text{if } q < \frac{3}{7} \\ p \in [0, 1] & \text{if } q = \frac{3}{7} \\ p = 0 & \text{if } q > \frac{3}{7}. \end{cases}$$

Similarly using the payoff functions $v_2(p, C)$ and $v_2(p, R)$ we have

$$BR_2(p) = \begin{cases} q = 1 & \text{if } p < \frac{1}{6} \\ q \in [0, 1] & \text{if } p = \frac{1}{6} \\ q = 0 & \text{if } p > \frac{1}{6}. \end{cases}$$

We can draw the two best-response correspondences as they appear in Figure 6.4. Notice that all three Nash equilibria are revealed in Figure 6.4: $(p, q) \in \{(1, 0), \left(\frac{1}{6}, \frac{3}{7}\right), (0, 1)\}$ are all Nash equilibria, where $(p, q) = (1, 0)$ corresponds to the pure strategy (M, R), and $(p, q) = (0, 1)$ corresponds to the pure strategy (D, C).

6.3 IESDS and Rationalizability Revisited

By introducing mixed strategies we offered two advancements: players can have richer beliefs, and players can choose a richer set of actions. This can be useful when we reconsider the concepts of IESDS and rationalizability, and in fact present them in their precise form using mixed strategies. In particular we can now state the following two definitions:

Definition 6.7 Let $\sigma_i \in \Delta S_i$ and $s_i' \in S_i$ be possible strategies for player i. We say that s_i' is **strictly dominated** by σ_i if

$$v_i(\sigma_i, s_{-i}) > v_i(s_i', s_{-i}) \ \forall s_{-i} \in S_{-i}.$$

Definition 6.8 A strategy $\sigma_i \in \Delta S_i$ is **never a best response** if there are no beliefs $\sigma_{-i} \in \Delta S_{-i}$ for player i for which $\sigma_i \in BR_i(\sigma_{-i})$.

the A, while if they are equally good then the professor will toss a fair coin to decide who gets the A. The other student will get a B. Since both wish to continue on to graduate school, a grade of A is worth 3 while a grade of B is worth 0.

a. Write down this game in matrix form.
b. Are there any strictly dominated strategies? Are there any weakly dominated strategies?
c. Find the unique mixed-strategy Nash equilibrium.

6.8 **Market Entry:** Three firms are considering entering a new market. The payoff for each firm that enters is $\frac{150}{n}$, where n is the number of firms that enter. The cost of entering is 62.

a. Find all the pure-strategy Nash equilibria.
b. Find the symmetric mixed-strategy equilibrium in which all three players enter with the same probability.

6.9 **Discrete All-Pay Auction:** In Section 6.1.4 we introduced a version of an all-pay auction that worked as follows: Each bidder submits a bid. The highest bidder gets the good, but *all bidders pay their bids.* Consider an auction in which player 1 values the item at 3 while player 2 values the item at 5. Each player can bid either 0, 1, or 2. If player i bids more than player j then i wins the good and both pay. If both players bid the same amount then a coin is tossed to determine who gets the good, but again both pay.

a. Write down the game in matrix form. Which strategies survive IESDS?
b. Find the Nash equilibria for this game.

6.10 **Continuous All-Pay Auction:** Consider an all-pay auction for a good worth 1 to each of the two bidders. Each bidder can choose to offer a bid from the unit interval so that $S_i = [0, 1]$. Players care only about the expected value they will end up with at the end of the game (i.e., if a player bids 0.4 and expects to win with probability 0.7 then his payoff is $0.7 \times 1 - 0.4$).

a. Model this auction as a normal-form game.
b. Show that this game has no pure-strategy Nash equilibrium.
c. Show that this game cannot have a Nash equilibrium in which each player is randomizing over a finite number of bids.
d. Consider mixed strategies of the following form: Each player i chooses an interval $[\underline{x}_i, \overline{x}_i]$ with $0 \leq \underline{x}_i < \overline{x}_i \leq 1$ together with a cumulative distribution $F_i(x)$ over the interval $[\underline{x}_i, \overline{x}_i]$. (Alternatively you can think of each player choosing $F_i(x)$ over the interval $[0, 1]$ with two values $\underline{x}_i$ and $\overline{x}_i$ such that $F_i(\underline{x}_i) = 0$ and $F_i(\overline{x}_i) = 1$.)
    i. Show that if two such strategies are a mixed-strategy Nash equilibrium then it must be that $\underline{x}_1 = \underline{x}_2$ and $\overline{x}_1 = \overline{x}_2$.
    ii. Show that if two such strategies are a mixed-strategy Nash equilibrium then it must be that $\underline{x}_1 = \underline{x}_2 = 0$.
    iii. Using your answers to (i) and (ii), argue that if two such strategies are a mixed-strategy Nash equilibrium then both players must be getting an expected payoff of zero.
    iv. Show that if two such strategies are a mixed-strategy Nash equilibrium then it must be that $\overline{x}_1 = \overline{x}_2 = 1$.
    v. Show that $F_i(x)$ being uniform over $[0, 1]$ is a symmetric Nash equilibrium of this game.

8

Credibility and Sequential Rationality

Consider the simple extensive-form coordination game described in Figure 8.1, which is represented in normal form in the following matrix:

		Player 2	
		e	*p*
Player 1	*E*	0, 0	0, 0
	P	0, 0	1, 1

Each player can exit (E for player 1 and e for player 2), or proceed (P and p, respectively). If the players coordinate on proceeding, then they both receive a payoff of 1 while any other profile of strategies yields each a payoff of 0.

There is a good reason to assume that the players will successfully coordinate on the Nash equilibrium profile (P, p): the exit strategy is weakly dominated by the choice to proceed. However, a quick look at the matrix shows that there is another Nash equilibrium: (E, e). Using the terminology of Section 7.3, the players stick to the equilibrium path of player 1 choosing E because of player 2's threat to choose e off the equilibrium path.

There is therefore something unappealing about the logic of Nash equilibrium in extensive-form games. The concept asks only for players to act rationally

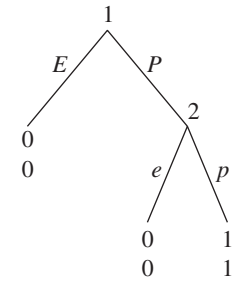


FIGURE 8.1 A coordination game.

on the equilibrium path given their beliefs about what will transpire both *on and off* the equilibrium path. Nevertheless it puts no restrictions on the beliefs of players off the equilibrium path, nor on how they should consider such beliefs. In fact the normal-form representation of a sequential game is not able to address such a requirement, which adds constraints on what we would tolerate as "rational behavior."

We would, however, expect rational players to play optimally in response to their beliefs *whenever* they are called to move. In the simple coordination game in Figure 8.1 this would require that player 2 commit to play p when it is his turn to move. Applying this reasoning to the sequential-move Battle of the Sexes game in Figure 7.11 suggests that of the three Nash equilibria of the game, two are unappealing. The equilibria (O, oo) and (F, ff) have player 2 committing to a strategy that, despite being a best response to player 1's strategy, would not have been optimal were player 1 to deviate from his strategy and cause the game to move off the equilibrium path. In what follows we will introduce a natural requirement that will result in more refined predictions for dynamic games. These will indeed rule out equilibria such as (E, e) in the coordination game and (O, oo) and (F, ff) in the sequential-move Battle of the Sexes game.

8.1 Sequential Rationality and Backward Induction

To address the critique we posed regarding the incredible nature of the equilibria (O, oo) and (F, ff) in the sequential-move Battle of the Sexes game, we will directly criticize the behavior of player 2 in the event that player 1 did not follow his prescribed strategy. We will insist that a player use strategies that are optimal at *every information set in the game tree.* We call this principle *sequential rationality,* because it implies that players are playing rationally at every stage in the sequence of the game, whether it is on or off the equilibrium path of play. Formally, we have

Definition 8.1 Given strategies $\sigma_{-i} \in \Delta S_{-i}$ of i's opponents, we say that σ_i is **sequentially rational** if and only if i is playing a best response to σ_{-i} in each of his information sets.

Using this definition we can revisit the sequential-move Battle of the Sexes game and ask: what are player 2's best responses in each of his information sets? The answer is obvious: if player 1 played O then player 2 should play o, and if player 1 played F then player 2 should play f. Any other strategy of player 2 is not a best response in at least one of these information sets, and this implies that a sequentially rational player 2 should be playing the pure strategy of.

Now move back to the root of the game where player 1 has to choose between O and F. Taking into account the sequential rationality of player 2, player 1 should conclude that playing O will result in the payoffs (2, 1) while playing F will result in the payoffs (1, 2). Now applying sequential rationality to player 1 requires that player 1, who is *correctly predicting* the behavior of player 2, should choose O. Hence the unique prediction from this process is the path of play that begins with player 1 choosing O followed by player 2 choosing o. Furthermore the process predicts what would happen if players *deviate from the path of play:* if player 1 chooses F then player 2 will choose f. We conclude that the Nash equilibrium (O, of) is the unique pair of strategies that survives our requirement of sequential rationality.

This type of procedure, which starts at nodes that directly precede the terminal nodes at the end of the game and then inductively moves backward through the game

tree, is also known as **backward induction in games.** This is exactly the multiperson dynamic programming version of the single decision maker's process described in Section 2.4.1. As the example just given suggests, when we apply this procedure to *finite games of perfect information* it will result in a prescription of strategies for each player that are sequentially rational, as the following proposition states:

Proposition 8.1 *Any finite game of perfect information has a backward induction solution that is sequentially rational. Furthermore if no two terminal nodes prescribe the same payoffs to any player then the backward induction solution is unique.*

The proof is rather straightforward and is based on the simple idea of backward induction. By definition a finite game of perfect information has a finite set of terminal nodes. Consider the nodes that immediately precede a terminal node, and call the set of these nodes "level 1" nodes. Select for every player at a level 1 node an action that yields him his highest payoff. (There can be more than one if there are ties in the payoffs.)[1] Now similarly define "level 2" nodes as those that immediately precede a level 1 node, and let the players at these nodes choose the action that maximizes their payoff given that level 1 players will choose their action as previously specified. This process continues iteratively until we reach the root of the game (because the game has finite length) and results in a specification of moves that are sequentially rational.

Remark This proposition is often referred to as *Zermelo's Theorem,* named after the mathematician Ernst Zermelo. In 1913 Zermelo published a paper roughly stating that in games like chess, a player can guarantee at least a tie in a finite number of moves. Interestingly neither backward induction nor the notion of strategies—and surely not sequential rationality—was used in Zermelo's original paper, despite widespread belief to the contrary. For an excellent summary of the history of this line of research see Schwalbe and Walker (2001).

Notice that, by the construction of the backward induction procedure, each player will necessarily play a best response to the actions of the other players who come after him (the best responses are constructed for every information set). This results in the following corollary:

Corollary 8.1 *Any finite game of perfect information has at least one sequentially rational Nash equilibrium in pure strategies. Furthermore if no two terminal nodes prescribe the same payoffs to any player then the game has a unique sequentially rational Nash equilibrium.*

As we saw in the sequential-move Battle of the Sexes example, other pure-strategy Nash equilibria may exist.

8.2 Subgame-Perfect Nash Equilibrium: Concept

As the previous section argues, we expect rational players to play in ways that are sequentially rational. Furthermore we established that backward induction is a useful method to find a sequentially rational Nash equilibrium in finite games of

1. This is where the qualifier "if no two terminal nodes prescribe the same payoffs to any player" plays a role. If this is the case then there is a unique action that maximizes each player's payoff. If not, we can select any one of the maximizing actions.

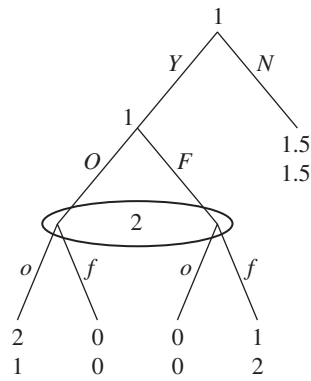


FIGURE 8.2 The voluntary Battle of the Sexes game.

perfect information. Things become a bit trickier when we try to expand our reach to suggest solutions for games of imperfect information, in which backward induction as previously defined encounters some serious problems.

Consider, for example, a game in which player 1 decides whether or not to play a Battle of the Sexes game with player 2. He can decide yes (Y), in which case they play a simultaneous-move Battle of the Sexes game, or no (N), in which case both players get a payoff of 1.5. The game is described in Figure 8.2. To try to solve this game using backward induction we need to first identify the set of "last players" that precede terminal nodes and then choose actions that would maximize their payoff at this stage. In this game, however, this is not possible, because player 2 has an information set before the terminal nodes that is not a singleton. His best response is therefore not well defined without assigning a belief to this player about what player 1 actually chose to do, and these beliefs are not part of the backward induction process.

This example shows that backward induction cannot be applied to games of imperfect information. Interestingly there is another important class of games for which we cannot apply the procedure of backward induction, and these are games that do not necessarily end in finite time. Intuitively games that do not necessarily end in a finite number of moves may not have a finite set of terminal nodes, and without such a set we cannot begin the backward induction procedure. At first thought such games may seem a bit bizarre—what kind of game will never end? As we will see in Chapters 10 and 11, such games not only are interesting in their own right but also will offer critical ways to model realistic situations. Hence our goal is to find a natural way in which to extend the concept of sequential rationality to games of imperfect information and to games that have an infinite sequence of moves.

Let's start with the problem posed by imperfect information. For example, the problem we encountered in the game depicted in Figure 8.2 is that the best response of player 2 in his information set depends on the node where he is, or on his beliefs about where he is in the information set. This of course depends on the action of player 1, which in turn depends on what player 1 believes that player 2 will do. Thus it seems that sequential rationality will have to cope with both of these decisions being interdependent. For this reason we advance the following definition:

Definition 8.2 A proper subgame G of an extensive-form game Γ consists of only a single node and all its successors in Γ with the property that if $x \in G$ and $x' \in h(x)$

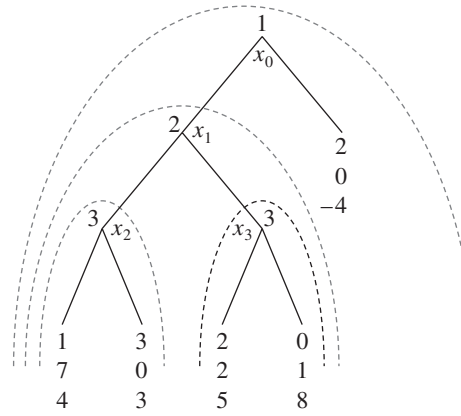


FIGURE 8.3 Subgames in a game with perfect information.

then $x' \in G$. The subgame G is itself a *game tree* with its information sets and payoffs inherited from Γ.

The idea of a proper subgame (which we will often just call a subgame) is simple and allows us to "dissect" an extensive-form game into a sequence of smaller games, an approach that in turn will allow us to apply the concept of sequential rationality to games of imperfect information. To be able to do this, however, we will require that every such smaller game be an extensive-form game in its own right, which means that it must have a unique root and follow the structure that we defined in Section 7.1.1.

In every game of perfect information, every node is a singleton and hence can be a root of a subgame. This implies that in games of perfect information every node, together with all the nodes that succeed it, forms a proper subgame. As an example, consider the game depicted in Figure 8.3. The two "smallest" subgames start at nodes x_2 and x_3. A"larger" subgame starts at x_1, and it includes the two subgames that start at x_2 and x_3. Finally the "largest" subgame starts at the original game's root, x_0, and includes all the other subgames.

Now let's return to the voluntary Battle of the Sexes game depicted again in Figure 8.4. There are only two proper subgames in this game: the whole game (which

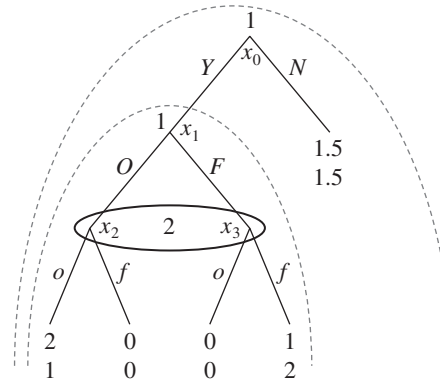


FIGURE 8.4 Proper subgames in the voluntary Battle of the Sexes game.

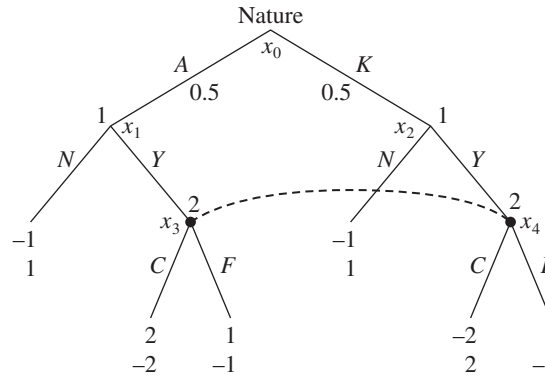


FIGURE 8.5 A game of cards.

is always, by definition, a proper subgame) and the subgame starting at the node x_1. The reason these are the only two subgames follows from definition 8.2. Note that x_2 and x_3 cannot be roots of a subgame because they belong to the same information set. From definition 8.2, if x_2 belongs to a subgame then x_3 must belong to that subgame too. In addition, from the definition we know that any subgame must begin with a single node, and thus the information set that contains x_2 and x_3 cannot begin a subgame. Hence, aside from the whole game, the only other proper subgame starts at x_1.

Another interesting example appears in Figure 8.5, which describes the following game. Players 1 and 2 put a dollar each in a pot, and player 1 pulls a card out of a deck of kings and aces, with an equal probability of getting a king (K) or an ace (A). Player 1 observes his card and then decides whether not to play the game (N), and forfeit his dollar to player 2, or proceed with the game (Y). If player 1 proceeds with the game, then without knowing which card player 1 drew (hence the information set denoted by the dashed line) player 2 can fold (F) and forfeit his dollar to player 1 or call (C), in which case each player must add another dollar to the pot. After this, if player 1 has a king then player 2 wins the pot, while if player 1 has an ace then player 1 wins the pot.

Now ask yourself: what are the proper subgames of this game? Clearly neither x_3 nor x_4 can be a root of a subgame because they belong to the same information set. Can x_1 be the root of a subgame? If it could be then x_3 must be in its subgame because x_1 precedes x_3. But by definition, if x_3 is in the subgame then $x_4 \in h(x_3)$, and therefore it too should be in the subgame (as well as all the relevant terminal nodes that follow after x_1, x_3, and x_4). But this would not be a proper subgame because x_1 does not precede x_4. A similar argument implies that x_2 cannot be the root of the subgame. We conclude that for this card game the only proper subgame is the whole game.

By now you may have realized why a subgame, which is a stand-alone game within the whole game, will prove to be useful to apply the concept of sequential rationality to extensive-form games. In particular, at any node or information set within a subgame G, a player's best response depends only on his beliefs about what the other players are doing *within* the subgame G, and not at nodes that are outside the subgame.

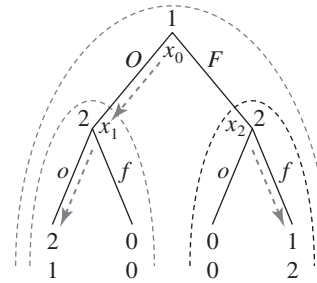


FIGURE 8.6 Subgame-perfect equilibrium in the sequential-move Battle of the Sexes game.

As demonstrated using the voluntary Battle of the Sexes game in Figure 8.4, the problem with backward induction was that we could not apply it to the end of the game because there was an information set that included two nodes that preceded terminal nodes, namely x_2 and x_3. However, what we can do is restrict attention to the *subgame* that begins at the node x_1 and require rational play in this subgame. Then we can use rational play in this "endgame" to move backward and apply sequential rationality. This is the idea behind a powerful solution concept defined as follows:

Definition 8.3 Let Γ be an n-player extensive-form game. A behavioral strategy profile $\sigma^* = (\sigma_1^*, \sigma_2^*, \ldots, \sigma_n^*)$ is a **subgame-perfect (Nash) equilibrium** if for every proper subgame G of Γ the restriction of σ^* to G is a Nash equilibrium in G.

This important concept was introduced by Reinhard Selten (1975), who was the second of the three Nobel Laureates sharing the prize in 1994 for the development of game theory. This equilibrium concept brings sequential rationality into the static Nash equilibrium solution concept. Using the terminology developed in Section 7.3, subgame perfection requires not only that a Nash equilibrium profile of strategies be a combination of best responses on the equilibrium path, which is a necessary condition of a Nash equilibrium, but also that the profile of strategies consist of mutual best responses *off the equilibrium path.* This is precisely what follows from the requirement that the restriction of the strategy profile σ^* be a Nash equilibrium in every proper subgame, including those subgames that *are not reached* in equilibrium.

Notice that, by the definition of a subgame-perfect equilibrium, every subgame-perfect equilibrium is a Nash equilibrium. However, not all Nash equilibria are necessarily subgame-perfect equilibria, implying that subgame-perfect equilibrium *refines* the set of Nash equilibria, yielding more refined predictions on behavior. To see this consider again the sequential-move Battle of the Sexes game in Figure 8.6 and its corresponding normal-form, given as:

		Player 2			
		oo	of	fo	ff
Player 1	O	$\underline{\overline{2, 1}}$	$\underline{\overline{2, 1}}$	$\underline{0, 0}$	$0, 0$
	F	$0, 0$	$\overline{\underline{1, 2}}$	$\underline{0, 0}$	$\overline{\underline{1, 2}}$

As we saw earlier, there are three pure-strategy Nash equilibria, (O, oo), (F, ff), and (O, of). Of these three, only (O, of) is the unique subgame-perfect equilibrium.

This follows from the fact that in the subgame beginning at x_1, the only Nash equilibrium is player 2 choosing o, because he is the only player in that subgame and he must choose a best response to his belief, which must be "I am at x_1." Similarly in the subgame beginning at x_2, the only Nash equilibrium is player 2 choosing f. Anticipating this, player 1 must choose O at x_0. Thus of the three Nash equilibria of the game, only (O, of) satisfies the condition that its restriction is a Nash equilibrium for every proper subgame of the whole game, and hence (O, oo) and (F, ff) are Nash equilibria that are not subgame perfect.

Notice that the game we just analyzed is a finite game of perfect information. For these games we have an easy and familiar way to find the subgame-perfect equilibria, which is using the procedure of backward induction because

Fact For any finite game of perfect information, the set of subgame-perfect Nash equilibria coincides with the set of Nash equilibria that survive backward induction.

This is a very useful fact because it gives us an operational procedure to find all the subgame-perfect equilibria in finite games of perfect information. However, because this procedure does not apply to games of imperfect information, we need to analyze the game using a revised version of backward induction that takes into account the proper subgames as the relevant stages in the backward induction process.

As an example, consider the voluntary Battle of the Sexes game depicted in Figure 8.4. In this game player 1 has four pure strategies. Using the previous convention, the strategy set for player 1 is $S_1 = \{YO, YF, NO, NF\}$, where, for example, YO means that player 1 plans to play Y at x_0 and O at x_1. Note that if player 1 plays either NO or NF then, regardless of player 2's strategy, the game will end after player 1's choice of N. Player 2, on the other hand, has only two strategies, $S_2 = \{o, f\}$, because he must make his choice without knowing the choice of player 1. Hence the matrix-form representation of this game is as follows:

		Player 2	
		o	f
Player 1	YO	$\underline{\overline{2, 1}}$	0, 0
	YF	0, 0	$\overline{1, 2}$
	NO	$\overline{1.5, 1.5}$	$\underline{1.5, 1.5}$
	NF	$\overline{1.5, 1.5}$	$\underline{1.5, 1.5}$

As we can see, there are three pure-strategy Nash equilibria in this game, given by the set $E^{\text{Nash}} = \{(YO, o), (NO, f), (NF, f)\}$. Of these only two pairs of strategies form a subgame-perfect equilibrium, so the set of subgame-perfect equilibria profiles is $E^{\text{SPE}} = \{(YO, o), (NF, f)\}$. The reason, as shown in Figure 8.7, is that in the subgame that starts at the node x_1, the only pairs of restricted strategies that form a Nash equilibrium are (O, o) and (F, f). Hence the profile (NO, f) is not a Nash equilibrium when we restrict attention to the subgame that starts at x_1. This game has one mixed-strategy subgame-perfect equilibrium that you are asked to find in exercise 8.1.

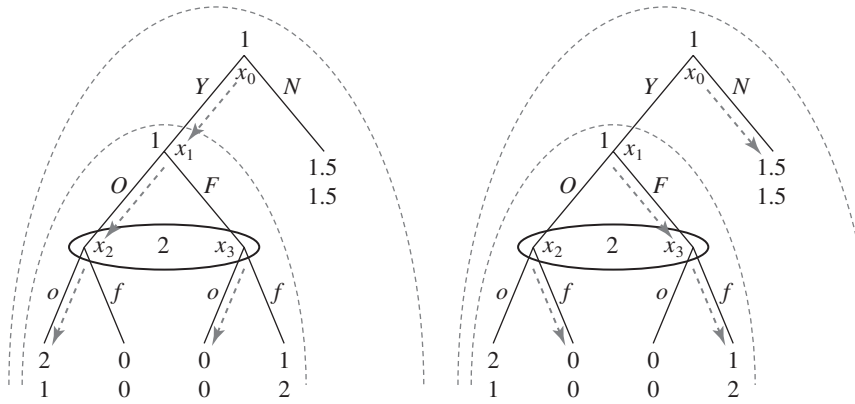


FIGURE 8.7 Subgame-perfect equilibria in the voluntary Battle of the Sexes game.

8.3 Subgame-Perfect Nash Equilibrium: Examples

This section presents some well-known examples of games and their corresponding subgame-perfect equilibria.

8.3.1 The Centipede Game

Consider the perfect-information game depicted in Figure 8.8. The game should be read from left to right as follows: Player 1 can terminate the game immediately by choosing N in his first information set or can continue by choosing C. Then player 2 faces the same choice (using lowercase letters for his choices), and if player 2 chooses to continue then the ball is back in player 1's court, who again can terminate or continue to player 2, at which stage player 2 concludes the game by choosing n or c for the second time.

It would be nice for the players to be able to continue through the game to reach the payoffs of (3, 3). However, backward induction indicates that this will not happen. At his last information set, player 2 will choose n to get 4 instead of 3. Anticipating this a step earlier, player 1 will choose N to get 2 instead of 1, and the logic follows until player 1's first information set, at which he will choose N and both players will receive a payoff of 1.

Notice that this game has an interesting structure: as long as the players continue, the sum of their payoffs goes up by 1. You can easily see that we can continue with the payoffs in this way with (2, 5), (4, 4), (3, 6), . . . and make the payoff from reaching the end of the game extremely large. Nevertheless the "curse of rationality," so to speak, predicts a unique outcome: at the last stage the last player will, by being selfish,

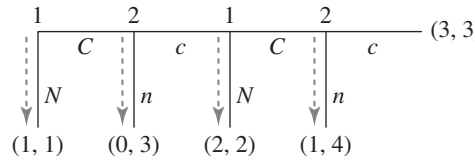


FIGURE 8.8 The Centipede Game.

prefer to stop short of the payoff that maximizes the sum of the players' payoffs, and backward induction implies that this decision is anticipated and acted on by the player before him, and so on as the actions unravel to the bleak outcome of (1, 1).

Remark The Centipede Game was first introduced by Rosenthal (1981), who used it as an example of a game in which the unique backward induction solution is extremely unappealing and goes against every intuition one might have about how the players will play. Indeed this game has been taken into the lab by letting pairs of players play it, and the experimental evidence of many studies, starting with McKelvey and Palfrey (1992), suggests that players do not play in the way predicted by backward induction. There are at least two reasons that players will not play the backward induction solution. One is that they actually care about each other's payoffs. This explanation may be less convincing for cases in which the players are anonymous, but it is not easy to rule out. The other reason is that players do not share a common knowledge of rationality. Palacios-Huerta and Volij (2009) try to put this second hypothesis to a test by taking the game to a chess tournament and having highly ranked chess players play it. Contrary to previous evidence, their results show that 69% of chess players choose to end the game immediately, and when grand masters are playing, all of them end the game immediately! This striking result suggests that when players are expected to share common knowledge of rationality they indeed play the backward induction solution.

8.3.2 Stackelberg Competition

The Stackelberg duopoly is a game of perfect information that is a sequential-moves variation on the Cournot duopoly model of competition. It was introduced and analyzed by Heinrich von Stackelberg (1934). It illustrates an important point: the order of moves might matter, and rational actors will take this into account.

Consider our familiar Cournot game with demand $p = 100 - q$, $q = q_1 + q_2$, and $c_i(q_i) = 10q_i$ for $i \in \{1, 2\}$. We have already solved for the best-response functions of each firm by maximizing their profit functions *when each takes the other's quantity as fixed* (the simultaneous-move game), so that each i solves

$$\max_{q_i}(100 - q_i - q_j)q_i - 10q_i,$$

and the best response is given by the first-order condition, which can be written as

$$q_i(q_j) = \frac{90 - q_j}{2}.$$

We now change a small but important detail of the game and assume that player 1 will choose q_1 first, and that player 2 will observe the choice made by player 1 before it makes its choice of q_2. We begin by analyzing the backward induction solution of this game.

From the fact that player 2 maximizes its profit when q_1 is already known it should be clear that player 2 will follow its best-response function, and hence sequential rationality implies that

$$q_2(q_1) = \frac{90 - q_1}{2}. \tag{8.1}$$

This follows from not only the fact that player 2 has a *belief* about q_1 but also that this belief is in reality due to the actual observation of q_1. This is precisely the backward induction conclusion of considering firm 2 first, because its moves are those that precede the terminal nodes. Notice that in this game there are infinitely many terminal nodes and infinitely many information sets preceding the terminal nodes. In particular, every different choice of q_1 is a different information set of player 2. Nevertheless, because there is perfect information, and because there is a well-defined optimal choice of player 2 for every choice of player 1, the backward induction procedure works.

Now assuming common knowledge of rationality, what should player 1 do? It would be rather naive to maximize its profits based on some fixed belief about q_2, because firm 1 knows *exactly* how a rational player 2 would respond to its choice of q_1: player 2 will choose q_2 using equation (8.1). This in turn means that a rational firm 1 would replace the "fixed" q_2 in its profit function with the best response of firm 2 and choose q_1 to solve

$$\max_{q_1} \left[100 - q_1 - \left(\frac{90 - q_1}{2} \right) \right] q_1 - 10q_1. \tag{8.2}$$

There is a fundamental difference between this maximization problem and the one we solved for the Cournot example in Section 3.1.2, and it depends on *what the firms know when they make their choices*. In the Cournot game neither firm *knew* what the other was choosing, so each set a belief and maximized its profit function. Here, in contrast, firm 2 *knows* what firm 1 has produced when it makes its choice, and as a result firm 1 *knows,* by common knowledge of rationality, that firm 2 will choose q_2 rationally, that is, by using its best-response function, given in (8.1).

The solution for firm 1 is then given by the first-order condition of (8.2), which is

$$100 - 2q_1 - 45 + q_1 - 10 = 0,$$

yielding $q_1 = 45$. Then, using firm 2's best response, we have $q_2 = 22.5$. The resulting profits are $\pi_1 = (100 - 67.5) \times 45 - 10 \times 45 = 1012.5$, and $\pi_2 = (100 - 67.5) \times 22.5 - 10 \times 22.5 = 506.25$. Recall that in the original (simultaneous-move) Cournot example the quantities and profits were $q_1 = q_2 = 30$ and $\pi_1 = \pi_2 = 900$. We conclude that when two firms are competing by setting quantities, if one firm can somehow commit to move first, it will enjoy a *first-mover advantage.*

We know already that if a profile of strategies survives backward induction then this profile is also a subgame-perfect equilibrium, and in particular a Nash equilibrium. There is a simple yet very important point worth emphasizing here. When we write down the pair of strategies as a pure-strategy Nash equilibrium *we must be careful to specify the strategies correctly:* player 2 has a continuum of information sets, each being a particular choice of q_1. This implies that the backward induction solution yields the following Nash equilibrium: $(q_1, q_2(q_1)) = \left(45, \frac{90-q_1}{2}\right)$. Writing down $(q_1, q_2) = (45, 22.5)$ is *not* identifying a Nash equilibrium because $q_1 = 45$ is not a best response to $q_2 = 22.5$. We must specify q_2 for *every information set* for it to be a well-defined strategy in the Stackelberg game, and in this case we defined it to be $q_2(q_1) = \frac{90-q_1}{2}$.

To illustrate the application of subgame perfection in this game with continuous strategy sets, consider a heuristic game tree described in Figure 8.9. Player 1 has an infinite number (a continuum) of choices, depicted by the arc, and each choice

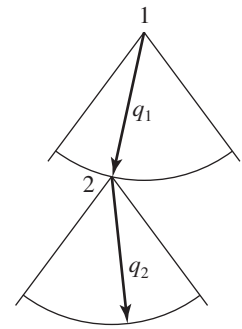


FIGURE 8.9 The Stackelberg duopoly game.

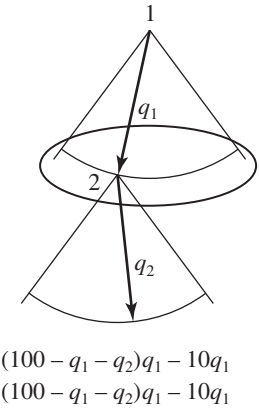


FIGURE 8.10 The Cournot duopoly game.

results in a proper subgame. Every choice q_1 of player 1 is an information set for player 2 that precedes an infinite number (a continuum) of terminal nodes associated with the specific q_1 together with player 2's choice of q_2. The payoff for player i is $(100 - q_1 - q_2)q_i - 10q_i$.

Similarly we can heuristically depict the simultaneous-move Cournot game in Figure 8.10. We draw player 2's information set to include all the choices of q_1 that player 1 can make to try to describe the simultaneous nature of the Cournot game in that player 2 makes its choice without observing the choice of player 1.

In the Cournot game we know from the analysis in Section 3.1.2 that $q_1 = q_2 = 30$ is the unique Nash equilibrium. Because every subgame-perfect equilibrium is a Nash equilibrium, and because the whole game is the only unique proper subgame in the Cournot example, then $q_1 = q_2 = 30$ is the unique subgame-perfect equilibrium in the Cournot game.

An interesting question is whether the outcome $q_1 = q_2 = 30$ could be the outcome of a Nash equilibrium in the Stackelberg game. The answer, perhaps surprisingly, is yes. To see this consider the following strategies in the Stackelberg game: $q_1 = 30$ and $q_2(q_1) = 30$ for *any* choice q_1. These two strategies are mutual best responses,

implying that they constitute a Nash equilibrium. Notice two things: First, we have to specify q_2 for *every information set* for it to be a well-defined strategy in the Stackelberg game, and in this case we defined it to be $q_2(q_1) = 30$ for every q_1. Second, it is a pair of mutual best responses *only on the equilibrium path.* That is, conditional on $q_1 = 30$, $q_2 = 30$ is a best response, and conditional on $q_2(q_1) = 30$ for every q_1, $q_1 = 30$ is a best response. This requirement, mutual best responses on the equilibrium path, is all that Nash equilibrium requires.

However, as we have shown using backward induction, the unconditional strategy $q_2 = 30$ for player 2 is not sequentially rational because it is never a best response *off the equilibrium path* for $q_1 \neq 30$. The only strategy for player 2 that satisfies sequential rationality is

$$q_2(q_1) = \frac{90 - q_1}{2},$$

which in turn implies that $q_1 = 45$ and $q_2 = \frac{90-q_1}{2}$ constitute the unique subgame-perfect equilibrium of the Stackelberg game.

Remark Interestingly there are many other Nash equilibria that appear when the game is a perfect-information sequential-move game like the Stackelberg model. To see this, consider the following strategy for player 2:

$$q_2'(q_1) = \begin{cases} 35 & \text{if } q_1 = 20 \\ 100 & \text{if } q_1 \neq 20. \end{cases}$$

What would be the best response of firm 1? Clearly $q_1 = 20$ yields positive profits, while any other choice yields nonpositive profits if player 1 believes that player 2 will follow its strategy. Thus $q_1 = 20$ is the best response. Then notice that $q_2 = 40$ is a best response to $q_1 = 20$ from player 2's best-response function in (8.1), implying that $q_1 = 20$ and $q_2'(q_1)$ constitute a Nash equilibrium. At this stage you should be able to convince yourself that there are many Nash equilibria—in fact, a continuum of Nash equilibria! To see this, fix any $\widetilde{q}_1 \in [0, 90]$ and replace the 35 in $q_2'(q_1)$ with player 2's best response $\widetilde{q}_2 = \frac{90-\widetilde{q}_1}{2}$.

8.3.3 Mutually Assured Destruction

A very nice example for subgame-perfect equilibrium analysis in a game with imperfect information appears in Gardner (2003, p. 165), who uses a game theoretic analysis to represent the Cuban missile crisis of 1962. The crisis started with the United States' discovery of Soviet nuclear missiles in Cuba, after which the United States escalated the crisis by quarantining Cuba. The USSR then backed down, agreeing to remove its missiles from Cuba, which suggests that the United States had a credible threat along the lines of "if you don't back off we both pay dearly." Could this indeed be a credible threat?

To analyze this as a game, consider two superpowers, player 1 (the United States) and player 2 (the USSR), that have engaged in a provocative incident.[2] The game starts with player 1's choice to either ignore the incident (I), resulting in maintenance of the

2. The pregame phase—the USSR placing the missiles in Cuba—is assumed to be a situation that materialized without taking into account the continuing game between the superpowers.

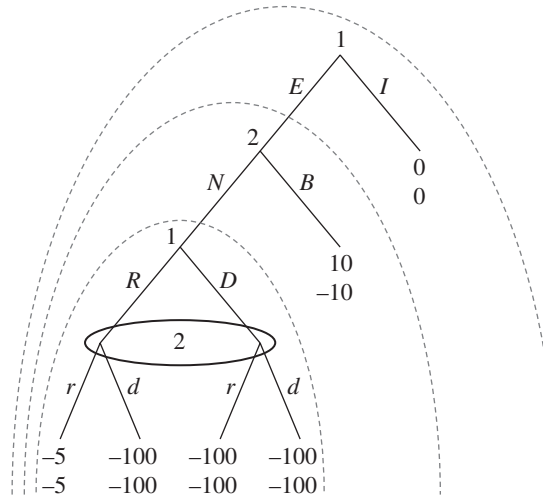


FIGURE 8.11 Mutually assured destruction.

status quo with payoffs (0, 0), or escalate the situation (E). Following escalation by player 1, player 2 can back down (B), causing it to lose face and resulting in payoffs of (10, −10), or it can choose to proceed to a nuclear confrontation (N). Upon this choice, the players play a simultaneous-move game in which they can either retreat (R for player 1, r for player 2) or choose Doomsday (D for player 1, d for player 2), in which the world is all but destroyed. If both call things off and retreat then they suffer a small loss due to the mobilization process and payoffs are (−5, −5), while if either party chooses Doomsday then the world destructs and payoffs are (−100, −100).

The extensive form of this game is depicted in Figure 8.11. Before solving for the subgame-perfect equilibria, we begin by solving for the Nash equilibria. To do this it is convenient to transform the extensive-form game into a normal-form game. Each player has two information sets, each with two actions, implying that each player has four pure strategies. Following our earlier convention, the strategy set for player 1 is $S_1 = \{IR, ID, ER, ED\}$ and that for player 2 is $S_2 = \{Br, Bd, Nr, Nd\}$ (where IR means that player 1 plays I in his first information set and R in his second one, and the other strategies are defined similarly). The normal form can be represented by the following matrix:

		Player 2			
		Br	Bd	Nr	Nd
Player 1	IR	$\overline{0, 0}$	$\overline{0, 0}$	$\underline{\overline{\mathbf{0, 0}}}$	$\overline{0, 0}$
	ID	$\overline{0, 0}$	$\overline{0, 0}$	$\underline{0, 0}$	$\underline{0, 0}$
	ER	$\underline{10, -10}$	$\underline{10, -10}$	$-5, -5$	$-100, -100$
	ED	$\underline{\overline{10, -10}}$	$\underline{\overline{\mathbf{10, -10}}}$	$-100, -100$	$-100, -100$

In this game there are six pure-strategy Nash equilibria, and these are the profiles in the set $E^{\text{Nash}} = \{(IR, Nr), (IR, Nd), (ID, Nr), (ID, Nd), (ED, Br), (ED, Bd)\}$. (We will ignore the mixed-strategy Nash equilibria in this example. There are too many!)

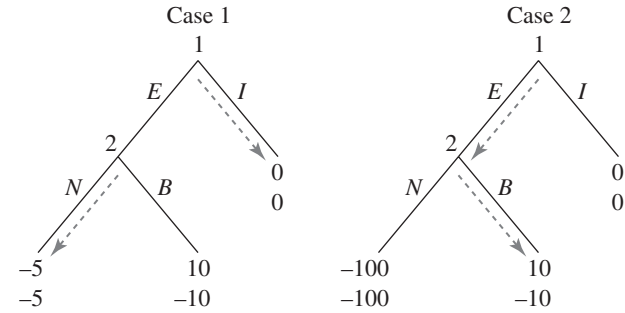


FIGURE 8.12 Reductions of the mutually assured destruction game.

To solve for the subgame-perfect equilibria, we perform backward induction using the subgames as we did earlier for the voluntary Battle of the Sexes game. There are three proper subgames depicted in Figure 8.11. The first is the whole game; the second subgame has its root at the point where player 2 chooses between B and N and includes all the moves that follow; and the third subgame starts where player 1 chooses between R and D and includes all the moves that follow.

It is easy to see that the third subgame is a simultaneous-move game, which we can call the "war stage" of the game. (Notice that we can switch the order of play in the third subgame because it is a simultaneous-move subgame.) This war-stage game can be described by the following matrix:

		Player 2	
		r	d
Player 1	R	$\underline{-5, -5}$	$\underline{-100, -100}$
	D	$\underline{-100, -100}$	$\underline{-100, -100}$

The war-stage game has two pure-strategy Nash equilibria:[3]

NE1: (R, r) that yields a payoff of $(-5, -5)$.

NE2: (D, d) that yields a payoff of $(-100, -100)$.

We can now proceed in one of two possible ways to continue our backward induction analysis. Case 1 will perform backward induction with the belief that if the war stage is reached then the players will play NE1, and case 2 will perform backward induction with the belief that if the war stage is reached then the players will play NE2. Thus we are analyzing one of the two games depicted in Figure 8.12:

Case 1: From the left panel of Figure 8.12 we can see that for case 1 player 2 would prefer to choose N over B because losing face is worse than the consequence of mutual retreat. Moving backward another step shows that player 1 would prefer I to E because, anticipating retreat, it is better to ignore the whole situation.

3. There is a mixed-strategy Nash equilibrium that you are asked to find, and use as part of a subgame-perfect equilibrium in exercise 8.2.

Case 2: From the right panel of Figure 8.12 we can see that for case 2 player 2 would prefer to choose B over N because of the dire consequences of war, and moving backward another step shows that player 1 would prefer E to I because he anticipates that player 2 will back off.

Thus we have found two different (pure-strategy) subgame-perfect equilibria given by the set $E^{\text{SPE}} = \{(IR, Nr), (ED, Bd)\}$. We have two different subgame-perfect equilibria because in the final war-stage subgame there are two *very distinct* pure-strategy Nash equilibria, NE1 and NE2. These two Nash equilibria have different implications for the behavior of player 2 in the preceding stage of the game, which in turn affects the behavior of player 1 at the beginning of the game.

There is a nice qualitative difference between the two equilibria. The first equilibrium, (ED, Bd), represents the case in which player 1 escalates because he believes that player 2 will treat this as a signal that player 1 is willing to "go all the way." More precisely in game theoretic terms, player 1 *believes* that both will play the Doomsday equilibrium NE2 in the war stage, and in turn player 2, who shares these beliefs, will back off. The second equilibrium, (IR, Nr), is one in which player 1 chooses to ignore the incident because he believes that if he were to escalate then two things would happen: player 2 would go nuclear and they would then retreat together, resulting in a payoff of -5 instead of 0. It is interesting that in both these equilibria the war game is *off the equilibrium path.* Nonetheless it is the expected behavior in this last and final stage that dictates how players will play.

8.3.4 Time-Inconsistent Preferences

In Section 2.5.2 we analyzed the decision problem of a single player who needs to allocate a scarce resource over two periods. We assumed that the player maximizes his discounted sum of payoffs, as we defined in Section 2.4.2. We explained that a player who expects to receive a stream of payments $x_1, x_2, \ldots, x_T$ over the periods $t = 1, 2, \ldots, T$, and who evaluates per-period payments using the utility function $u(x)$, will evaluate his *discounted sum of future payoffs* in period $t = 1$ as

$$v(x_1, x_2, \ldots, x_T) = u(x_1) + \delta u(x_2) + \cdots + \delta^{T-1} u(x_T) = \sum_{t=1}^{T} \delta^{t-1} u(x_t), \quad (8.3)$$

where $\delta \in (0, 1)$ is the player's *discount factor,* which he uses to discount future payoffs.

Consider a player with $u(x) = \ln(x)$ who needs to allocate a fixed budget K across three periods. Since he will not waste any of his budget, it follows that $x_1 = K - x_2 - x_3$ and the player solves the following problem:

$$\max_{x_2, x_3} \; v(K - x_2 - x_3, x_2, x_3) = \ln(K - x_2 - x_3) + \delta \ln(x_2) + \delta^2 \ln(x_3).$$

To solve this problem we have to solve the two first-order conditions

$$\frac{\partial v}{\partial x_2} = -\frac{1}{K - x_2 - x_3} + \frac{\delta}{x_2} = 0$$

and

$$\frac{\partial v}{\partial x_3} = -\frac{1}{K - x_2 - x_3} + \frac{\delta^2}{x_3} = 0.$$

Solving these two equations yields the solutions

$$x_3 = K\frac{\delta^2}{1+\delta+\delta^2} \tag{8.4}$$

and

$$x_2 = K\frac{\delta}{1+\delta+\delta^2}\,, \tag{8.5}$$

and using $x_1 = K - x_2 - x_3$ gives

$$x_1 = K\frac{1}{1+\delta+\delta^2}. \tag{8.6}$$

The player's solution is similar to the solution we saw in Section 2.5.2. The player chooses to consume more in earlier periods because he is equating the marginal utility from consumption across time, taking into account that future periods are discounted. An interesting question is the following: if the player planned out his consumption over time, would he choose to stick to his plan of action after he consumes x_1? To answer this, imagine that the player has already consumed his planned amount x_1 given in (8.6), which implies that in period $t = 2$ he is left with a budget of

$$K_2 = K - K\frac{1}{1+\delta+\delta^2} = K\frac{\delta+\delta^2}{1+\delta+\delta^2}. \tag{8.7}$$

Now consider the player's maximization problem in period $t = 2$, where he is left with a budget of K_2. The problem is the same two-period problem as the one we saw in Section 2.5.2:

$$\max_{x_2}\ \ln(x_2) + \delta\ln(K_2 - x_2),$$

and the solution to the player's problem will be

$$\frac{1}{x_2} = \frac{\delta}{K_2 - x_2}$$

or

$$x_2 = \frac{K_2}{1+\delta}\,. \tag{8.8}$$

We can now substitute K_2 from (8.7) into (8.8) to solve for the choice of x_2 in terms of the original budget K to obtain

$$x_2 = \frac{K_2}{1+\delta} = K\frac{\delta+\delta^2}{1+\delta+\delta^2} \times \frac{1}{1+\delta} = K\frac{\delta}{1+\delta+\delta^2}\,,$$

which is identical to the player's original choice of x_2 given by (8.5).

We call this kind of behavior *time consistent* because a player will choose to follow his original plan. The discounting rule described by (8.3) is called *exponential discounting,* and it is generally known in economic analysis that a player who uses this formula to evaluate his present-value utility will always stick to his plan of action; therefore his behavior will be time consistent. Strotz (1956) suggested that players who use discounting rules other than exponential discounting will exhibit

time-inconsistent behavior in that, as time goes by, they will want to deviate from plans of action that they originally set.

To model this idea consider a different discounting rule known as *hyperbolic discounting*. The player uses the discount rate δ for the future periods as in exponential discounting, but he uses an *additional* discount factor $\beta \in (0, 1)$ to discount all of the future compared to present consumption. Using our three-period example, the player's modified discounted present-value problem is given by

$$\max_{x_2,x_3} \; v(K - x_2 - x_3, x_2, x_3) = \ln(K - x_2 - x_3) + \beta\delta \ln(x_2) + \beta\delta^2 \ln(x_3). \quad (8.9)$$

Simply put, when the player is looking toward the future, the discount factor he uses between periods $t = 1$ and $t = 2$ is stronger than the one he uses between periods $t = 2$ and $t = 3$. In particular, at any period and looking forward, the player uses $\beta\delta < \delta$ to discount between the current period and the next, while he uses only δ to discount between any two consecutive future periods.[4] As we will now see, hyperbolic discounting will cause problems of self-control, in that a player will plan to do one thing but later choose to revise his plan.

Imagine that the player is very well aware of his self-control problem—at period $t = 1$ he discounts between periods $t = 2$ and $t = 3$ at the rate of δ, but he knows that when tomorrow ($t = 2$) comes around, he will discount $t = 3$ at a rate of $\beta\delta$. In other words, the player at $t = 1$ (call him player 1) is aware that he is playing a game with his "future self" at $t = 2$ (call him player 2), who will exhibit different preferences over time. In this case a rational, forward-looking player 1 will solve his problem using backward induction; in other words he will solve for the subgame-perfect equilibrium of the implied dynamic game.

To make things simple, fix $\delta = 1$. Recall from equations (8.4)–(8.6) that a player who uses exponential discounting with $\delta = 1$ will choose to equalize his consumption over time, so that $x_t = \frac{K}{3}$ for all $t \in \{1, 2, 3\}$. Now consider a sophisticated player with hyperbolic discounting with $\delta = 1$ and $\beta = \frac{1}{2}$ (you are asked to solve for the more general problem in exercise 8.15). Looking forward to his future self, he knows that player 2, who is left with a budget of K_2, will solve the following problem (remember that $\delta = 1$ and $\beta = \frac{1}{2}$):

$$\max_{x_2} \; v_2(x_2, K_2 - x_2) = \ln(x_2) + \tfrac{1}{2} \ln(K_2 - x_2),$$

for which the first-order condition is

$$\frac{dv_2}{dx_2} = \frac{1}{x_2} - \frac{0.5}{K_2 - x_2} = 0,$$

which in turn implies that player 2's best-response function is

$$x_2(K_2) = \tfrac{2}{3} K_2.$$

As a consequence $x_3(K_2) = \frac{1}{3} K_2$. Hence the choices $x_2(K_2)$ and $x_3(K_2)$ made by player 2 are the solution to any of the (infinitely many) subgames that start at the

4. More generally the discounted present value of a player who uses hyperbolic discounting is given by $u(x_1) + \beta \sum_{t=2}^{T} \delta^{t-1} u(x_t)$.

node at which player 2 is left with a budget of K_2. Now moving up to the root of the game, where player 1 decides how much to allocate between his own consumption and that of player 2, he must solve the following problem:

$$\max_{x_1} \; v_1\left(x_1, \tfrac{2}{3}(K - x_1), \tfrac{1}{3}(K - x_1)\right)$$
$$= \ln(x_1) + \tfrac{1}{2}\ln\left(\tfrac{2}{3}(K - x_1)\right) + \tfrac{1}{2}\ln\left(\tfrac{1}{3}(K - x_1)\right),$$

for which the first-order condition is

$$\frac{dv_1}{dx_1} = \frac{1}{x_1} - \frac{2}{3} \times \frac{0.5}{K - x_1} - \frac{1}{3} \times \frac{0.5}{K - x_1} = 0\,,$$

which in turn implies that player 1's best-response function is

$$x_1(K) = \tfrac{2}{3}K.$$

This results in a very different solution than our exponential discounting benchmark, in which (with $\delta = 1$) the player split his resources equally across the three periods. Here player 1 realizes that player 2 will choose to overconsume resources compared to what player 1 would have preferred his future self to choose. As a consequence player 1 himself overconsumes to leave a lower budget for his future self, player 2, by spending more in period 1.

Problems of self-control and time-inconsistent behavior are at the center of the field known as behavioral economics. Models of hyperbolic discounting have been used to explain insufficient savings (Laibson, 1997), procrastination (O'Donoghue and Rabin, 1999), and other related phenomena. Scholars in this literature often distinguish between players who are not aware of their self-control problem (they do not perform backward induction), referring to them as naive, and players who are aware of their self-control problem, referring to them as sophisticated. What is interesting about sophisticated players is that they would choose to constrain their own future behavior in order to achieve a higher net present value from their stream of consumption. You are asked to show this in exercise 8.16.

8.4 Summary

- Extensive-form games will often have some Nash equilibria that are not sequentially rational, yet we expect rational players to choose sequentially rational strategies.
- In games of perfect information backward induction will result in sequentially rational Nash equilibria. If there are no two payoffs at terminal nodes that are the same then there will be a unique sequentially rational Nash equilibrium.
- Subgame-perfect equilibrium is the more general construct of backward induction for games of imperfect information.
- At least one of the Nash equilibria in a game will be a subgame-perfect equilibrium.

8.5 Exercises

8.1 **Mixed-Strategy Subgame-Perfect Equilibrium:** Find the mixed-strategy subgame-perfect equilibrium of the sequential-move Battle of the Sexes game depicted in Figure 8.2.

8.2 **Mutually Assured Destruction Revisited:** Consider the game in Section 8.3.3.

a. Find the mixed-strategy Nash equilibrium of the war-stage game and argue that it is unique.
b. What is the unique subgame-perfect equilibrium that includes the mixed strategy you found in (a)?

8.3 **Brothers Revisited:** Find all the subgame-perfect equilibria in the "Brothers" exercise (exercise 7.8) from the previous chapter.

8.4 **Industry Leader:** Three oligopolists operate in a market with inverse demand given by $P(Q) = a - Q$, where $Q = q_1 + q_2 + q_3$, and q_i is the quantity produced by firm i. Each firm has a constant marginal cost of production, c, and no fixed cost. The firms choose their quantities dynamically as follows: Firm 1, which is the industry leader, chooses $q_1 \geq 0$. Firms 2 and 3 observe q_1 and then simultaneously choose q_2 and q_3, respectively.

a. How many proper subgames does this dynamic game have? Explain briefly.
b. Is it a game of perfect or imperfect information? Explain briefly.
c. What is the subgame-perfect equilibrium of this game? Show that it is unique.
d. Find a Nash equilibrium that is not a subgame-perfect equilibrium.

8.5 **Technology Adoption:** During the adoption of a new technology a CEO (player 1) can design a new task for a division manager. The new task can be either high level (H) or low level (L). The manager simultaneously chooses to invest in good training (G) or bad training (B). The payoffs from this interaction are given by the following matrix:

		Player 2	
		G	B
Player 1	H	5, 4	−5, 2
	L	2, −2	0, 0

a. Present the game in extensive form (a game tree) and solve for all the Nash equilibria and subgame-perfect equilibria.
b. Now assume that before the game is played the CEO can choose not to adopt this new technology, in which case the payoffs are (1, 1), or to adopt it, in which case the game is played. Present the *entire* game in extensive form. How many proper subgames does it have?
c. Solve for all the Nash equilibria and subgame-perfect equilibria of the game described in (b).

8.6 **Investment in the Future:** Consider two firms that play a Cournot competition game with demand $p = 100 - q$ and costs for each firm given by $c_i(q_i) = 10q_i$. Imagine that before the two firms play the Cournot game firm

1 can invest in cost reduction. If it invests the costs of firm 1 will drop to $c_1(q_1) = 5q_1$. The cost of investment is $F > 0$. Firm 2 does not have this investment opportunity.

a. Find the value F^* for which the unique subgame-perfect equilibrium involves firm 1 investing.
b. Assume that $F > F^*$. Find a Nash equilibrium of the game that is not subgame perfect.

8.7 **Debt and Repayment:** A project costing \$100 yields a gross return of \$110. A lender (player 1) is approached by a debtor (player 2) requesting a standard loan contract to complete the project. If the lender chooses *not to offer* a loan, then both parties earn nothing. If the lender chooses to *offer* a loan of \$100, the debtor can realize the project's gains and is obliged by contract to repay \$105. For simplicity assume that money is continuous and that the debtor can choose to return any amount of money $x \leq 110$. Ignore the time value of money. Assume first that no legal system is in place that can compel the debtor to repay, so that default on the loan (less than full repayment) carries no repercussions for the debtor.

a. Model this as an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game. Is it unique?
b. Now assume that there is a legal system in place that allows the lender to *voluntarily* choose whether or not to sue when the debtor defaults and repays an amount $x < 105$. Furthermore assume that it is *costless* to use the legal system (it is supplied by the state), and if the lender sues a debtor that defaulted, the lender will get the \$105 repaid in full. After paying the lender, the borrower will pay a fine of \$5 to the court, above and beyond the repayment. Model this as an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game. Is it unique?
c. Are there Nash equilibria in the game described in (b) that are not subgame-perfect equilibria?
d. Now assume that using the legal system is *costly:* if the lender sues, he pays lawyers a *legal fee* of \$105 (this is the lawyers' price; it is unrelated to the contract). The rest of the game proceeds as before. (If the lender sues a debtor that defaulted, the lender will get repaid in full; after paying the lender, the borrower will pay a fine of \$5 above and beyond the repayment.) Model this as an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game. Is it unique?
e. Are there Nash equilibria in the game described in (d) that are not subgame-perfect equilibria?
f. Now assume that a change in the law is proposed: upon default, if a debtor is sued he has to first repay the lender \$105 and then pay the legal fees of \$105 above and beyond repayment of the loan, and no extra fine is imposed. Should the lender be willing to pay to have this change in the law enacted? If so, how much?
g. If you were a social planner, would you implement the suggested law?

8.8 **Entry Deterrence 1:** NSG Corporation is considering entry into the local phone market in the San Francisco Bay Area. The incumbent, S&P Corporation, predicts that a price war will result if NSG enters. If NSG stays out,

S&P earns monopoly profits valued at \$10 million (net present value of profits) while NSG earns zero. If NSG enters, it must incur irreversible entry costs of \$2 million. If there is a price war, each firm earns \$1 million. S&P always has the option of accommodating entry (i.e., not starting a price war). In such a case both firms earn \$4 million (net present value). Suppose that the timing is such that NSG first has to choose whether or not to enter the market. Then S&P decides whether to "accommodate entry" or "engage in a price war." What is the subgame-perfect equilibrium outcome to this sequential game? Set up a game tree.

8.9 **Entry Deterrence 2:** Consider the Cournot duopoly game with demand $p = 100 - (q_1 + q_2)$ and variable costs $c_i(q_i) = 0$ for $i \in \{1, 2\}$. The twist is that there is now a fixed cost of production $k > 0$ that is the same for both firms.

a. Assume first that both firms choose their quantities simultaneously. Model this as a normal-form game.
b. Write down the firm's best-response function for $k = 1000$ and solve for a pure-strategy Nash equilibrium. Is it unique?
c. Now assume that firm 1 is a "Stackelberg leader" in the sense that it moves first and chooses q_1. Then after observing q_1 firm 2 chooses q_2. Also assume that if firm 2 cannot make strictly positive profits then it will not produce at all. Model this as an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game for $k = 25$. Is it unique?
d. How does your answer in (c) change for $k = 225$?

8.10 **Playing It Safe:** Consider the following dynamic game: Player 1 can choose to play it safe (denote this choice by S), in which case both he and player 2 get a payoff of 3 each, or he can risk playing a game with player 2 (denote this choice by R). If he chooses R then they play the following simultaneous-move game:

		Player 2	
		A	B
Player 1	C	8, 0	0, 2
	D	6, 6	2, 2

a. Draw a game tree that represents this game. How many proper subgames does it have?
b. Are there other game trees that would work? Explain briefly.
c. Construct the matrix representation of the normal form of this dynamic game.
d. Find all the Nash and subgame-perfect equilibria of the dynamic game.

8.11 **Resident Assistant Selection with a Twist:** Two staff managers in the ΠΒΦ sorority, the house manager (player 1) and kitchen manager (player 2), must select a resident assistant from a pool of three candidates: $\{a, b, c\}$. Player 1 prefers a to b, and b to c. Player 2 prefers b to a, and a to c. The process that is imposed on them is as follows: First, the house manager vetoes one of the candidates and announces the veto to the central office for staff selection and to the kitchen manager. Next the kitchen manager vetoes one of the remaining two candidates and announces it to the central office. Finally the director of

the central office assigns the remaining candidate to be a resident assistant at ΠΒΦ.

a. Model this as an extensive-form game (using a game tree) in which a player's most-preferred candidate gives a payoff of 2, the second most-preferred candidate gives a payoff of 1, and the last candidate gives 0.
b. Find the subgame-perfect equilibrium for this game. Is it unique?
c. Are there Nash equilibria that are not subgame-perfect equilibria?
d. Now assume that before the two players play the game, player 2 can send an alienating e-mail to one of the candidates, which would result in that candidate withdrawing her application. Would player 2 choose to do this, and if so, with which candidate?

8.12 **Agenda Setting:** An agenda-setting game is described as follows. The "issue space" (set of possible policies) is an interval $X = [0, 5]$. An agenda setter (player 1) proposes an alternative $x \in X$ against the status quo $q = 4$. After player 1 proposes x, the legislator (player 2) observes the proposal and selects between the proposal x and the status quo q. Player 1's most-preferred policy is 1, and for any final policy $y \in X$ his payoff is given by

$$v_1(y) = 10 - |y - 1|,$$

where $|y - 1|$ denotes the absolute value of $(y - 1)$. Player 2's most preferred policy is 3, and for any final policy $y \in X$ his payoff is given by

$$v_2(y) = 10 - |y - 3|.$$

That is, each player prefers policies that are closer to his most-preferred policy.

a. Write the game down as a normal-form game. Is this a game of perfect or imperfect information?
b. Find a subgame-perfect equilibrium for this game. Is it unique?
c. Find a Nash equilibrium that is not subgame perfect. Is it unique? If yes, explain. If not, show all the Nash equilibria for this game.

8.13 **Junk Mail Advertising:** Suppose a single good is owned by a single seller who values it at $c > 0$ (he can consume the good and get a payoff of c). There is a single buyer who has a small transportation cost $k > 0$ to get to and back from the seller's store, and he values the good at $v > c + k$. The buyer first decides whether to make the commute or stay at home, not buy the good, and receive a payoff of 0. If the buyer commutes to the store, the seller can then make the buyer a take-it-or-leave-it price offer $p \geq 0$. The buyer can then accept the offer, pay p, and get the good, or he can walk out and not buy the good. Assume that c, v, and k are common knowledge.

a. As best you can, draw the extensive form of this game. What is the best response of the buyer at the node at which he decides whether to accept or reject the seller's offer?
b. Find the subgame-perfect equilibrium of the game and show that it is unique. Is it Pareto optimal?
c. Are there Nash equilibria that yield a higher payoff to both players as compared to the subgame-perfect equilibrium you found in (b)?

d. Now assume that before the game is played the seller can, at a small cost $\varepsilon < (v - c - k)$, send the buyer a postcard that commits the seller to a certain price at which the buyer can buy the good (e.g., "bring this coupon and get the good at a price p"). Would the seller choose to do so? Justify your answer with an equilibrium analysis.

8.14 **Hyperbolic Discounting:** Consider the three-period example of a player with hyperbolic discounting described in Section 8.3.4 with utility $\ln(x)$ in each of the three periods and with discount factors $0 < \delta < 1$ and $0 < \beta < 1$.

a. Solve the optimal choice of player 2, the second-period self, as a function of his budget K_2, δ, and β.
b. Solve the optimal choice of player 1, the first-period self, as a function of K, δ, and β.

8.15 **Time Inconsistency:** Consider the three-period example of a player with hyperbolic discounting described in Section 8.3.4 with utility $\ln(x)$ in each of the three periods, with initial budget K and with discount factors $\delta = 1$ and $\beta = \frac{1}{2}$.

a. Solve the optimal plan of action of a naive player 1 who does not take into account how his future self, player 2, will alter the plan. What is player 1's optimal plan x_1^*, x_2^*, and x_3^* as a function of K?
b. Let K_2 be the amount left from the solution to (a) after player 1 consumes his planned choice of x_1^*. Given K_2, what is the optimal plan of player 2? In what way does it differ from the optimal plan set out by player 1?

8.16 **The Value of Commitment:** Consider the three-period example of a player with hyperbolic discounting described in Section 8.3.4 with utility $\ln(x)$ in each of the three periods and with discount factors $\delta = 1$ and $\beta = \frac{1}{2}$. We solved the optimal consumption plan of a sophisticated player 1.

a. Imagine that an external entity can enforce any plan of action that player 1 chooses in $t = 1$ and will prevent player 2 from modifying it. What is the plan that player 1 would choose to enforce?
b. Assume that $K = 90$. Up to how much of his initial budget K will player 1 be willing to pay the external entity in order to enforce the plan you found in (a)?

9

Multistage Games

The normal-form game was a model of players choosing their actions simultaneously, without observing the moves of their opponents. The extensive form enriches the set of situations we can model to include those in which players learn something about the choices of other players who moved earlier, in which case a player can condition his behavior on the moves of other players in the game.

The extensive-form games analyzed so far were characterized by having payoffs delayed until the game reached a terminal node that was associated with the game's end. In reality dynamic play over time may be more complex, and it may not be correctly modeled by one "grand" game that unfolds over time with payoffs distributed at the end of the game. For instance, consider two firms that compete in one market, with the flow of profits that results from their behavior; after some time they choose to compete in another market, with a new flow of payoffs. Similarly a group of elected lawmakers may serve on a committee, on which their behavior and votes will affect the likelihood of their being reelected. Those who are reelected may sit on another committee together, with different choices and different consequences.

The broad message of these examples is that players can play one game that is followed by another, or maybe even several other games and receive some payoffs after each one of the games in this sequence is played. In such cases two natural questions emerge. First, if the players are rational and forward looking, should they not view this sequence of games as one grand game? Second, if they do view these as one grand game, should we expect that their actions in the later stages will depend on the outcomes of earlier stages? In particular, will the players be destined to play a sequence of action profiles that are Nash equilibria in each stage-game, or will they be able to use future games to support behavior in the earlier stages that is not consistent with Nash equilibrium in those early stages?

This chapter answers these questions. Before doing so, Sections 9.1–9.3 formally show how such multistage games can be modeled as extensive-form games so that we can use the tools developed in the previous chapter. It will be possible to answer the two questions just posed. First, the players should indeed anticipate future games and use them to create a richer environment for themselves. Second, they will benefit from using future play to create incentives that constrain their behavior in earlier stages. If players have the opportunity to condition future behavior on past outcomes, they may be able to create credible, self-enforcing incentives to follow behavior in earlier-stage games that would not be possible if there were no continuation games following. This

is the core insight of *multistage games:* if players can condition future behavior on past outcomes then this may lead to a richer set of self-enforcing outcomes.

9.1 Preliminaries

We define a **multistage game** as a finite sequence of normal-form **stage-games,** in which each stage-game is an independent, well-defined game of complete but imperfect information (a simultaneous-move game). These stage-games are played *sequentially* by the *same players,* and the *total payoffs* from the sequence of games are evaluated using the sequence of outcomes in the games that are played. We adopt the convention that each game is played in a *distinct period,* so that game 1 is played in period $t = 1$, game 2 in period $t = 2$, and so on, up until period $t = T$, which will be the last stage in the game.[1] We will also assume that, after each stage is completed, all the players observe the outcome of that stage, and that this information structure is common knowledge.[2]

In each stage-game, players have a set of actions from which they can choose, and the profiles of actions lead to payoffs for *that specific game,* which is then followed by another game, and another, until the sequence of games is over. A sequence of normal-form games of complete information with the same players is not too hard to conceive. Consider again the examples used earlier in this chapter, such as firms in common markets or lawmakers interacting on a sequence of committees. To make things concrete, let's start with a stylized example, which is a small twist on the familiar Prisoner's Dilemma game.

Consider a two-stage game that we call the Prisoner-Revenge Game. Suppose that in a first period labeled $t = 1$ two players from different local neighborhoods play the familiar Prisoner's Dilemma with pure actions mum and fink (uppercase for player 1 and lowercase for player 2) and with the payoff matrix

		Player 2	
		m	*f*
Player 1	*M*	4, 4	−1, 5
	F	5, −1	1, 1

Now imagine that after this game is over the same two players play the following Revenge Game in a second period $t = 2$. Each player can choose whether to join his local neighborhood gang or remain a "loner." If the players remain loners then they go their separate ways, regardless of the outcome of the first game, and obtain second-period payoffs of 0 each. If both join gangs, then because of the nature of neighborhood gangs they will fight each other and suffer a loss, causing each to receive a payoff of −3. Finally if player i joins a gang while player j remains a loner then

1. It is possible to consider an infinite sequence of games, but we restrict attention to a finite sequence of games in this chapter. In the next chapter we will consider such infinite sequences, in which the stage-games are always the same game that repeats itself again and again.

2. It is possible to imagine games in which some stages are played before the outcomes of previous stages are learned. In the extreme case of no information being revealed, this is no different than a simultaneous-move game. If some but not all the information is revealed, the analysis becomes more complicated, and such games are beyond the scope of this text.

the loner suffers dearly since he has no gang to defend him, thus receiving a payoff of -4, while the new gang member suffers much less and receives a payoff of -1. The Revenge Game is described by the following matrix:

		Player 2	
		l	g
Player 1	L	0, 0	$-4, -1$
	G	$-1, -4$	$-3, -3$

From our previous analysis of static games of complete information, we know how to analyze these two games independently. In the first-stage game, the Prisoner's Dilemma, there is a unique Nash equilibrium, (F, f). In the second-stage game, the Revenge Game, there are two pure-strategy Nash equilibria, (L, l) and (G, g), and a mixed-strategy equilibrium in which each player plays loner with probability $\frac{1}{2}$ (verify this!). In what follows, we will see that once these two games are played in sequence, equilibrium behavior can be more interesting.

9.2 Payoffs

How should we evaluate the *total payoffs* from a sequence of outcomes in each of the sequentially played stage-games? We adopt the well-defined notion of present value, which we used to analyze the single-person decision problems of consumption over time described in Sections 2.5.2 and 8.3.4.

In particular at any period t we will collapse any sequence of payments from the games played starting at period t onward into a *single value* that can be evaluated at period t. To do this we take the payoffs derived from each of the individual games and add them up with the added assumption (as described and motivated in Section 2.5.2) that future payoffs are discounted at a rate $0 \leq \delta \leq 1$. A higher discount factor δ means that the players are more patient and will care more about future payoffs. An extreme case of impatience occurs when $\delta = 0$, which means that only the payoffs of the current stage-game matter.

Consider a multistage game in which there are T stage-games played in each of the periods $1, 2, \ldots, T$. Let v_i^t be player i's payoff from the anticipated outcome in the stage-game played in period t. We denote by v_i the *total payoff* of player i from playing the sequence of games in the multistage game and define it as

$$v_i = v_i^1 + \delta v_i^2 + \delta^2 v_i^3 + \cdots + \delta^{T-1} v_i^T = \sum_{t=1}^{T} \delta^{t-1} v_i^t,$$

which is the *discounted sum* of payoffs that the player expects to get in the sequence of games. The payoffs one period away are discounted once, those two periods away are discounted twice (hence the δ^2), and so on. For example, if in the Prisoner-Revenge Game we have the sequence of play be (F, m) followed by (L, g) then, given a discount factor δ for each player, the payoffs for players 1 and 2 are $v_1 = 5 + \delta(-4)$ and $v_2 = -1 + \delta(-1)$.

Using the concept of discounted payoffs we can write down the extensive form of the multistage Prisoner-Revenge Game to demonstrate all the possible outcomes, and the total sum of discounted payoffs, as shown in Figure 9.1. Notice that this way of

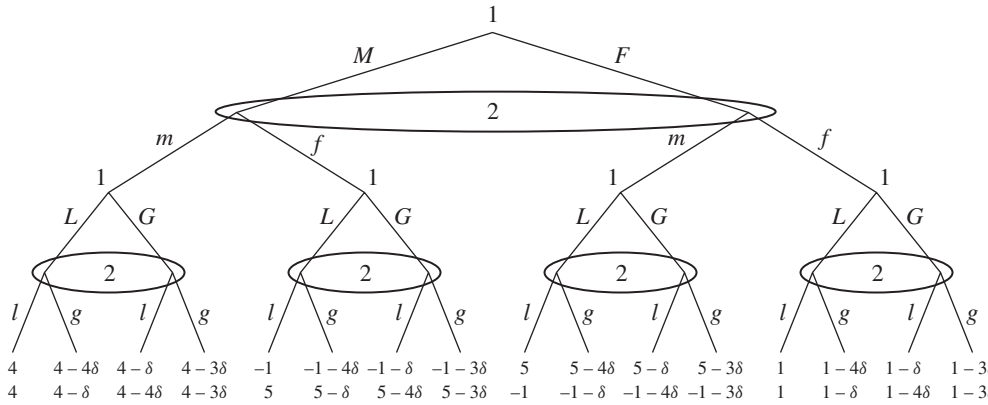


FIGURE 9.1 The multistage Prisoner-Revenge Game.

representing the game does not seem to capture the fact that payoffs are accumulated as the stages of the game proceed. That said, this is not important from a strategic point of view because we have incorporated the payoffs into the extensive form of the multistage game by accurately taking into account the sum of discounted payoffs that are a consequence of any combination of play in the two stage-games.

9.3 Strategies and Conditional Play

Imagine that the players are playing a multistage game that consists of T stage-games. If we naively treat each game independently, so that players do not make any attempt to link their behavior in a given stage-game to the outcomes of previous stage-games, then strategies would be extremely simple: each player treats each stage-game independently and commits to some action for each of the stage-games that is independent of how the games unfold over time.

More sophisticated players may benefit from using strategies that create a *strategic link* between the stage-games in ways that are no different than what children do in preschool: "If you play with the dolls this way now, I will play hide-and-seek with you later, and if you don't then I won't play with you at all." More generally players can use strategies of the form "If such-and-such happens in games 1, 2, . . . , $t - 1$ then I will choose action a_i in game t." In the Prisoner-Revenge Game, for example, player 1 can use the following strategy: "I will play F in the first game and I will play L in the second game only if player 2 played m in the first game. If player 2 played f in the first game then I will play G in the second."

A convenient way to introduce such strategies is by writing down the extensive form of the complete multistage game as we did in Figure 9.1 and focus on what the players can do at every stage of the game. It is important to determine correctly the information sets of each player. Because the outcomes of every stage are revealed before the next stage is played, the number of information sets at any stage t must be equal to the number of possible outcomes from the previously played stage-games, 1, 2, . . . , $t - 1$.

For example, in the multistage Prisoner-Revenge Game, the first-stage game has four outcomes. This implies that in the second-stage game each player should have four information sets, each associated with one of the four outcomes of the first-stage game. Note, however, that in each stage-game the players do not know what their

opponent is choosing *at that stage,* which means that we must have some information sets with more than one node in them, as demonstrated in Figure 9.1. In the first stage each player chooses between mum and fink without knowing what his opponent is choosing at that stage. In the second stage he chooses between loner and gang without knowing what his opponent is choosing at that stage, but each player knows exactly how the first-stage game concluded: with either Mm, Mf, Fm, or Ff.

We can therefore define a strategy for player i in the multistage Prisoner-Revenge Game as a quintuple, $s_i = (s_i^1, s_i^2(Mm), s_i^2(Mf), s_i^2(Fm), s_i^2(Ff))$, where s_i^1 is what player i will do in the first-stage Prisoner's Dilemma game and $s_i^2(ab)$ is what player i will do in the second-stage Revenge Game, if player 1 played $a \in \{M, F\}$ and player 2 played $b \in \{m, f\}$ in the first-stage game. Notice that each element of the quintuple that represents each player's strategy can take on two values (e.g., $s_1^1 \in \{M, F\}$ and $s_1^2(Mm) \in \{L, G\}$). Hence each player has 32 pure strategies, which means that if we try to draw a matrix it will be of size 32×32 and will have 1024 pure-strategy combinations! For this reason it is most useful to use the extensive form to analyze such a game.

More generally, in a multistage game that consists of T stage-games, a pure strategy of player i will be a list of *conditional* pure strategies of the following form: $S_i = \{s_i^1, s_i^2(h_1), \ldots, s_i^t(h_{t-1}), \ldots, s_i^T(h_{T-1})\}$, where h_{t-1} is a particular outcome that occurred up to period t, not including period t, and $s_i^t(h_{t-1})$ is an action for player i from the tth stage-game. We can conveniently think of h_{t-1} as a particular *history* of events that occurred up to period t, from the set of all possible histories (or outcomes) H_{t-1}. The different histories that occurred before stage-game t are associated with different information sets in the stage-game of period t, each history being associated with a distinct sequence of outcomes from the previous stage-games in periods $1, 2, \ldots, t-1$ combined.

To solidify this idea consider a game with n firms that are choosing prices in a sequence of markets. At stage $t = 1$ they choose prices in market 1, profits in this market are determined, and this is followed by $T - 1$ more market games, each with its own demand structure. If each firm selects its price p_i^t in period t (which is stage-game t), then a pure strategy for firm i is a list of prices $p_i^1, p_i^2(h_1), \ldots, p_i^t(h_{t-1}), \ldots, p_i^T(h_{T-1})$, where $p_i^1 \geq 0$ and $p_i^t(h_{t-1}) \geq 0$ for all $t \in \{2, \ldots, T\}$. Each history h_t is the sequence of previously chosen prices: $h_1 = (p_1^1, p_2^1, \ldots, p_n^1)$, $h_2 = ((p_1^1, p_2^1, \ldots, p_n^1), (p_1^2, p_2^2, \ldots, p_n^2))$, and generally

$$h_{t-1} = ((p_1^1, p_2^1, \ldots, p_n^1), (p_1^2, p_2^2, \ldots, p_n^2), \ldots, (p_1^{t-1}, p_2^{t-1}, \ldots, p_n^{t-1})).$$

Notice that for this example a pure strategy for player i is a list of continuous functions, each from the set of continuously many histories to a single price.

Just as we have defined pure strategies for multistage games, we can define mixed behavioral strategies in which every player can mix between his or her pure actions in every information set. The formal extension is straightforward: in a multistage game that consists of T stage-games a behavioral strategy of player i will be a list of conditional randomizations of the following form: $\sigma_i = \{\sigma_i^1, \sigma_i^2(h_1), \ldots, \sigma_i^t(h_{t-1}), \ldots, \sigma_i^T(h_{T-1})\}$, where h_{t-1} is the set of outcomes that occurred up to period t, not including period t, and $\sigma_i^t(h_{t-1})$ is a randomization over player i's actions in the stage-game of period t.

In summary the definition of strategies for multistage games uses the extensive form to represent the game's structure. Furthermore, as in extensive-form games, strategies are defined as a complete list of (mixed or pure) actions for each player

at each of his information sets. In multistage games, each player's information set is associated with the history of play from the previous stage-games.

9.4 Subgame-Perfect Equilibria

Because multistage games are dynamic in nature, and because past play is revealed over time, we turn to the notion of a subgame-perfect equilibrium as a solution concept. In particular, rational players should play sequentially rational strategies, which justifies the use of the subgame-perfect equilibrium concept. The question then is, how do we find a subgame-perfect equilibrium for such a game? The following result will offer a natural first step:

Proposition 9.1 *Consider a multistage game with T stages, and let* σ^{t*} *be a Nash equilibrium strategy profile for the t*th *stage-game. There exists a subgame-perfect equilibrium in the multistage game in which the equilibrium path coincides with the path generated by* $\sigma^{1*}, \sigma^{2*}, \ldots, \sigma^{T*}$.

Proof Fix a sequence of Nash equilibria for each stage-game, and let σ^{t*} denote the profile of strategies that form the Nash equilibrium for the stage-game t, where σ_i^{t*} is player i's strategy in the profile σ^{t*}. Consider the following *unconditional* strategies for each player i: in each stage $t \in \{1, 2, \ldots, T\}$ play σ_i^{t*} *regardless* of the history of plays before stage t (i.e., $\sigma_i^1 = \sigma_i^{1*}$ and $\sigma_i^t(h_{t-1}) = \sigma_i^{t*}$ for any $h_{t-1} \in H_{t-1}$). By construction, for subgames that start at stage T this is a Nash equilibrium. Now consider all the subgames that start at stage $T-1$. The total payoffs from play in stage $T-1$ are equal to the payoffs from the $T-1$ stage-game plus a constant that has two components. One is the fixed payoff resulting from the previous games, and the second is the expected payoff from the game at stage T that does not depend on the outcomes of stage $T-1$. This implies that σ_{T-1}^* followed by σ_T^* in any subgame at stage T is a Nash equilibrium in any subgame starting at time $T-1$. Using backward induction, this argument implies that at any stage t, current play does not affect future play, and this inductive argument for a finite number of stages in turn implies that in any subgame these strategies constitute a Nash equilibrium. ■

The proof follows from a very intuitive argument. To see this consider the Prisoner-Revenge Game, described earlier, and focus on the sequence of stage-game Nash equilibria, (F, f) followed by (L, l). Now consider the following multistage strategies for the two players:

$$s_1 = (s_1^1, s_1^2(Mm), s_1^2(Mf), s_1^2(Fm), s_1^2(Ff)) = (F, L, L, L, L)$$

$$s_2 = (s_2^1, s_2^2(Mm), s_2^2(Mf), s_2^2(Fm), s_2^2(Ff)) = (f, l, l, l, l).$$

That is, each player plays fink in the first-stage game and then plays loner in the second-stage game *regardless of what was played in the first stage.* Now observe that with this pair of strategies the players are clearly playing a Nash equilibrium in each of the four subgames in the second stage. This follows because the first-period payoffs are set in stone and do not depend on the second-period choices, and in each of these second-period subgames both players playing loner is a Nash equilibrium. Then, moving back to the subgame that begins with the first stage-game (which is the whole subgame), because the payoffs in the second period are a constant equal to 0 for each player, regardless of the play in the first stage-game, it follows that each player has

a dominant strategy, which is to play fink.[3] Thus both playing fink in the first-stage game is part of the Nash equilibrium in the whole game. It is easy to see that this argument immediately implies that there is another (pure-strategy) subgame-perfect equilibrium of the multistage game in which the players play

$$s_1 = (s_1^1, s_1^2(Mm), s_1^2(Mf), s_1^2(Fm), s_1^2(Ff)) = (F, G, G, G, G)$$

$$s_2 = (s_2^1, s_2^2(Mm), s_2^2(Mf), s_2^2(Fm), s_2^2(Ff)) = (f, g, g, g, g).$$

The logic is identical because both players playing gang is a Nash equilibrium of the Revenge Game.[4]

The proposition should therefore be quite consistent with what we expect. It basically says that if we look at a sequence of plays that is a Nash equilibrium in each game *independently* then players should be content playing these stage-game strategies in each stage-game. By considering such unconditional strategies, we are removing any *strategic link* between stages, and thus each stage can be treated as if it were an independent game.

The interesting question that remains to be answered is whether we are ignoring behavior that can be supported as a subgame-perfect equilibrium if we actually use a strategic linkage between the games. In other words, if players were to condition their future play on past play in a sequentially rational manner, would we be able to support behavior in early stages that is not a Nash equilibrium in the early stage-games?

The answer is yes, and this approach yields some interesting insights, which we discuss subsequently. The trick for strategically linking the stage-games will be to *condition the behavior in later stage-games* on the *actions taken in earlier stage-games.* As we will see, this will be possible only when some of the stage-games in later periods have multiple Nash equilibria. As a first step toward understanding this requirement it is easy to establish the following proposition:

Proposition 9.2 *If σ^* is a Nash equilibrium of the multistage game consisting of stage-games $G_1, G_2, \ldots, G_T$ then the restriction of σ^* to the stage-game in period T must be a Nash equilibrium of that stage-game.*

The proposition states that for any finite stage-game of length T, in the last stage-game G_T the players *must play a Nash equilibrium* of that stage-game. The reasoning is simple: because there is no future that can depend on the actions taken at stage T, the only thing that determines what is best for any player at that stage is to play a best response to what he believes the other players are doing at that stage. This insight implies another important fact:

Proposition 9.3 *If a finite multistage game consists of stage-games that each have a* unique Nash equilibrium *then the multistage game has a* unique subgame-perfect equilibrium.

3. Note that adding a constant to the payoffs of players at every terminal node of an extensive-form game will not change the players' ranking of the outcomes and hence will not change the set of equilibria.

4. Because there is also a mixed-strategy Nash equilibrium in the Revenge Game, there is a third subgame-perfect equilibrium that can be constructed in a similar way: fink in the first-stage game followed by the mixed-strategy Nash equilibrium in the Revenge Game, which is played unconditional on the outcome of the first-stage game.

The proof of this simple observation follows immediately from backward induction. In the last stage, T, the players must play the unique Nash equilibrium of that game. In the stage before last, $T-1$, they cannot condition future (T-stage) behavior on current ($T-1$)-stage outcomes, so they must play the unique Nash equilibrium of the stage-game in period $T-1$, continued by the unique Nash equilibrium of the stage-game in period T. This induction argument continues until the first stage of the game.

We now use the Prisoner-Revenge Game as an illustration of what players can do when the stage-game played at the end of a multistage game has more than one Nash equilibrium. It would be interesting, and surely appealing for the players, to try to support a subgame-perfect equilibrium in which the players will play (M, m) in the first period. This would be interesting because, as we know, in the stand-alone Prisoner's Dilemma, playing (M, m) cannot be supported as an equilibrium. However, given the appended Revenge Game, we may be able to use history-dependent strategies to support the cooperative behavior in the first stage. As a starting point, proposition 9.2 implies that as part of any pure-strategy subgame-perfect equilibrium of the multistage Prisoner-Revenge Game the players will have to play (L, l) or (G, g) in any of the subgames that include the second-stage game (or the unique mixed-strategy Nash equilibrium of the second-stage game). Intuitively sequential rationality implies that players *must play a Nash equilibrium* in the second-stage game.

To support cooperative behavior (M, m) as part of an equilibrium path of play in a subgame-perfect equilibrium, the players must find a way to give themselves incentives to stick to (M, m) and resist the temptation to deviate to fink. Notice that in the second-stage Revenge Game the two pure-strategy Nash equilibria are very different. The Nash equilibrium (L, l) yields each player a payoff of 0, which we can call the "friendly" equilibrium, and (G, g) yields each player a payoff of -3, which we can call the "gang" equilibrium. To provide the incentives for the first-stage game, the players can agree to play the friendly equilibrium of the Revenge Game if (M, m) was played in the first stage, while they can *punish themselves* by switching to the gang equilibrium of the Revenge Game if anything other than (M, m) was played in the first-stage game. That is, each player will play the following strategy in the multistage game:

Player 1: Play M in stage 1. In stage 2 play L if (M, m) was played in stage 1, and play G if anything but (M, m) was played in stage 1 (or $s_1^* = (s_1^1, s_1^2(Mm), s_1^2(Mf), s_1^2(Fm), s_1^2(Ff)) = (M, L, G, G, G)$).

Player 2: Play m in stage 1. In stage 2 play l if (M, m) was played in stage 1, and play g if anything but (M, m) was played in stage 1 (or $s_2^* = (s_2^1, s_2^2(Mm), s_2^2(Mf), s_2^2(Fm), s_2^2(Ff)) = (m, l, g, g, g)$).

We are left to check if this pair of strategies is a subgame-perfect equilibrium. It is easy to see that in each of the subgames beginning at stage 2 the players are playing a Nash equilibrium (either both play loner following (M, m) or both play gang following other histories of play). Hence to check that this is a subgame-perfect equilibrium we need to check that players would not want to deviate from mum in the first stage of the game. In other words, is mum a best response in period 1, given what each player believes about his opponent, and given the continuation play in the second-period stage-games? Consider player 1 and observe that

$$v_1(M, s_2) = 4 + 0 \times \delta \qquad \text{and} \qquad v_1(F, s_2) = 5 + (-3) \times \delta,$$

which implies that M is a best response if and only if $4 \geq 5 - 3\delta$, or

$$\delta \geq \tfrac{1}{3}.$$

Thus if the discount factor is not too small then we can support the behavior of (M, m) in the first-stage game even though it is not a Nash equilibrium in the stand-alone Prisoner's Dilemma. The fact that the Revenge Game has two different Nash equilibria, one of which is significantly better than the other, allows the players to offer a self-enforcing incentive scheme that supports cooperative behavior in the first-stage game.

This example is illuminating because it sheds light on two requirements that are crucial to supporting behavior in the first stage (or early periods in general) that is *not* a Nash equilibrium at this first stage. These requirements are as follows:

1. There must be at least two distinct equilibria in the second stage: a "stick" and a "carrot."
2. The discount factor has to be large enough for the difference in payoffs between the "stick" and the "carrot" to have enough impact in the first stage of the game.

The first requirement is necessary to offer the players the possibility of "reward and punishment" strategies, which help support first-period play that is not a Nash equilibrium in the stand-alone first-stage game. To see this, consider the following: If we are trying to support behavior in the first period that is not a Nash equilibrium in that period, it must be the case that at least one player would like to deviate in the first stage (because at least one is not playing a best response). The player would indeed deviate from the proposed play in the first period if there were no future consequences to such a deviation. In other words, in the first stage some players would benefit in the short run from deviating from the proposed path of play, and to keep them on the path of play we must guarantee that such deviations will be met with credible "punishments." For these punishments to be credible, they must take the form of moving from an equilibrium in which the player gets a high payoff to an equilibrium in which he gets a low payoff. That is, the players use *long-term losses* to deter themselves from pursuing *short-term gains.*

This is where the second requirement has bite: it must be that these long-term losses are enough to deter the players from deviating toward their short-term gains. For this to be the case it must be that the players value the payoffs that they will receive in the second period, which can only happen if they do not discount the future too much. Thus the *effective punishment* from deviating will depend first on the difference in payoffs of the two equilibria, the "reward" and the "punishment," and second on the discount factor. A more figurative way to think about this is that the friendly and gang continuation equilibria are the "carrot" and the "stick," respectively. Then a smaller discount factor makes the carrot smaller and the stick less painful, which in turn makes deterrence harder.

To see another example of a similarly constructed subgame-perfect equilibrium in the Prisoner-Revenge Game, we can support the path of play (F, m) followed by (L, l). To do this we use the following strategies for each player: Player 1 plays F in stage 1, and player 2 plays m. In stage 2 both players play loner if (F, m) was played in stage 1, while they play gang if anything but (F, m) was played in

stage 1 (formally, $s_1 = (s_1^1, s_1^2(Mm), s_1^2(Mf), s_1^2(Fm), s_1^2(Ff)) = (F, G, G, L, G)$, and $s_2 = (s_2^1, s_2^2(Mm), s_2^2(Mf), s_2^2(Fm), s_2^2(Ff)) = (m, g, g, l, g)$). Clearly player 1 would not want to deviate from F in the first period because he's playing a dominant strategy in that period, followed by the best possible second-stage equilibrium. For player 2 things are different. In the first-stage game his payoffs from m and f, respectively, are:

$$v_2(s_1, m) = -1 + 0 \times \delta \qquad \text{and} \qquad v_2(s_1, f) = 1 - 3 \times \delta,$$

which implies that m is a best response for player 2 in the first-stage game if and only if $-1 \geq 1 - 3\delta$, or if

$$\delta \geq \tfrac{2}{3}.$$

As we can see, to support this outcome we need a higher discount factor compared with supporting (M, m) in the first period. Compared to the earlier path of (M, m) followed by (L, l), here the future loss of 3 is enough to deter player 2's deviation only when the future is not discounted as much, because the short-term gains from deviating are equal to 2. These are higher than the gains from deviating from (M, m), which were only 1.

The foregoing analysis of the multistage Prisoner-Revenge Game captures most of what is interesting about multistage games. For this reason it is not too useful to consider more involved games. The next section offers a result that is quite technical in nature but that is also very useful in proving which strategies are or are not part of a subgame-perfect equilibrium. To summarize: in order to check if a strategy profile $(\sigma_i^*, \sigma_{-i}^*)$ is a subgame-perfect equilibrium in a multistage (or any finite extensive-form) game, all we need to show is that for every player i, given σ_{-i}^*, player i does not have a *single information set* from which he would want to deviate. In the next chapter this result will prove to be even more useful.

9.5 The One-Stage Deviation Principle

As an illustration of multistage games, the Prisoner-Revenge Game was not hard to analyze. In particular, to check that a certain proposed strategy profile was a subgame-perfect equilibrium, we needed only to check that no player wanted to deviate in the first stage, because we constructed our candidates for subgame-perfect equilibria to consist of Nash equilibria in each of the second-stage subgames. It might seem that if we had, say, a five-stage game it would be more complicated to check that a profile of strategies constitutes a subgame-perfect equilibrium, and that in particular we would have to check that the players do not want to deviate at any single stage-game. That is, maybe they could gain by combining several deviations in separate stages of the game.

This is where the **one-stage deviation principle** comes in to simplify what seems like a rather daunting task. Interestingly the principle is based on an idea that was originally formulated by David Blackwell (1965) in the context of dynamic programming for a single-player decision problem. The relation to a single-player decision problem comes from the fact that when we want to check if a player i is playing a best response to σ_{-i} in every subgame, then all we are really doing is checking to see whether he is playing an optimal action in each of his information sets, *taking the actions of all other players σ_{-i} as given.* Thus once the strategies σ_{-i} of other players in the extensive form are considered to be fixed, player i solves a standard dynamic programming problem like the ones discussed in Section 2.4. This in turn

implies that we can formulate the one-stage deviation principle within the context of a single-person decision tree.

Consider an extensive-form game tree of finite length (every path ends after a finite number of moves) with payoffs associated with each terminal node, and fix the strategies of all players that are not i. The strategy σ_i assigns to each information set h_i a probability distribution over i's actions $A_i(h_i)$, and starting from any information set h_i a strategy profile (σ_i, σ_{-i}) induces a probability distribution over paths of play and terminal nodes.

Thus if we treat every information set h_i as a node in the single-player decision tree induced by σ_{-i}, we can define $v_i(\sigma_i, h_i)$ to be the expected payoff of player i from the information set h_i onward by playing σ_i (recall that σ_{-i} is considered to be fixed). We say that a strategy σ_i is *optimal* if there is no strategy σ_i' and information set h_i such that $v_i(\sigma_i', h_i) > v_i(\sigma_i, h_i)$.

Given a strategy σ_i, define the strategy σ_i^{a,h_i} as the strategy that is identical to σ_i everywhere except at h_i, and at h_i we substitute the prescribed (possibly random) action of σ_i with the action $a \in A_i(h_i)$. We introduce the following definition:

Definition 9.1 A strategy σ_i is **one-stage unimprovable** if there is no information set h_i, action $a \in A_i(h_i)$, and corresponding strategy σ_i^{a,h_i} such that $v_i(\sigma_i^{a,h_i}, h_i) > v_i(\sigma_i, h_i)$.

Clearly an optimal strategy is one-stage unimprovable. It is the converse that is stated in the following theorem:

Theorem 9.1 *A one-stage unimprovable strategy must be optimal.*

Proof Suppose in negation that σ_i is one-stage unimprovable but it is not optimal. Then there exist σ_i' and information set h_i^1 such that $v_i(\sigma_i', h_i^1) > v_i(\sigma_i, h_i^1)$. Because stochastic strategies do not affect the highest attainable payoffs, this is equivalent to saying that there is a path starting from h_i^1 through a finite number n of i's information sets (the last being a terminal node) such that i's payoff from this path is greater than $v_i(\sigma_i, h_i^1)$. Call these n information sets that define this path $h_i^1, h_i^2, \ldots, h_i^n$. This implies that a finite number of one-stage deviations at the information sets h_i^1 through h_i^{n-1}, with σ_i used everywhere else in the tree, will offer i a higher payoff than σ_i. Define a collection of $n-1$ different strategies σ_i^t, for $t = 1, \ldots, n-1$, such that σ_i^t chooses the action $a_i \in A_i(h_i^r)$ that induces the information set h_i^{r+1}, for every r between 1 and t, and coincides with σ_i at every other information set of i. Then our earlier assertion that σ_i is not optimal implies that

$$v_i(\sigma_i^{n-1}, h_i^1) > v_i(\sigma_i, h_i^1). \tag{9.1}$$

Notice that σ_i^{n-2} fully coincides with σ_i from the information set h_i^{n-1} up to h_i^n, while σ_i^{n-1} is a one-stage deviation from σ_i at h_i^{n-1}. Because σ_i is one-stage unimprovable by assumption, it must be that

$$v_i(\sigma_i^{n-2}, h_i^{n-1}) = v_i(\sigma_i, h_i^{n-1}) \geq v_i(\sigma_i^{n-1}, h_i^{n-1}).$$

Because σ_i^{n-2} and σ_i^{n-1} share the same information sets h_i^1 through h_i^n along every path generated by the two, it must be that

$$v_i(\sigma_i^{n-2}, h_i^1) \geq v_i(\sigma_i^{n-1}, h_i^1). \tag{9.2}$$

Combining (9.1) and (9.2) we conclude that

$$v_i(\sigma_i^{n-2}, h_i^1) > v_i(\sigma_i, h_i^1).$$

Using an induction argument we must conclude that

$$v_i(\sigma_i^1, h_i^1) > v_i(\sigma_i, h_i^1), \tag{9.3}$$

but σ_i^1 is a one-stage deviation from σ_i, implying that (9.3) contradicts the assertion that σ_i is one-stage unimprovable. Therefore if σ_i is one-stage unimprovable then it must be optimal. ■

Though the formal proof is a bit tedious, the intuition is not. If the strategy σ_i were not a best response to σ_{-i} (not "optimal" in the proof just given), then given that the game is finite, there must be a finite number of deviations to improve upon σ_i. We can look for the "last" such deviation in the sequence, and if we restrict attention to the subgame that starts at that information set (or at the closest possible root that precedes it, if it is not a singleton) then in that subgame player i has a one-stage deviation that makes him better off.

9.6 Summary

- Multistage games are defined by a series of normal-form stage-games that are played in sequence, in which players obtain payoffs after every stage-game and future payoffs are discounted.
- Any sequence of stage-game Nash equilibrium play can be supported as a subgame-perfect equilibrium in the multistage game, regardless of the discount factor.
- Players can use credible threats and promises in later stages to provide long-term incentives for short-term actions that may not be self-enforcing in the earlier-period stage-games.
- Because future payoffs are discounted, the effectiveness of long-term incentives will depend on how patient the players are.
- The set of outcomes that can be supported by a subgame-perfect equilibrium will often depend on the discount factor.

9.7 Exercises

9.1 **Multistage Practice:** Consider the following simultaneous-move game that is played twice (the players observe the first-period outcome prior to the second-period play):

		Player 2		
		L	*C*	*R*
Player 1	*T*	10, 10	2, 12	0, 13
	M	12, 2	5, 5	0, 0
	B	13, 0	0, 0	1, 1

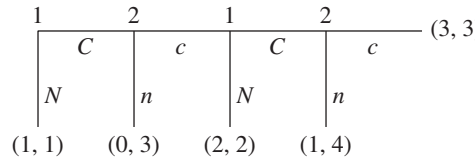


FIGURE 9.2 The Centipede Game.

a. Find all the pure-strategy subgame-perfect equilibria with no discounting ($\delta = 1$). Be precise in defining history-contingent strategies for both players.
b. For each of the equilibria you found in (a), find the smallest discount factor that supports it.

9.2 **Centipedes Revisited:** Two players are playing two consecutive games. First, they play the Centipede Game described in Figure 9.2. After the Centipede Game they play the following coordination game:

		Player 2	
		a	*b*
Player 1	*A*	1, 1	0, 0
	B	0, 0	3, 3

a. What are the Nash equilibria of each stage-game?
b. How many pure strategies does each player have in the multistage game?
c. Find all the pure-strategy subgame-perfect equilibria with extreme discounting ($\delta = 0$). Be precise in defining history-contingent strategies for both players.
d. Now let $\delta = 1$. Find a subgame-perfect equilibrium for the two-stage game in which the players receive the payoffs (2, 2) in the first stage-game.
e. What is the lowest value of δ for which the subgame-perfect equilibrium you found in (d) survives?
f. For δ greater than the value you found in (e), are there other outcomes of the first-stage Centipede Game that can be supported as part of a subgame-perfect equilibrium?

9.3 **Campaigning Revisited:** Two political candidates are scheduled to campaign in two states, in one in period $t = 1$ and in the other in $t = 2$. In each state each candidate can choose either a positive campaign that promotes his own agenda (P for player 1, p for player 2) or a negative one that attacks his opponent (N for player 1, n for player 2). Residents of the first-period state do not mind negative campaigns, which are generally effective, and payoffs in this state are given by the following matrix:

		Player 2	
		p	*n*
Player 1	*P*	2, 2	0, 5
	N	5, 0	3, 3

In the second-period state, residents dislike negative campaigns despite their effectiveness, and the payoffs are given by the following matrix:

		Player 2	
		p	n
Player 1	P	4, 4	1, 0
	N	0, 1	2, 2

a. What are the Nash equilibria of each stage-game? Find all the pure-strategy subgame-perfect equilibria with extreme discounting ($\delta = 0$). Be precise in defining history-contingent strategies for both players.
b. Now let $\delta = 1$. Find a subgame-perfect equilibrium for the two-stage game in which the players choose (P, p) in the first stage-game.
c. What is the lowest value of δ for which the subgame-perfect equilibrium you found in (b) survives?
d. Can you find a subgame-perfect equilibrium for this game in which the players play something other than (P, p) or (N, n) in the first stage?

9.4 **Online Gaming:** Consider a two-stage game between two firms that produce online games. In the first stage, they play a Cournot competition game (each chooses a quantity q_i) with demand function $p = 100 - q$ and zero marginal production costs ($c_i(q_i) = 0$ for $i = 1, 2$). In the second stage, after observing the pair (q_1, q_2) and after profits have been distributed, the players play a simultaneous-move "access" game in which each can open its platform to allow players on the other platform to play online with players on its platform (O for player 1, o for player 2), or choose to keep its platform noncompatible (N for player 1, n for player 2), in which case each platform's players can play only with others on the same platform. If they choose (N, n) then second-stage payoffs are (0, 0). If only one firm chooses to open its platform, it bears a cost of -10 with no benefit, since the other firm did not allow open access. Finally if both firms choose (O, o) then each firm gets many more eyeballs for advertising, and payoffs for each are 2500. Both players use the same discount factor δ to discount future payoffs.

a. Find the unique Nash equilibrium in the first-stage Cournot game and all of the pure-strategy Nash equilibria of the second-stage access game. Find all the pure-strategy subgame-perfect equilibria with extreme discounting ($\delta = 0$). Be precise in defining history-contingent strategies for both players.
b. Now let $\delta = 1$. Find a subgame-perfect equilibrium for the two-stage game in which the players choose the monopoly (total profit–maximizing) quantities and split them equally (a symmetric equilibrium).
c. What is the lowest value of δ for which the subgame-perfect equilibrium you found in (b) survives?
d. Now let $\delta = 0.4$. Can you support a subgame-perfect equilibrium for the two-stage game in which the players choose the monopoly quantities and split them equally? If not, what are the highest profits that the firms can make in a symmetric equilibrium?

9.5 **Campaign Spending:** Two political candidates are destined to play the following two-stage game. Assume throughout that there is no discounting

11

Strategic Bargaining

Of the many possible applications of strategic interaction among a small number of players, one ubiquitous situation surely comes to mind: bargaining. Examples are plentiful: a firm and a union bargaining over wages and benefits; a local municipality bargaining with a private provider over the terms of service; the head of a political party bargaining with other party members over campaign issues; and even the mundane bargaining that occurs between a buyer and a merchant at a street bazaar. In all of these examples, deals are struck by parties bargaining over terms, payments, and other aspects that all boil down to one basic fact: each party wants to get the best deal it can.

How can we model and analyze bargaining through the lens of game theory? As demonstrated by the examples just mentioned, the issue is often about a surplus that has to be split among the parties, and it involves the parties making proposals, responding to them, and trying to settle on an agreement. Following some early work by Ståhl (1972, 1977), Rubinstein (1982) offered a particular stylized model of bargaining that has become the standard in most applications. The simple framework considers two players that need to split a "pie" (representing the surplus from an agreement or the gains from trade). The pie is assumed to have a total value that is normalized to equal 1, and then the parties bargain on how to split the pie.

In order to analyze this process as a strategic game, the bargaining process itself needs to be structured to include actions, outcomes that result from actions, and preferences over the outcomes, as with any other game we can imagine. Ståhl suggested a prespecified procedure in which the game starts with player 1 making an offer to split the pie and player 2 either accepting or rejecting the offer. If the offer is accepted, the split proposed by player 1 is implemented, and the game ends. If player 2 rejects player 1's offer, the players then switch roles: player 2 makes the offer and player 1 responds by accepting or rejecting the offer. The game can either continue this way until some exogenous deadline arrives or simply go on forever.

Yet, as the saying goes, "time is money," and delay ought to impose some loss on the players. There are two ways to impose this loss. First, there may be a fixed cost of progressing from one round to another, so that a constant piece of the pie is removed every time a rejection occurs. Alternatively we can impose discounting in the same way we did for repeated games: after every period of rejection, the total size of the pie is only a fraction of what it was before. This second approach is the one we will be following here, and we can summarize the game as follows.

In the first round:

- Player 1 offers shares $(x, 1-x)$, where player 1 receives x and player 2 receives $1-x$.
- Player 2 then chooses to accept, causing the game to end with payoffs $v_1 = x$ and $v_2 = 1-x$, or reject, causing the game to move to the second round.

In the second round:

- A share $1-\delta$ of the pie is dissipated (the pie is discounted with discount factor $0 < \delta < 1$).
- Player 2 offers shares $(x, 1-x)$, where player 1 receives x and player 2 receives $1-x$.
- Player 1 chooses to accept, causing the game to end with payoffs $v_1 = \delta x$ and $v_2 = \delta(1-x)$, or reject, causing the game to move to the third round.

In the third round and beyond:

- The game continues in the same way, where following rejection in an odd period player 2 gets to offer in the next even period, and vice versa.
- Each period has further discounting of the pie, so that in period t the total pie is worth δ^{t-1}.

From the specification of the bargaining procedure we see that once a player agrees to the offer made by his opponent, the game will end with an agreement. If instead the players keep rejecting offers then the game may continue either indefinitely or until some prespecified deadline is reached. Assume for now that the game ends in some prespecified final period T, which will be reached if the players did not reach an agreement in the $T-1$ periods that preceded T. Furthermore assume that period T is a "hard deadline," so that both players receive a payoff of zero if they fail to reach an agreement in this last period. This, for example, can be the case with a fisherman bargaining with a restaurateur over a fresh catch. If they wait more than a few hours, the fish will go bad and there will no longer be gains from trade. The heuristic extensive form of this game with an odd number of periods T (so that player 1 is first and last) is depicted in Figure 11.1.

Notice that this game has an interesting structure that is different from what we have seen so far. On the one hand, it has some features of a finitely repeated game as follows: If we think of an "odd" round as one in which player 1 proposes and player 2 responds, and an "even" round as the converse, then we can treat each *pair of rounds* as *one stage* that repeats itself as long as an agreement is not reached. Furthermore as rounds proceed the value of the pie that these players need to split is shrinking according to the discount factor δ. On the other hand, there are two features that are not part of the repeated-game structure we developed in Chapter 10. First, the game can end during any round if the proposal is accepted, and second, payoffs are obtained only when the game actually ends and not as a flow of stage-game payoffs.

Before proceeding with the analysis, an obvious question is whether this stylized bargaining game is a reasonable caricature of reality. We need to convince ourselves that this game is, at least at some level, representative of what we believe bargaining is about. To a large extent, sequences of offers and counteroffers are natural components of any type of bargaining, and clearly once agreements are reached the bargaining stage is over. However, the fixed end date T, at which an all-or-nothing agreement

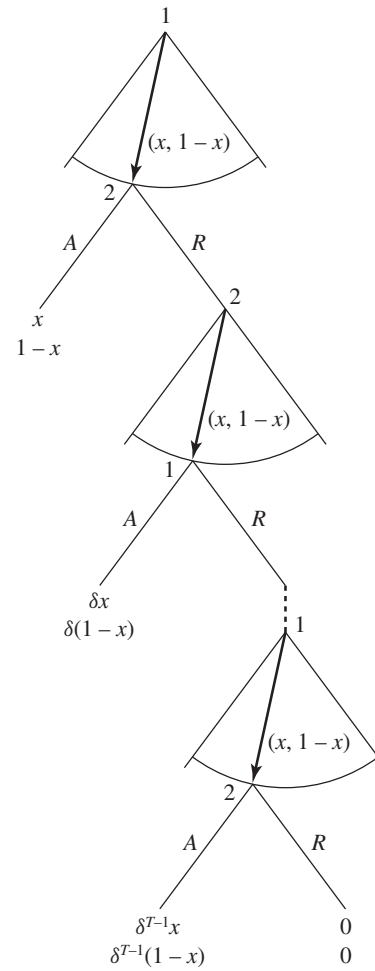


FIGURE 11.1 An odd T-period alternating-offer bargaining game.

is reached, is not extremely appealing. For simplicity, we will first analyze the game with a finite end date; later we will extend the analysis to eliminate this artificial termination stage.

11.1 One Round of Bargaining: The Ultimatum Game

We begin the analysis with an easy case in which there is only one round of bargaining, so that $T = 1$. This is often referred to in the literature as an **ultimatum game:** player 1 makes an ultimatum take-it-or-leave-it offer to player 2, and player 2 either accepts it, in which case the pie is split between the players, or rejects it, in which case both players receive a payoff of zero. Because this is a game of perfect information (player 2 sees the offer of player 1 before he needs to accept or reject it), we can apply backward induction to find which paths of play can be supported by a subgame-perfect equilibrium.

As a benchmark, we begin the analysis by finding which paths of play can be supported by a Nash equilibrium without requiring sequential rationality. The result is quite striking:

Proposition 11.1 *In the bargaining game if* $T = 1$ *then* any division of surplus $x^* \in [0, 1]$, $(v_1, v_2) = (x^*, 1 - x^*)$, *can be supported as a Nash equilibrium.*

Proof We will construct a pair of strategies that are mutual best responses and that lead to $(x^*, 1 - x^*)$ as the division of surplus. Let player 1's strategy be "I propose x^*," and let player 2's strategy be "I accept any offer $x \leq x^*$ and reject any offer $x > x^*$." It is easy to see that these two strategies are mutual best responses independent of the value of $x^* \in [0, 1]$. ■

This proposition tells us that the concept of Nash equilibrium has no bite for this simple one-stage bargaining game because it can rationalize *any division of surplus.* In other words, we cannot predict in any meaningful way what the outcome of such a game will be if we require only that people choose strategies that are mutual best responses.

That said, a close observation of the strategies constructed to support an arbitrary division of surplus will immediately reveal that sequential rationality is almost always violated: Player 2 is saying "there is a minimum that I am willing to accept." But what if player 1 offers him less? In particular what if he offers him $(1 - x) = \varepsilon > 0$ with ε being a very small number? If player 2 rejects this offer he will get 0, while if he accepts it he will get a payoff of $\varepsilon > 0$, implying that his best response is to accept *any strictly positive payoff.* Anticipating this, player 1 should offer player 2 the smallest possible amount. This logic leads to the following result:

Proposition 11.2 *The bargaining game with* $T = 1$ *admits a unique subgame-perfect equilibrium in which player 1 offers* $x = 1$ *and player 2 accepts any offer* $x \leq 1$.

Proof We have established that player 2 must accept any positive share ($x < 1$). Player 2 is indifferent between accepting or rejecting $x = 1$, so the proposed strategy is sequentially optimal, and the unique best response of player 1 to player 2's strategy is to offer $x = 1$. The only other sequentially rational strategy for player 2 is to accept any strictly positive share ($x < 1$) and reject getting 0 ($x = 1$). But player 1 does not have a best response to this strategy,[1] and therefore it cannot be part of a subgame-perfect equilibrium. ■

This result is very stark, especially when compared with the analysis of Nash equilibrium. Requiring only mutual best responses yields no meaningful prediction, but sequential rationality predicts a unique and extreme outcome: player 2 should accept anything, and thus player 1 has an extreme form of take-it-or-leave-it first-mover advantage, giving him the whole pie of surplus.

Remark Many experiments have been performed by researchers to see what behavior actually prevails for the ultimatum game, in which player 1 can offer to split a sum of money with player 2. If player 2 accepts then the split is realized and both players receive the proposal of player 1. If player 2 rejects then the players receive

1. This is similar to what happens in the asymmetric-cost Bertrand competition model presented in Section 5.2.4, in which player 1's payoff function is discontinuous, and hence he does not have a well-defined best response. To see it in this context, imagine that player 2's strategy is to accept any offer $x < 1$ and reject $x = 1$. Clearly $x = 1$ is not a best response because any offer $x < 1$ is better. The problem is that no offer $x < 1$ is a best response either. If player 1 offers some $x' < 1$, then the offer $x = x' + \frac{1-x'}{2}$ is better than x'. The conclusion is that player 1's payoff function is discontinuous at $x = 1$, and he does not have a best response to player 2's strategy.

nothing. Contrary to the theory, and maybe not too surprisingly, the experiments show that those players in the role of player 1 offer significant shares to those in the role of player 2, typically between 25% and 50% of the total value. Furthermore, when player 2 is offered a small amount, typically less than 20–25% of the total value, then he will reject these offers. This suggests that subjects in the role of player 2 act with some "spite" when they are offered low shares because they prefer getting nothing to an outcome in which player 1 gets a large fraction of the total value. See Camerer (2003) for an excellent summary of the experimental results and some theories that predict this behavior.

11.2 Finitely Many Rounds of Bargaining

When we consider bargaining games that extend beyond one period but that will still end at some specified date $T < \infty$, the Nash equilibrium concept continues to have no predictive power. In particular we can easily construct the kind of strategies used in the proof of proposition 11.1 to *any* horizon, *including an infinite horizon.* To construct a Nash equilibrium that results in an arbitrary division of surplus $x^* \in [0, 1]$ we can use the following time-independent strategies: In every odd round in which player 1 proposes, player 1 will propose x^* and player 2 will accept proposals of $x \leq x^*$. In every even round in which player 2 proposes, player 2 will propose x^* and player 1 will accept proposals of $x \geq x^*$. It is easy to see that these strategies are mutual best responses, and an agreement is reached in the first round with the division of surplus $(x^*, 1 - x^*)$.

What is even more disappointing is that, using the Nash concept, we cannot even pin down when an agreement will be reached. For example, we can adopt the strategies that result in any $(x^*, 1 - x^*)$ with the following modification: in the first period player 1 offers $x = 1$ and player 2 rejects *any* offer. It is easy to see that these form a pair of best responses, resulting in a Nash equilibrium in which an agreement is reached in the second period. In exercise 11.1, you are asked to show that similar strategies can be constructed to support an agreement of *any* division of surplus in *any* period.

Imposing sequential rationality will, once again, offer a stark contrast. Indeed it should not be surprising that applying sequential rationality and subgame perfection to any finite-length bargaining game will result in a unique outcome, because backward induction will apply in a similar way that it did for the case of one round of bargaining. In the last round the proposer will offer to keep the entire pie to himself, the responder will agree, and the solution will proceed sequentially using backward induction.

In the one-round ultimatum game analyzed previously, player 1 got all the surplus in the unique subgame-perfect equilibrium because by construction the only round is the last round. Sequential rationality implies that in the last round the player making the offer has all the "bargaining power," which in turn implies that he will get the whole surplus. We can ask then, what will happen in a game with two rounds of bargaining?

As Figure 11.2 demonstrates, the second round is a one-round bargaining game, which implies that *if it is reached* then player 2 gets the whole pie in any subgame-perfect equilibrium. Continuing with backward induction, and assuming that $\delta \leq 1$, we can find the subgame-perfect equilibrium for the two-round game as follows. In the first round player 2 knows that in the next (and last) round he will receive the whole pie, which will be worth only δ because of discounting. Thus if he is offered less than δ for himself (player 1 offers $x > 1 - \delta$) then player 2 should reject the offer and get the whole pie in the next period. Hence the unique sequentially rational strategy for

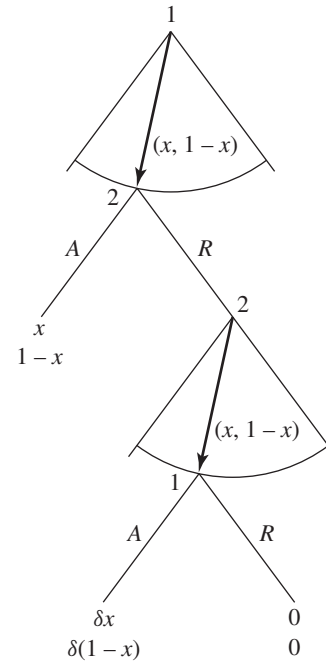


FIGURE 11.2 An alternating-offer bargaining game with two periods.

player 2 at the accept/reject stage of the first round is to accept any offer of $x \leq 1 - \delta$. Moving up the tree to the node at which player 1 makes his first offer, he realizes that any offer $x > 1 - \delta$ will be rejected, while any offer $x \leq 1 - \delta$ will be accepted. Thus player 1's best response is to offer $x = 1 - \delta$, leading to a division of surplus equal to $(v_1, v_2) = (1 - \delta, \delta)$.

Note that if $\delta = 1$ then player 2 gets the whole pie because he then has the ultimate form of a take-it-or-leave-it advantage, and with $\delta < 1$ player 1 has a bit of a first-mover advantage owing to the discounting. This observation is useful because it suggests an interesting pair of forces. Essentially the backward induction logic that we have used for the two-round bargaining model will generalize to any finite-round game. Therefore a tension will emerge between the ability to make a last take-it-or-leave-it offer and the first-mover advantage that is a result of discounting given $\delta < 1$. Whether player 1 or player 2 is the last mover will artificially depend on whether the game has an odd or even number of periods.

Consider the case with an odd number of rounds $T < \infty$, implying that player 1 has both the first-mover and last-mover advantages. The following backward induction argument applies:

- In period T player 2 accepts any offer, so player 1 offers $x = 1$ and payoffs are $v_1 = \delta^{T-1}$ and $v_2 = 0$.
- In period $T - 1$ (even period, player 2 offers), by backward induction player 1 should accept anything resulting in a payoff of $v_1 \geq \delta^{T-1}$. If player 1 is offered x in period $T - 1$ then $v_1 = \delta^{T-2}x$. This implies that in period $T - 1$ player 1 will accept any $x \geq \delta$ and by backward induction player 2 should offer $x = \delta$, which yields player 1 a payoff of $v_1 = \delta \times \delta^{T-2} = \delta^{T-1}$. Payoffs are then $v_1 = \delta^{T-1}$ and $v_2 = \delta^{T-2}(1 - \delta)$.

- In period $T-2$ (odd period), conditional on the analysis for $T-1$, player 2's best response is to accept any x that gives him $\delta^{T-3}(1-x) \geq \delta^{T-2}(1-\delta)$. Player 1's best response to this is to offer the largest x that satisfies this inequality, and solving it with equality yields player 1's best response: $x = 1-\delta+\delta^2$. This offer followed by player 2's acceptance yields $v_1 = \delta^{T-3}x = \delta^{T-3} - \delta^{T-2} + \delta^{T-1}$ and $v_2 = \delta^{T-2} - \delta^{T-1}$.
- In period $T-3$ (even period), using the same backward induction argument, player 1's best response is to accept any x that satisfies $\delta^{T-4}x \geq \delta^{T-3} - \delta^{T-2} + \delta^{T-1}$, and player 2 then makes an offer that satisfies this constraint with equality, $x = \delta - \delta^2 + \delta^3$, yielding $v_1 = \delta^{T-3} - \delta^{T-2} + \delta^{T-1}$ and $v_2 = \delta^{T-4}(1-x) = \delta^{T-4} - \delta^{T-3} + \delta^{T-2} - \delta^{T-1}$.

We can continue with this tedious exercise only to realize that a simple pattern emerges. If we consider the solution for an odd period $T-s$ (s being even because T is assumed to be odd) then the backward induction argument leads to the sequentially rational offer

$$x_{T-s} = 1 - \delta + \delta^2 - \delta^3 + \cdots + \delta^s,$$

while for an even period $T-s$ (s being odd) then the backward induction argument leads to the sequentially rational offer

$$x_{T-s} = \delta - \delta^2 + \delta^3 - \delta^4 + \cdots + \delta^s.$$

We can use this pattern to solve for the subgame-perfect equilibrium offer in the first period, x_1, which by backward induction must be accepted by player 2 and it is equal to

$$\begin{aligned} x_1 &= 1 - \delta + \delta^2 - \delta^3 + \delta^4 + \cdots + \delta^{T-1} \\ &= (1 + \delta^2 + \delta^4 + \cdots + \delta^{T-1}) - (\delta + \delta^3 + \delta^5 + \cdots + \delta^{T-2}) \\ &= \frac{1-\delta^{T+1}}{1-\delta^2} - \frac{\delta - \delta^T}{1-\delta^2} \\ &= \frac{1+\delta^T}{1+\delta}, \end{aligned}$$

and this in turn implies that

$$v_1^* = x_1 = \frac{1+\delta^T}{1+\delta} \qquad \text{and} \qquad v_2^* = (1-x_1) = \frac{\delta - \delta^T}{1+\delta}. \tag{11.1}$$

We can now offer some insights into this solution, which has some very appealing properties. The first observation is important:

Proposition 11.3 *Any subgame-perfect equilibrium must have the players reach an agreement in the first round.*

The proof follows a simple logic. If an agreement is reached in a later round with payoffs (v_1', v_2') then discounting implies that part of the surplus is wasted and that $v_1' + v_2' < 1$. But then player 1 could deviate and offer $x = 1 - v_2' - \varepsilon$ for some small $\varepsilon > 0$, which guarantees player 2 the payoff $v_2' + \varepsilon$ immediately in the first round. Sequential rationality implies that player 2 should accept this immediately, and for ε small enough this gives player 1 a payoff greater than v_1'.

A second observation follows from the fact that player 1 has both the last-mover take-it-or-leave-it advantage and the first-mover discounting advantage. This implies that $v_1^* > v_2^*$ for any discount factor $\delta \in [0, 1]$, as can be seen from the equilibrium values in (11.1). Because T is fixed, however, it is hard to isolate these two effects, especially if we wish to measure the way in which patience, represented by δ, affects the equilibrium payoffs. For example, if players are very impatient and $\delta = 0$ then the first-mover advantage kicks in with full force. Regardless of T, this is basically equivalent to a one-round game in which player 1 gets the whole surplus. On the other hand, if we consider the limit of $\delta = 1$ then $v_1^* = 1$ and $v_2^* = 0$, because no discounting implies that moving last guarantees the whole surplus, regardless of the length of the game T.

This last observation is somewhat disturbing: if players are very patient then the last mover's take-it-or-leave-it advantage just flows up the game tree no matter how long the game is! This rather counterintuitive result is precisely due to the artificial stopping period T. If $\delta = 1$ then the only thing that matters is who has the last word—there is no loss from waiting, and hence the forward-looking players know that the last person to make an offer holds all the bargaining power.

To overcome this artificially dictated outcome, we will first address the source of the problem by fixing the discount factor δ and observing what happens to the equilibrium payoffs as the game gets longer and longer. More precisely,

$$\lim_{T\to\infty} v_1^* = \lim_{T\to\infty} \frac{1+\delta^T}{1+\delta} = \frac{1}{1+\delta} \qquad \text{and} \qquad \lim_{T\to\infty} v_2^* = \lim_{T\to\infty} \frac{\delta-\delta^T}{1+\delta} = \frac{\delta}{1+\delta}. \tag{11.2}$$

As we can see from the limit payoffs in (11.2), the artificial end-period effect disappears. We can clearly observe the first-mover discounting advantage in that for any $0 < \delta < 1$ it must be that $\lim_{T\to\infty} v_1^* > \lim_{T\to\infty} v_2^*$. We can also use the limiting equilibrium payoffs to ask what happens as the patience of the players changes without the problem posed by an artificial end period T. For the extreme case of impatience with $\delta = 0$ we get the same result: the game is equivalent to a one-round ultimatum game. When the players become very patient, we now obtain a more sensible result,

$$\lim_{\delta\to 1}\lim_{T\to\infty} v_1^* = \lim_{\delta\to 1} \frac{1}{1+\delta} = \frac{1}{2} \qquad \text{and} \qquad \lim_{\delta\to 1}\lim_{T\to\infty} v_2^* = \lim_{\delta\to 1} \frac{\delta}{1+\delta} = \frac{1}{2}. \tag{11.3}$$

The limit payoffs in (11.3) suggest a rather appealing and intuitive result for very long horizons with very patient players: the very long horizon eliminates the last mover's take-it-or-leave-it advantage for any given discount factor δ, leaving only the first-mover advantage due to discounting. If we take the level of patience to its limit of $\delta = 1$ then this eliminates the first-mover advantage as well, and we have the two players splitting the pie equally in the unique subgame-perfect equilibrium.[2]

As exercise 11.3, you can solve for the subgame-perfect equilibrium when the number of periods is even, and see that at the limit we get the same results.

2. Note that to reach this appealing conclusion the order of limits matters. If we reverse the order of limits and take $\delta \to 1$ first, followed by $T \to \infty$ second, then $\lim v_1^* = 1$ and $\lim v_2^* = 0$, preserving the artificial take-it-or-leave-it advantage.

11.3 The Infinite-Horizon Game

We now analyze the infinite-horizon bargaining game in which, if an agreement is never reached, the players continue to alternate roles indefinitely. In this case the assumption that disagreement leads to zero payoffs is natural because of discounting: if the two players disagree forever then there is nothing left on which to agree. However, there is a crucial difference between analyzing the infinite-horizon game and considering the infinite limit of the finite-horizon game: the path of perpetual disagreement is of infinite length, and we cannot therefore apply a backward induction argument to find the subgame-perfect equilibrium.

There is, however, an interesting feature of the infinite-horizon bargaining game: the *stationary structure of the game* following a disagreement at any stage. Every odd period is the same, with player 1 making an offer, and the continuation game has a potentially infinite horizon. Similarly every even period is the same, with player 2 making the offer. This allows us to apply a rather appealing, and not too difficult, logic to solve for what turns out to be the unique subgame-perfect equilibrium of the infinite-horizon bargaining game.[3]

The first important observation is that, following the same logic we introduced earlier in proposition 11.3, sequential rationality implies that an agreement must be reached in the first period. The reason is that wasteful discounting will not be tolerated because it can be avoided in the first stage. Hence any subgame-perfect equilibrium must have agreement reached in the first period.

Now imagine that there is more than one subgame-perfect equilibrium. If this were the case then when player 1 makes an offer he must have a *best* subgame-perfect equilibrium, yielding him a value of $\overline{v}_1$, and a *worst* subgame-perfect equilibrium, yielding him a value of $\underline{v}_1$. Similarly when player 2 makes an offer after rejecting an offer made by player 1 then he too has a *best* subgame-perfect equilibrium, yielding him a value of $\overline{v}_2$, and a *worst* subgame-perfect equilibrium, yielding him a value of $\underline{v}_2$.

This is where the stationary structure of the game is useful, because it implies that $\overline{v}_1 = \overline{v}_2 = \overline{v}$ and $\underline{v}_1 = \underline{v}_2 = \underline{v}$. The reason is that in any odd period the game looks exactly the same as in any even period, just with the identities of the players switched. Hence if the structure of the game results in the existence of different subgame-perfect equilibria that support certain payoffs for player 1 starting in period 1, the same must be true for player 2 in period 2.

Now we take advantage of the fact that what one player gives up, the other player must receive (and recall that agreement must be reached in the first period), which results in a straightforward relationship between the best- and worst-payoff subgame-perfect equilibria of the game. What is worst for player 1 must be best for player 2 and vice versa. Hence in the subgame-perfect equilibrium that supports the payoff $\underline{v}$ for player 1 in period 1, it must be true that if player 2 rejects player 1's offer, he can secure himself a payoff of $\overline{v}$ when it is his turn to make the offer.

Immediate agreement implies that player 1 must offer player 2 a share of the pie that will deter player 2 from rejecting player 1's offer. If player 2 can get $\overline{v}$ after rejection, this implies that player 1 must offer player 2 a payoff of at least $\delta\overline{v}$ in the first period. Of course player 1 will offer the least he can to secure an agreement,

3. The method used here was developed by Shaked and Sutton (1984), but the infinite model was proposed, and the unique solution was identified, in the seminal paper by Rubinstein (1982).

which implies that he will offer player 2 a share exactly equal to $\delta\overline{v}$, which in turn implies that player 1's payoff from this subgame-perfect equilibrium must be

$$\underline{v} = 1 - \delta\overline{v}. \tag{11.4}$$

A symmetric argument applies for the subgame-perfect equilibrium that results in player 1 obtaining a payoff of $\overline{v}$ and player 2 obtaining a payoff of $\underline{v}$, which results in the equation

$$\overline{v} = 1 - \delta\underline{v}. \tag{11.5}$$

Taking (11.4) and (11.5) we obtain that

$$\overline{v} = \underline{v} = \frac{1}{1+\delta}.$$

This result characterizes the unique subgame-perfect equilibrium of the infinite-horizon bargaining game. Notice that it coincides with the limit of the unique subgame-perfect equilibrium of the finite-horizon bargaining game as shown in Section 11.2. Furthermore because we know that agreement is reached in the first round we can use this payoff characterization to spell out the strategies that together support this outcome. In the unique subgame-perfect equilibrium the strategies are as follows: In each odd round player 1 offers to keep $x = \frac{1}{1+\delta}$ to himself and player 2 accepts the offer if $x \leq \frac{1}{1+\delta}$. In each even round player 2 offers to give player 1 $x = \frac{\delta}{1+\delta}$ and player 1 accepts any $x \geq \frac{\delta}{1+\delta}$.

Remark As you might have imagined, bargaining games have received considerable attention in the literature, in both economics and political science, because bargaining situations are so ubiquitous. Many variants of the bargaining game described here have been analyzed, in which the relationship might break down, the players each have an "outside option" that is greater than zero, there are more than two players that can be matched randomly, . . . the list goes on and on. Nash (1950b) offered an analysis of the bargaining problem, but interestingly he did not follow the framework of noncooperative game theory that he himself had founded. Instead he offered an axiomatic approach to suggest a sensible solution to the two-person bargaining problem. The resulting "Nash bargaining solution" is widely used in economics. For a broad treatment of bargaining games and solutions see Osborne and Rubinstein (1990) and Muthoo (1999).

11.4 Application: Legislative Bargaining

The bargaining model analyzed here offers a reasonable depiction of many two-person bargaining situations. In particular there is the potential for give-and-take, and at the end an agreement has to be reached by both parties for the deal to be sealed. Another setting in which bargaining is ubiquitous is in legislatures, where lawmakers bargain over the allocation of surplus through either bills, budget agreements, or regulations. In most political settings the convention is that once a majority of players concur on an offer, the allocation is agreed upon despite the dismay of those who may still oppose it.

In an important contribution, Baron and Ferejohn (1989) build on the Rubinstein-Ståhl model of bargaining and offer a framework for analyzing multilateral bargaining

- The bargaining protocol and agreement rules will have a significant impact on how the surplus is shared. In finite games with prespecified alternating moves, the last player to move has a last-mover strategic advantage, while the first player has an advantage owing to discounting.
- When the game is not finite or when the role of proposer is determined randomly with equal probability, there is no last-mover advantage and only the first-mover advantage remains.

11.6 Exercises

11.1 **Disagreement:** Construct a pair of strategies for the ultimatum game ($T = 1$ bargaining game) that constitute a Nash equilibrium and together support the outcome that there is no agreement reached by the two players and the payoffs are zero to each. Show that this disagreement outcome can be supported by a Nash equilibrium regardless of the number of bargaining periods.

11.2 **Holdup:** Consider an ultimatum game ($T = 1$ bargaining game) in which before player 1 makes his offer to player 2, player 2 can invest in the size of the pie. If player 2 chooses a low level of investment (L) then the size of the pie is small, equal to v_L, while if player 2 chooses a high level of investment (H) then the size of the pie is large, equal to v_H. The cost to player 2 of choosing L is c_L, while the cost of choosing H is c_H. Assume that $v_H > v_L > 0$, $c_H > c_L > 0$, and $v_H - c_H > v_L - c_L$.

a. What is the unique subgame-perfect equilibrium of this game? Is it Pareto optimal?
b. Can you find a Nash equilibrium of the game that results in an outcome that is better for both players as compared to the unique subgame-perfect equilibrium?

11.3 **Even/Odd Symmetry:** In Section 11.2 we analyzed the alternating-offer bargaining game for a finite number of periods when T was odd. Repeat the analysis for T even.

11.4 **Constant Delay Cost:** Consider a two-player alternating-offer bargaining game in which instead of the pie shrinking by a discount factor $\delta < 1$, the players each pay a cost $c_i > 0$, $i \in \{1, 2\}$, to advance from one period to the next. So if player i receives a share of the pie that gives him a value of x_i in period t then his payoff is $v_i = x_i - (t - 1)c_i$. If the game has T periods then a sequence of rejections results in each player receiving $v_i = -(T - 1)c_i$.

a. Assume that $T = 2$. Find the subgame-perfect equilibrium of the game and show in which way it depends on the values of c_1 and c_2.
b. Are there Nash equilibria in the two-period game that are not subgame perfect?
c. Assume that $T = 3$. Find the subgame-perfect equilibrium of the game and show in which way it depends on the values of c_1 and c_2.

11.5 **Asymmetric Patience 1:** Consider a three-period sequential (alternating-offer) bargaining model in which two players have to split a pie worth 1 (starting with player 1 making the offer). Now the players have different discount factors, δ_1 and δ_2.

a. Compute the outcome of the unique subgame-perfect equilibrium.

b. Show that when $\delta_1 = \delta_2$ player 1 has an advantage.
c. What conditions on δ_1 and δ_2 give player 2 an advantage? Why?

11.6 **Asymmetric Patience 2:** Consider the analysis of the infinite-horizon bargaining model in Section 11.3 and assume that the players have different discount factors δ_1 and δ_2. Find the unique subgame-perfect equilibrium using the same techniques, and show that as δ_1 and δ_2 become closer in value, the solution you found converges to the solution derived in Section 11.3.

11.7 **Legislative Bargaining Revisited:** Consider a finite T-period version of the Baron and Ferejohn legislative bargaining game with an odd number N of players and with a closed rule as described in Section 11.4.1.

a. Find the unique subgame-perfect equilibrium for $T = 1$. Also find a Nash equilibrium that is not subgame perfect.
b. Find the unique subgame-perfect equilibrium for $T = 2$ with a discount factor $0 < \delta \leq 1$. Also find a Nash equilibrium that is not subgame perfect.
c. Compare what the first period's proposer receives in the subgame-perfect equilibrium you found in (b) to what a first-period proposer receives in the two-period, two-person Rubinstein-Ståhl bargaining game. What intuitively accounts for the difference?
d. Compare the subgame-perfect equilibrium you found in (b) to the solution of the infinite-horizon model in Section 11.4.1. What intuitively accounts for the similarity?

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- Normal form; IESDS (block 1): §1.1.A–B (p. 2–7)
- Cournot duopoly (block 2): §1.2.A (p. 14–21)

happen"). We then motivate and define *Nash equilibrium*—a solution concept that produces much tighter predictions in a very broad class of games.

In Section 1.2 we analyze four applications, using the tools developed in the previous section: Cournot's (1838) model of imperfect competition, Bertrand's (1883) model of imperfect competition, Farber's (1980) model of final-offer arbitration, and the problem of the commons (discussed by Hume [1739] and others). In each application we first translate an informal statement of the problem into a normal-form representation of the game and then solve for the game's Nash equilibrium. (Each of these applications has a unique Nash equilibrium, but we discuss examples in which this is not true.)

In Section 1.3 we return to theory. We first define the notion of a *mixed strategy*, which we will interpret in terms of one player's uncertainty about what another player will do. We then state and discuss Nash's (1950) Theorem, which guarantees that a Nash equilibrium (possibly involving mixed strategies) exists in a broad class of games. Since we present first basic theory in Section 1.1, then applications in Section 1.2, and finally more theory in Section 1.3, it should be apparent that mastering the additional theory in Section 1.3 is not a prerequisite for understanding the applications in Section 1.2. On the other hand, the ideas of a mixed strategy and the existence of equilibrium do appear (occasionally) in later chapters.

This and each subsequent chapter concludes with problems, suggestions for further reading, and references.

1.1 Basic Theory: Normal-Form Games and Nash Equilibrium

1.1.A Normal-Form Representation of Games

In the normal-form representation of a game, each player simultaneously chooses a strategy, and the combination of strategies chosen by the players determines a payoff for each player. We illustrate the normal-form representation with a classic example — *The Prisoners' Dilemma.* Two suspects are arrested and charged with a crime. The police lack sufficient evidence to convict the suspects, unless at least one confesses. The police hold the suspects in

separate cells and explain the consequences that will follow from the actions they could take. If neither confesses then both will be convicted of a minor offense and sentenced to one month in jail. If both confess then both will be sentenced to jail for six months. Finally, if one confesses but the other does not, then the confessor will be released immediately but the other will be sentenced to nine months in jail—six for the crime and a further three for obstructing justice.

The prisoners' problem can be represented in the accompanying bi-matrix. (Like a matrix, a bi-matrix can have an arbitrary number or rows and columns; "bi" refers to the fact that, in a two-player game, there are two numbers in each cell—the payoffs to the two players.)

		Prisoner 2	
		Mum	Fink
Prisoner 1	Mum	$-1, -1$	$-9, \ 0$
	Fink	$0, -9$	$-6, -6$

The Prisoners' Dilemma

In this game, each player has two strategies available: confess (or fink) and not confess (or be mum). The payoffs to the two players when a particular pair of strategies is chosen are given in the appropriate cell of the bi-matrix. By convention, the payoff to the so-called row player (here, Prisoner 1) is the first payoff given, followed by the payoff to the column player (here, Prisoner 2). Thus, if Prisoner 1 chooses Mum and Prisoner 2 chooses Fink, for example, then Prisoner 1 receives the payoff -9 (representing nine months in jail) and Prisoner 2 receives the payoff 0 (representing immediate release).

We now turn to the general case. The *normal-form representation* of a game specifies: (1) the players in the game, (2) the strategies available to each player, and (3) the payoff received by each player for each combination of strategies that could be chosen by the players. We will often discuss an n-player game in which the players are numbered from 1 to n and an arbitrary player is called player i. Let S_i denote the set of strategies available to player i (called i's *strategy space*), and let s_i denote an arbitrary member of this set. (We will occasionally write $s_i \in S_i$ to indicate that the

strategy s_i is a member of the set of strategies S_i.) Let $(s_1, \ldots, s_n)$ denote a combination of strategies, one for each player, and let u_i denote player i's *payoff function*: $u_i(s_1, \ldots, s_n)$ is the payoff to player i if the players choose the strategies $(s_1, \ldots, s_n)$. Collecting all of this information together, we have:

Definition *The* ***normal-form representation*** *of an n-player game specifies the players' strategy spaces $S_1, \ldots, S_n$ and their payoff functions $u_1, \ldots, u_n$. We denote this game by $G = \{S_1, \ldots, S_n; u_1, \ldots, u_n\}$.*

Although we stated that in a normal-form game the players choose their strategies simultaneously, this does not imply that the parties necessarily *act* simultaneously: it suffices that each choose his or her action without knowledge of the others' choices, as would be the case here if the prisoners reached decisions at arbitrary times while in their separate cells. Furthermore, although in this chapter we use normal-form games to represent only static games in which the players all move without knowing the other players' choices, we will see in Chapter 2 that normal-form representations can be given for sequential-move games, but also that an alternative—the *extensive-form* representation of the game—is often a more convenient framework for analyzing dynamic issues.

1.1.B Iterated Elimination of Strictly Dominated Strategies

Having described one way to represent a game, we now take a first pass at describing how to solve a game-theoretic problem. We start with the Prisoners' Dilemma because it is easy to solve, using only the idea that a rational player will not play a strictly dominated strategy.

In the Prisoners' Dilemma, if one suspect is going to play Fink, then the other would prefer to play Fink and so be in jail for six months rather than play Mum and so be in jail for nine months. Similarly, if one suspect is going to play Mum, then the other would prefer to play Fink and so be released immediately rather than play Mum and so be in jail for one month. Thus, for prisoner i, playing Mum is dominated by playing Fink—for each strategy that prisoner j could choose, the payoff to prisoner i from playing Mum is less than the payoff to i from playing Fink. (The same would be true in any bi-matrix in which the payoffs 0, -1, -6,

and −9 above were replaced with payoffs T, R, P, and S, respectively, provided that $T > R > P > S$ so as to capture the ideas of temptation, reward, punishment, and sucker payoffs.) More generally:

Definition *In the normal-form game* $G = \{S_1, \ldots, S_n; u_1, \ldots, u_n\}$, *let* s_i' *and* s_i'' *be feasible strategies for player* i *(i.e.,* s_i' *and* s_i'' *are members of* S_i*). Strategy* s_i' *is* ***strictly dominated*** *by strategy* s_i'' *if for each feasible combination of the other players' strategies, i's payoff from playing* s_i' *is strictly less than i's payoff from playing* s_i''*:*

$$u_i(s_1, \ldots, s_{i-1}, s_i', s_{i+1}, \ldots, s_n) < u_i(s_1, \ldots, s_{i-1}, s_i'', s_{i+1}, \ldots, s_n) \quad \text{(DS)}$$

for each $(s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n)$ *that can be constructed from the other players' strategy spaces* $S_1, \ldots, S_{i-1}, S_{i+1}, \ldots, S_n$.

Rational players do not play strictly dominated strategies, because there is no belief that a player could hold (about the strategies the other players will choose) such that it would be optimal to play such a strategy.[1] Thus, in the Prisoners' Dilemma, a rational player will choose Fink, so (Fink, Fink) will be the outcome reached by two rational players, even though (Fink, Fink) results in worse payoffs for both players than would (Mum, Mum). Because the Prisoners' Dilemma has many applications (including the arms race and the free-rider problem in the provision of public goods), we will return to variants of the game in Chapters 2 and 4. For now, we focus instead on whether the idea that rational players do not play strictly dominated strategies can lead to the solution of other games.

Consider the abstract game in Figure 1.1.1.[2] Player 1 has two strategies and player 2 has three: $S_1 = \{\text{Up}, \text{Down}\}$ and $S_2 = \{\text{Left}, \text{Middle}, \text{Right}\}$. For player 1, neither Up nor Down is strictly

[1] A complementary question is also of interest: if there is no belief that player i could hold (about the strategies the other players will choose) such that it would be optimal to play the strategy s_i, can we conclude that there must be another strategy that strictly dominates s_i? The answer is "yes," *provided* that we adopt appropriate definitions of "belief" and "another strategy," both of which involve the idea of mixed strategies to be introduced in Section 1.3.A.

[2] Most of this book considers economic applications rather than abstract examples, both because the applications are of interest in their own right and because, for many readers, the applications are often a useful way to explain the underlying theory. When introducing some of the basic theoretical ideas, however, we will sometimes resort to abstract examples that have no natural economic interpretation.

		Player 2		
		Left	Middle	Right
Player 1	Up	1,0	1,2	0,1
	Down	0,3	0,1	2,0

Figure 1.1.1.

dominated: Up is better than Down if 2 plays Left (because $1 > 0$), but Down is better than Up if 2 plays Right (because $2 > 0$). For player 2, however, Right is strictly dominated by Middle (because $2 > 1$ and $1 > 0$), so a rational player 2 will not play Right. Thus, if player 1 knows that player 2 is rational then player 1 can eliminate Right from player 2's strategy space. That is, if player 1 knows that player 2 is rational then player 1 can play the game in Figure 1.1.1 *as if* it were the game in Figure 1.1.2.

		Player 2	
		Left	Middle
Player 1	Up	1,0	1,2
	Down	0,3	0,1

Figure 1.1.2.

In Figure 1.1.2, Down is now strictly dominated by Up for player 1, so if player 1 is rational (and player 1 knows that player 2 is rational, so that the game in Figure 1.1.2 applies) then player 1 will not play Down. Thus, if player 2 knows that player 1 is rational, *and* player 2 knows that player 1 knows that player 2 is rational (so that player 2 knows that Figure 1.1.2 applies), then player 2 can eliminate Down from player 1's strategy space, leaving the game in Figure 1.1.3. But now Left is strictly dominated by Middle for player 2, leaving (Up, Middle) as the outcome of the game.

This process is called *iterated elimination of strictly dominated strategies*. Although it is based on the appealing idea that rational players do not play strictly dominated strategies, the process has two drawbacks. First, each step requires a further assumption

		Player 2	
		Left	Middle
Player 1	Up	1,0	1,2

Figure 1.1.3.

about what the players know about each other's rationality. If we want to be able to apply the process for an arbitrary number of steps, we need to assume that it is *common knowledge* that the players are rational. That is, we need to assume not only that all the players are rational, but also that all the players know that all the players are rational, and that all the players know that all the players know that all the players are rational, and so on, *ad infinitum*. (See Aumann [1976] for the formal definition of common knowledge.)

The second drawback of iterated elimination of strictly dominated strategies is that the process often produces a very imprecise prediction about the play of the game. Consider the game in Figure 1.1.4, for example. In this game there are no strictly dominated strategies to be eliminated. (Since we have not motivated this game in the slightest, it may appear arbitrary, or even pathological. See the case of three or more firms in the Cournot model in Section 1.2.A for an economic application in the same spirit.) Since all the strategies in the game survive iterated elimination of strictly dominated strategies, the process produces no prediction whatsoever about the play of the game.

	L	*C*	*R*
T	0,4	4,0	5,3
M	4,0	0,4	5,3
B	3,5	3,5	6,6

Figure 1.1.4.

We turn next to Nash equilibrium—a solution concept that produces much tighter predictions in a very broad class of games. We show that Nash equilibrium is a stronger solution concept

statements of these two observations are: there exists s_i in S_i such that

$$u_i(s_1^*, \ldots, s_{i-1}^*, s_i^*, s_{i+1}^*, \ldots, s_n^*) < u_i(s_1^*, \ldots, s_{i-1}^*, s_i, s_{i+1}^*, \ldots, s_n^*); \tag{1.1.3}$$

and there exists s_i' in the set of player i's strategies remaining at some stage of the process such that

$$u_i(s_1, \ldots, s_{i-1}, s_i, s_{i+1}, \ldots, s_n) < u_i(s_1, \ldots, s_{i-1}, s_i', s_{i+1}, \ldots, s_n) \tag{1.1.4}$$

for each $(s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n)$ that can be constructed from the strategies remaining in the other players' strategy spaces at that stage of the process. Since the other players' strategies $(s_1^*, \ldots, s_{i-1}^*, s_{i+1}^*, \ldots, s_n^*)$ are never eliminated, one of the implications of (1.1.4) is

$$u_i(s_1^*, \ldots, s_{i-1}^*, s_i, s_{i+1}^*, \ldots, s_n^*) < u_i(s_1^*, \ldots, s_{i-1}^*, s_i', s_{i+1}^*, \ldots, s_n^*). \tag{1.1.5}$$

If $s_i' = s_i^*$ (that is, if s_i^* is the strategy that strictly dominates s_i) then (1.1.5) contradicts (1.1.3), in which case the proof is complete. If $s_i' \neq s_i^*$ then some other strategy s_i'' must later strictly dominate s_i', since s_i' does not survive the process. Thus, inequalities analogous to (1.1.4) and (1.1.5) hold with s_i' and s_i'' replacing s_i and s_i', respectively. Once again, if $s_i'' = s_i^*$ then the proof is complete; otherwise, two more analogous inequalities can be constructed. Since s_i^* is the only strategy from S_i to survive the process, repeating this argument (in a finite game) eventually completes the proof.

1.2 Applications

1.2.A Cournot Model of Duopoly

As noted in the previous section, Cournot (1838) anticipated Nash's definition of equilibrium by over a century (but only in the context of a particular model of duopoly). Not surprisingly, Cournot's work is one of the classics of game theory; it is also one of the cornerstones of the theory of industrial organization. We consider a

very simple version of Cournot's model here, and return to variations on the model in each subsequent chapter. In this section we use the model to illustrate: (a) the translation of an informal statement of a problem into a normal-form representation of a game; (b) the computations involved in solving for the game's Nash equilibrium; and (c) iterated elimination of strictly dominated strategies.

Let q_1 and q_2 denote the quantities (of a homogeneous product) produced by firms 1 and 2, respectively. Let $P(Q) = a - Q$ be the market-clearing price when the aggregate quantity on the market is $Q = q_1 + q_2$. (More precisely, $P(Q) = a - Q$ for $Q < a$, and $P(Q) = 0$ for $Q \geq a$.) Assume that the total cost to firm i of producing quantity q_i is $C_i(q_i) = cq_i$. That is, there are no fixed costs and the marginal cost is constant at c, where we assume $c < a$. Following Cournot, suppose that the firms choose their quantities simultaneously.[6]

In order to find the Nash equilibrium of the Cournot game, we first translate the problem into a normal-form game. Recall from the previous section that the normal-form representation of a game specifies: (1) the players in the game, (2) the strategies available to each player, and (3) the payoff received by each player for each combination of strategies that could be chosen by the players. There are of course two players in any duopoly game—the two firms. In the Cournot model, the strategies available to each firm are the different quantities it might produce. We will assume that output is continuously divisible. Naturally, negative outputs are not feasible. Thus, each firm's strategy space can be represented as $S_i = [0, \infty)$, the nonnegative real numbers, in which case a typical strategy s_i is a quantity choice, $q_i \geq 0$. One could argue that extremely large quantities are not feasible and so should not be included in a firm's strategy space. Because $P(Q) = 0$ for $Q \geq a$, however, neither firm will produce a quantity $q_i > a$.

It remains to specify the payoff to firm i as a function of the strategies chosen by it and by the other firm, and to define and

[6]We discuss Bertrand's (1883) model, in which firms choose prices rather than quantities, in Section 1.2.B, and Stackelberg's (1934) model, in which firms choose quantities but one firm chooses before (and is observed by) the other, in Section 2.1.B. Finally, we discuss Friedman's (1971) model, in which the interaction described in Cournot's model occurs repeatedly over time, in Section 2.3.C.

solve for equilibrium. We assume that the firm's payoff is simply its profit. Thus, the payoff $u_i(s_i, s_j)$ in a general two-player game in normal form can be written here as[7]

$$\pi_i(q_i, q_j) = q_i[P(q_i + q_j) - c] = q_i[a - (q_i + q_j) - c].$$

Recall from the previous section that in a two-player game in normal form, the strategy pair (s_1^*, s_2^*) is a Nash equilibrium if, for each player i,

$$u_i(s_i^*, s_j^*) \geq u_i(s_i, s_j^*) \qquad \text{(NE)}$$

for every feasible strategy s_i in S_i. Equivalently, for each player i, s_i^* must solve the optimization problem

$$\max_{s_i \in S_i} u_i(s_i, s_j^*).$$

In the Cournot duopoly model, the analogous statement is that the quantity pair (q_1^*, q_2^*) is a Nash equilibrium if, for each firm i, q_i^* solves

$$\max_{0 \leq q_i < \infty} \pi_i(q_i, q_j^*) = \max_{0 \leq q_i < \infty} q_i[a - (q_i + q_j^*) - c].$$

Assuming $q_j^* < a - c$ (as will be shown to be true), the first-order condition for firm i's optimization problem is both necessary and sufficient; it yields

$$q_i = \frac{1}{2}(a - q_j^* - c). \qquad (1.2.1)$$

Thus, if the quantity pair (q_1^*, q_2^*) is to be a Nash equilibrium, the firms' quantity choices must satisfy

$$q_1^* = \frac{1}{2}(a - q_2^* - c)$$

and

$$q_2^* = \frac{1}{2}(a - q_1^* - c).$$

[7]Note that we have changed the notation slightly by writing $u_i(s_i, s_j)$ rather than $u_i(s_1, s_2)$. Both expressions represent the payoff to player i as a function of the strategies chosen by all the players. We will use these expressions (and their n-player analogs) interchangeably.

Solving this pair of equations yields

$$q_1^* = q_2^* = \frac{a-c}{3},$$

which is indeed less than $a - c$, as assumed.

The intuition behind this equilibrium is simple. Each firm would of course like to be a monopolist in this market, in which case it would choose q_i to maximize $\pi_i(q_i, 0)$—it would produce the monopoly quantity $q_m = (a - c)/2$ and earn the monopoly profit $\pi_i(q_m, 0) = (a - c)^2/4$. Given that there are two firms, aggregate profits for the duopoly would be maximized by setting the aggregate quantity $q_1 + q_2$ equal to the monopoly quantity q_m, as would occur if $q_i = q_m/2$ for each i, for example. The problem with this arrangement is that each firm has an incentive to deviate: because the monopoly quantity is low, the associated price $P(q_m)$ is high, and at this price each firm would like to increase its quantity, in spite of the fact that such an increase in production drives down the market-clearing price. (To see this formally, use (1.2.1) to check that $q_m/2$ is *not* firm 2's best response to the choice of $q_m/2$ by firm 1.) In the Cournot equilibrium, in contrast, the aggregate quantity is higher, so the associated price is lower, so the temptation to increase output is reduced—reduced by just enough that each firm is just deterred from increasing its output by the realization that the market-clearing price will fall. See Problem 1.4 for an analysis of how the presence of n oligopolists affects this equilibrium trade-off between the temptation to increase output and the reluctance to reduce the market-clearing price.

Rather than solving for the Nash equilibrium in the Cournot game algebraically, one could instead proceed graphically, as follows. Equation (1.2.1) gives firm i's best response to firm j's *equilibrium* strategy, q_j^*. Analogous reasoning leads to firm 2's best response to an *arbitrary* strategy by firm 1 and firm 1's best response to an arbitrary strategy by firm 2. Assuming that firm 1's strategy satisfies $q_1 < a - c$, firm 2's best response is

$$R_2(q_1) = \frac{1}{2}(a - q_1 - c);$$

likewise, if $q_2 < a - c$ then firm 1's best response is

$$R_1(q_2) = \frac{1}{2}(a - q_2 - c).$$

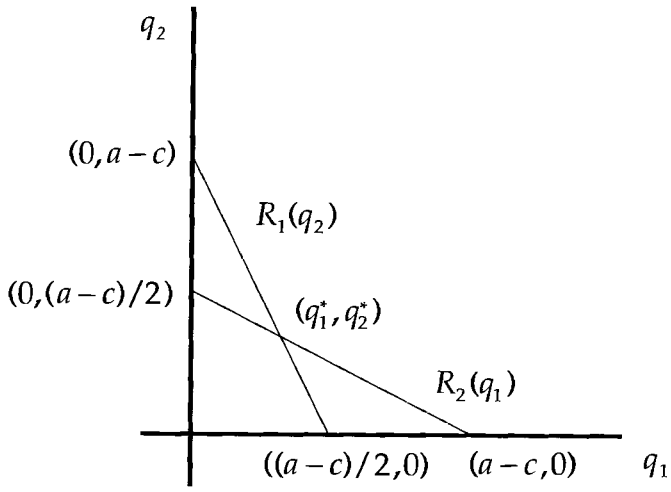


Figure 1.2.1.

As shown in Figure 1.2.1, these two best-response functions intersect only once, at the equilibrium quantity pair (q_1^*, q_2^*).

A third way to solve for this Nash equilibrium is to apply the process of iterated elimination of strictly dominated strategies. This process yields a unique solution—which, by Proposition A in Appendix 1.1.C, must be the Nash equilibrium (q_1^*, q_2^*). The complete process requires an infinite number of steps, each of which eliminates a fraction of the quantities remaining in each firm's strategy space; we discuss only the first two steps. First, the monopoly quantity $q_m = (a - c)/2$ strictly dominates any higher quantity. That is, for any $x > 0$, $\pi_i(q_m, q_j) > \pi_i(q_m + x, q_j)$ for all $q_j \geq 0$. To see this, note that if $Q = q_m + x + q_j < a$, then

$$\pi_i(q_m, q_j) = \frac{a-c}{2}\left[\frac{a-c}{2} - q_j\right]$$

and

$$\pi_i(q_m + x, q_j) = \left[\frac{a-c}{2} + x\right]\left[\frac{a-c}{2} - x - q_j\right] = \pi_i(q_m, q_j) - x(x + q_j),$$

and if $Q = q_m + x + q_j \geq a$, then $P(Q) = 0$, so producing a smaller

quantity raises profit. Second, given that quantities exceeding q_m have been eliminated, the quantity $(a-c)/4$ strictly dominates any lower quantity. That is, for any x between zero and $(a-c)/4$, $\pi_i[(a-c)/4, q_j] > \pi_i[(a-c)/4 - x, q_j]$ for all q_j between zero and $(a-c)/2$. To see this, note that

$$\pi_i\left(\frac{a-c}{4}, q_j\right) = \frac{a-c}{4}\left[\frac{3(a-c)}{4} - q_j\right]$$

and

$$\begin{aligned}\pi_i\left(\frac{a-c}{4} - x, q_j\right) &= \left[\frac{a-c}{4} - x\right]\left[\frac{3(a-c)}{4} + x - q_j\right] \\ &= \pi_i(q_m, q_j) - x\left[\frac{a-c}{2} + x - q_j\right].\end{aligned}$$

After these two steps, the quantities remaining in each firm's strategy space are those in the interval between $(a-c)/4$ and $(a-c)/2$. Repeating these arguments leads to ever-smaller intervals of remaining quantities. In the limit, these intervals converge to the single point $q_i^* = (a-c)/3$.

Iterated elimination of strictly dominated strategies can also be described graphically, by using the observation (from footnote 1; see also the discussion in Section 1.3.A) that a strategy is strictly dominated if and only if there is no belief about the other players' choices for which the strategy is a best response. Since there are only two firms in this model, we can restate this observation as: a quantity q_i is strictly dominated if and only if there is no belief about q_j such that q_i is firm i's best response. We again discuss only the first two steps of the iterative process. First, it is never a best response for firm i to produce more than the monopoly quantity, $q_m = (a-c)/2$. To see this, consider firm 2's best-response function, for example: in Figure 1.2.1, $R_2(q_1)$ equals q_m when $q_1 = 0$ and declines as q_1 increases. Thus, for any $q_j \geq 0$, if firm i believes that firm j will choose q_j, then firm i's best response is less than or equal to q_m; there is no q_j such that firm i's best response exceeds q_m. Second, given this upper bound on firm j's quantity, we can derive a lower bound on firm i's best response: if $q_j \leq (a-c)/2$, then $R_i(q_j) \geq (a-c)/4$, as shown for firm 2's best response in Figure 1.2.2.[8]

[8]These two arguments are slightly incomplete because we have not analyzed

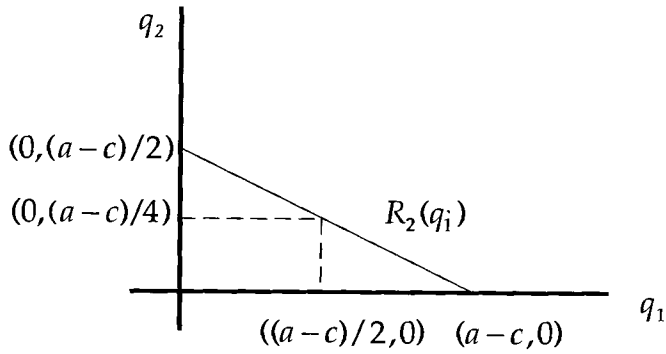


Figure 1.2.2.

As before, repeating these arguments leads to the single quantity $q_i^* = (a - c)/3$.

We conclude this section by changing the Cournot model so that iterated elimination of strictly dominated strategies does *not* yield a unique solution. To do this, we simply add one or more firms to the existing duopoly. We will see that the first of the two steps discussed in the duopoly case continues to hold, but that the process ends there. Thus, when there are more than two firms, iterated elimination of strictly dominated strategies yields only the imprecise prediction that each firm's quantity will not exceed the monopoly quantity (much as in Figure 1.1.4, where no strategies were eliminated by this process).

For concreteness, we consider the three-firm case. Let Q_{-i} denote the sum of the quantities chosen by the firms other than i, and let $\pi_i(q_i, Q_{-i}) = q_i(a - q_i - Q_{-i} - c)$ provided $q_i + Q_{-i} < a$ (whereas $\pi_i(q_i, Q_{-i}) = -cq_i$ if $q_i + Q_{-i} \geq a$). It is again true that the monopoly quantity $q_m = (a - c)/2$ strictly dominates any higher quantity. That is, for any $x > 0$, $\pi_i(q_m, Q_{-i}) > \pi_i(q_m + x, Q_{-i})$ for all $Q_{-i} \geq 0$, just as in the first step in the duopoly case. Since

---

firm i's best response when firm i is uncertain about q_j. Suppose firm i is uncertain about q_j but believes that the expected value of q_j is $E(q_j)$. Because $\pi_i(q_i, q_j)$ is linear in q_j, firm i's best response when it is uncertain in this way simply equals its best response when it is certain that firm j will choose $E(q_j)$—a case covered in the text.

there are two firms other than firm i, however, all we can say about Q_{-i} is that it is between zero and $a - c$, because q_j and q_k are between zero and $(a - c)/2$. But this implies that no quantity $q_i \geq 0$ is strictly dominated for firm i, because for each q_i between zero and $(a - c)/2$ there exists a value of Q_{-i} between zero and $a - c$ (namely, $Q_{-i} = a - c - 2q_i$) such that q_i is firm i's best response to Q_{-i}. Thus, no further strategies can be eliminated.

1.2.B Bertrand Model of Duopoly

We next consider a different model of how two duopolists might interact, based on Bertrand's (1883) suggestion that firms actually choose prices, rather than quantities as in Cournot's model. It is important to note that Bertrand's model is a *different game* than Cournot's model: the strategy spaces are different, the payoff functions are different, and (as will become clear) the behavior in the Nash equilibria of the two models is different. Some authors summarize these differences by referring to the Cournot and Bertrand equilibria. Such usage may be misleading: it refers to the difference between the Cournot and Bertrand games, and to the difference between the equilibrium behavior in these games, *not* to a difference in the equilibrium concept used in the games. *In both games, the equilibrium concept used is the Nash equilibrium defined in the previous section.*

We consider the case of differentiated products. (See Problem 1.7 for the case of homogeneous products.) If firms 1 and 2 choose prices p_1 and p_2, respectively, the quantity that consumers demand from firm i is

$$q_i(p_i, p_j) = a - p_i + bp_j,$$

where $b > 0$ reflects the extent to which firm i's product is a substitute for firm j's product. (This is an unrealistic demand function because demand for firm i's product is positive even when firm i charges an arbitrarily high price, provided firm j also charges a high enough price. As will become clear, the problem makes sense only if $b < 2$.) As in our discussion of the Cournot model, we assume that there are no fixed costs of production and that marginal costs are constant at c, where $c < a$, and that the firms act (i.e., choose their prices) simultaneously.

As before, the first task in the process of finding the Nash equilibrium is to translate the problem into a normal-form game. There

equilibrium to exist—there are many games that do not satisfy the hypotheses of the Theorem but nonetheless have one or more Nash equilibria.

1.4 Further Reading

On the assumptions underlying iterated elimination of strictly dominated strategies and Nash equilibrium, and on the interpretation of mixed strategies in terms of the players' beliefs, see Brandenburger (1992). On the relation between (Cournot-type) models where firms choose quantities and (Bertrand-type) models where firms choose prices, see Kreps and Scheinkman (1983), who show that in some circumstances the Cournot outcome occurs in a Bertrand-type model in which firms face capacity constraints (which they choose, at a cost, prior to choosing prices). On arbitration, see Gibbons (1988), who shows how the arbitrator's preferred settlement can depend on the information content of the parties' offers, in both final-offer and conventional arbitration. Finally, on the existence of Nash equilibrium, including pure-strategy equilibria in games with continuous strategy spaces, see Dasgupta and Maskin (1986).

1.5 Problems

Section 1.1

1.1. What is a game in normal form? What is a strictly dominated strategy in a normal-form game? What is a pure-strategy Nash equilibrium in a normal-form game?

1.2. In the following normal-form game, what strategies survive iterated elimination of strictly dominated strategies? What are the pure-strategy Nash equilibria?

	L	C	R
T	2.0	1,1	4,2
M	3,4	1,2	2,3
B	1,3	0,2	3,0

1.3. Players 1 and 2 are bargaining over how to split one dollar. Both players simultaneously name shares they would like to have, s_1 and s_2, where $0 \leq s_1, s_2 \leq 1$. If $s_1 + s_2 \leq 1$, then the players receive the shares they named; if $s_1 + s_2 > 1$, then both players receive zero. What are the pure-strategy Nash equilibria of this game?

Section 1.2

1.4. Suppose there are n firms in the Cournot oligopoly model. Let q_i denote the quantity produced by firm i, and let $Q = q_1 + \cdots + q_n$ denote the aggregate quantity on the market. Let P denote the market-clearing price and assume that inverse demand is given by $P(Q) = a - Q$ (assuming $Q < a$, else $P = 0$). Assume that the total cost of firm i from producing quantity q_i is $C_i(q_i) = cq_i$. That is, there are no fixed costs and the marginal cost is constant at c, where we assume $c < a$. Following Cournot, suppose that the firms choose their quantities simultaneously. What is the Nash equilibrium? What happens as n approaches infinity?

1.5. Consider the following two finite versions of the Cournot duopoly model. First, suppose each firm must choose either half the monopoly quantity, $q_m/2 = (a - c)/4$, or the Cournot equilibrium quantity, $q_c = (a - c)/3$. No other quantities are feasible. Show that this two-action game is equivalent to the Prisoners' Dilemma: each firm has a strictly dominated strategy, and both are worse off in equilibrium than they would be if they cooperated. Second, suppose each firm can choose either $q_m/2$, or q_c, or a third quantity, q'. Find a value for q' such that the game is equivalent to the Cournot model in Section 1.2.A, in the sense that (q_c, q_c) is a unique Nash equilibrium and both firms are worse off in equilibrium than they could be if they cooperated, but neither firm has a strictly dominated strategy.

1.6. Consider the Cournot duopoly model where inverse demand is $P(Q) = a - Q$ but firms have asymmetric marginal costs: c_1 for firm 1 and c_2 for firm 2. What is the Nash equilibrium if $0 < c_i < a/2$ for each firm? What if $c_1 < c_2 < a$ but $2c_2 > a + c_1$?

1.7. In Section 1.2.B, we analyzed the Bertrand duopoly model with differentiated products. The case of homogeneous products

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- Backward induction; Stackelberg (blocks 3–4): §2.1.A–B (p. 57–64)
- Two-stage games of imperfect information (block 4): §2.2.A (p. 71–73)

peated games, and we analyze Friedman's (1971) model of collusion between Cournot duopolists, Shapiro and Stiglitz's (1984) model of efficiency wages, and Barro and Gordon's (1983) model of monetary policy.

In Section 2.4 we introduce the tools necessary to analyze a general dynamic game of complete information, whether with perfect or imperfect information. We define the *extensive-form* representation of a game and relate it to the normal-form representation introduced in Chapter 1. We also define subgame-perfect Nash equilibrium for general games. The main point (of both this section and the chapter as a whole) is that a dynamic game of complete information may have many Nash equilibria, but some of these may involve noncredible threats or promises. The subgame-perfect Nash equilibria are those that pass a credibility test.

2.1 Dynamic Games of Complete and Perfect Information

2.1.A Theory: Backwards Induction

The grenade game is a member of the following class of simple games of complete and perfect information:

1. Player 1 chooses an action a_1 from the feasible set A_1.

2. Player 2 observes a_1 and then chooses an action a_2 from the feasible set A_2.

3. Payoffs are $u_1(a_1, a_2)$ and $u_2(a_1, a_2)$.

Many economic problems fit this description.[2] Two examples

[2]Player 2's feasible set of actions, A_2, could be allowed to depend on player 1's action, a_1. Such dependence could be denoted by $A_2(a_1)$ or could be incorporated into player 2's payoff function, by setting $u_2(a_1, a_2) = -\infty$ for values of a_2 that are not feasible for a given a_1. Some moves by player 1 could even end the game, without player 2 getting a move; for such values of a_1, the set of feasible actions $A_2(a_1)$ contains only one element, so player 2 has no choice to make.

(discussed later in detail) are Stackelberg's model of duopoly and Leontief's model of wages and employment in a unionized firm. Other economic problems can be modeled by allowing for a longer sequence of actions, either by adding more players or by allowing players to move more than once. (Rubinstein's bargaining game, discussed in Section 2.1.D, is an example of the latter.) The key features of a dynamic game of complete and perfect information are that (i) the moves occur in sequence, (ii) all previous moves are observed before the next move is chosen, and (iii) the players' payoffs from each feasible combination of moves are common knowledge.

We solve a game from this class by backwards induction, as follows. When player 2 gets the move at the second stage of the game, he or she will face the following problem, given the action a_1 previously chosen by player 1:

$$\max_{a_2 \in A_2} u_2(a_1, a_2).$$

Assume that for each a_1 in A_1, player 2's optimization problem has a unique solution, denoted by $R_2(a_1)$. This is player 2's *reaction* (or best response) to player 1's action. Since player 1 can solve 2's problem as well as 2 can, player 1 should anticipate player 2's reaction to each action a_1 that 1 might take, so 1's problem at the first stage amounts to

$$\max_{a_1 \in A_1} u_1(a_1, R_2(a_1)).$$

Assume that this optimization problem for player 1 also has a unique solution, denoted by a_1^*. We will call $(a_1^*, R_2(a_1^*))$ the *backwards-induction outcome* of this game. The backwards-induction outcome does not involve noncredible threats: player 1 anticipates that player 2 will respond optimally to *any* action a_1 that 1 might choose, by playing $R_2(a_1)$; player 1 gives no credence to threats by player 2 to respond in ways that will not be in 2's self-interest when the second stage arrives.

Recall that in Chapter 1 we used the normal-form representation to study static games of complete information, and we focused on the notion of Nash equilibrium as a solution concept for such games. In this section's discussion of dynamic games, however, we have made no mention of either the normal-form representation or Nash equilibrium. Instead, we have given a

verbal description of a game in (1)–(3), and we have defined the backwards-induction outcome as the solution to that game. In Section 2.4.A we will see that the verbal description in (1)–(3) is the extensive-form representation of the game. We will relate the extensive- and normal-form representations, but we will find that for dynamic games the extensive-form representation is often more convenient. In Section 2.4.B we will define subgame-perfect Nash equilibrium: a Nash equilibrium is subgame-perfect if it does not involve a noncredible threat, in a sense to be made precise. We will find that there may be multiple Nash equilibria in a game from the class defined by (1)–(3), but that the only subgame-perfect Nash equilibrium is the equilibrium associated with the backwards-induction outcome. This is an example of the observation in Section 1.1.C that some games have multiple Nash equilibria but have one equilibrium that stands out as the compelling solution to the game.

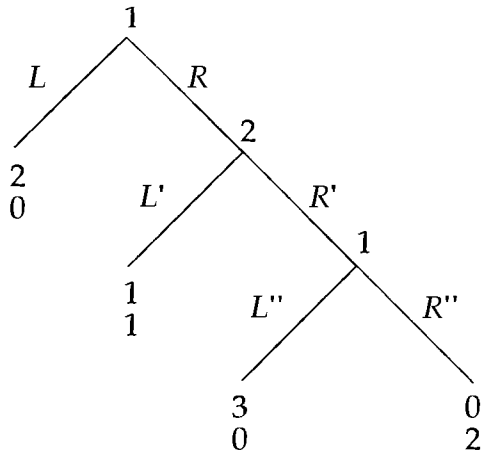
We conclude this section by exploring the rationality assumptions inherent in backwards-induction arguments. Consider the following three-move game, in which player 1 moves twice:

1. Player 1 chooses L or R, where L ends the game with payoffs of 2 to player 1 and 0 to player 2.

2. Player 2 observes 1's choice. If 1 chose R then 2 chooses L' or R', where L' ends the game with payoffs of 1 to both players.

3. Player 1 observes 2's choice (and recalls his or her own choice in the first stage). If the earlier choices were R and R' then 1 chooses L'' or R'', both of which end the game, L'' with payoffs of 3 to player 1 and 0 to player 2 and R'' with analogous payoffs of 0 and 2.

All these words can be translated into the following succinct game tree. (This is the extensive-form representation of the game, to be defined more generally in Section 2.4.) The top payoff in the pair of payoffs at the end of each branch of the game tree is player 1's, the bottom player 2's.



To compute the backwards-induction outcome of this game, we begin at the third stage (i.e., player 1's second move). Here player 1 faces a choice between a payoff of 3 from L'' and a payoff of 0 from R'', so L'' is optimal. Thus, at the second stage, player 2 anticipates that if the game reaches the third stage then 1 will play L'', which would yield a payoff of 0 for player 2. The second-stage choice for player 2 therefore is between a payoff of 1 from L' and a payoff of 0 from R', so L' is optimal. Thus, at the first stage, player 1 anticipates that if the game reaches the second stage then 2 will play L', which would yield a payoff of 1 for player 1. The first-stage choice for player 1 therefore is between a payoff of 2 from L and a payoff of 1 from R, so L is optimal.

This argument establishes that the backwards-induction outcome of this game is for player 1 to choose L in the first stage, thereby ending the game. Even though backwards induction predicts that the game will end in the first stage, an important part of the argument concerns what would happen if the game did not end in the first stage. In the second stage, for example, when player 2 anticipates that if the game reaches the third stage then 1 will play L'', 2 is assuming that 1 is rational. This assumption may seem inconsistent with the fact that 2 gets to move in the second stage only if 1 deviates from the backwards-induction outcome of the game. That is, it may seem that if 1 plays R in the first stage then 2 cannot assume in the second stage that 1 is rational, but this is not the case: if 1 plays R in the first stage then it cannot be common knowledge that both players are rational, but there

remain reasons for 1 to have chosen R that do not contradict 2's assumption that 1 is rational.[3] One possibility is that it is common knowledge that player 1 is rational but not that player 2 is rational: if 1 thinks that 2 might not be rational, then 1 might choose R in the first stage, hoping that 2 will play R' in the second stage, thereby giving 1 the chance to play L'' in the third stage. Another possibility is that it is common knowledge that player 2 is rational but not that player 1 is rational: if 1 is rational but thinks that 2 thinks that 1 might not be rational, then 1 might choose R in the first stage, hoping that 2 will think that 1 is not rational and so play R' in the hope that 1 will play R'' in the third stage. Backwards induction assumes that 1's choice of R could be explained along these lines. For some games, however, it may be more reasonable to assume that 1 played R because 1 is indeed irrational. In such games, backwards induction loses much of its appeal as a prediction of play, just as Nash equilibrium does in games where game theory does not provide a unique solution and no convention will develop.

2.1.B Stackelberg Model of Duopoly

Stackelberg (1934) proposed a dynamic model of duopoly in which a dominant (or leader) firm moves first and a subordinate (or follower) firm moves second. At some points in the history of the U.S. automobile industry, for example, General Motors has seemed to play such a leadership role. (It is straightforward to extend what follows to allow for more than one following firm, such as Ford, Chrysler, and so on.) Following Stackelberg, we will develop the model under the assumption that the firms choose quantities, as in the Cournot model (where the firms' choices are simultaneous, rather than sequential as here). We leave it as an exercise to develop the analogous sequential-move model in which firms choose prices, as they do (simultaneously) in the Bertrand model.

The timing of the game is as follows: (1) firm 1 chooses a quantity $q_1 \geq 0$; (2) firm 2 observes q_1 and then chooses a quantity

[3]Recall from the discussion of iterated elimination of strictly dominated strategies (in Section 1.1.B) that it is common knowledge that the players are rational if all the players are rational, and all the players know that all the players are rational, and all the players know that all the players know that all the players are rational, and so on, ad infinitum.

$q_2 \geq 0$; (3) the payoff to firm i is given by the profit function

$$\pi_i(q_i, q_j) = q_i[P(Q) - c],$$

where $P(Q) = a - Q$ is the market-clearing price when the aggregate quantity on the market is $Q = q_1 + q_2$, and c is the constant marginal cost of production (fixed costs being zero).

To solve for the backwards-induction outcome of this game, we first compute firm 2's reaction to an arbitrary quantity by firm 1. $R_2(q_1)$ solves

$$\max_{q_2 \geq 0} \pi_2(q_1, q_2) = \max_{q_2 \geq 0} q_2[a - q_1 - q_2 - c],$$

which yields

$$R_2(q_1) = \frac{a - q_1 - c}{2},$$

provided $q_1 < a - c$. The same equation for $R_2(q_1)$ appeared in our analysis of the simultaneous-move Cournot game in Section 1.2.A. The difference is that here $R_2(q_1)$ is truly firm 2's reaction to firm 1's observed quantity, whereas in the Cournot analysis $R_2(q_1)$ is firm 2's best response to a hypothesized quantity to be simultaneously chosen by firm 1.

Since firm 1 can solve firm 2's problem as well as firm 2 can solve it, firm 1 should anticipate that the quantity choice q_1 will be met with the reaction $R_2(q_1)$. Thus, firm 1's problem in the first stage of the game amounts to

$$\begin{aligned} \max_{q_1 \geq 0} \pi_1(q_1, R_2(q_1)) &= \max_{q_1 \geq 0} q_1[a - q_1 - R_2(q_1) - c] \\ &= \max_{q_1 \geq 0} q_1 \frac{a - q_1 - c}{2}, \end{aligned}$$

which yields

$$q_1^* = \frac{a - c}{2} \quad \text{and} \quad R_2(q_1^*) = \frac{a - c}{4}$$

as the backwards-induction outcome of the Stackelberg duopoly game.[4]

[4]Just as "Cournot equilibrium" and "Bertrand equilibrium" typically refer to the Nash equilibria of the Cournot and Bertrand games, references to

Recall from Chapter 1 that in the Nash equilibrium of the Cournot game each firm produces $(a-c)/3$. Thus, aggregate quantity in the backwards-induction outcome of the Stackelberg game, $3(a-c)/4$, is greater than aggregate quantity in the Nash equilibrium of the Cournot game, $2(a-c)/3$, so the market-clearing price is lower in the Stackelberg game. In the Stackelberg game, however, firm 1 could have chosen its Cournot quantity, $(a-c)/3$, in which case firm 2 would have responded with its Cournot quantity. Thus, in the Stackelberg game, firm 1 could have achieved its Cournot profit level but chose to do otherwise, so firm 1's profit in the Stackelberg game must exceed its profit in the Cournot game. But the market-clearing price is lower in the Stackelberg game, so aggregate profits are lower, so the fact that firm 1 is better off implies that firm 2 is worse off in the Stackelberg than in the Cournot game.

The observation that firm 2 does worse in the Stackelberg than in the Cournot game illustrates an important difference between single- and multi-person decision problems. In single-person decision theory, having more information can never make the decision maker worse off. In game theory, however, having more information (or, more precisely, having it known to the other players that one has more information) *can* make a player worse off.

In the Stackelberg game, the information in question is firm 1's quantity: firm 2 knows q_1, and (as importantly) firm 1 knows that firm 2 knows q_1. To see the effect this information has, consider the modified sequential-move game in which firm 1 chooses q_1, after which firm 2 chooses q_2 but does so without observing q_1. If firm 2 believes that firm 1 has chosen its Stackelberg quantity $q_1^* = (a-c)/2$, then firm 2's best response is again $R_2(q_1^*) = (a-c)/4$. But if firm 1 anticipates that firm 2 will hold this belief and so choose this quantity, then firm 1 prefers to choose its best response to $(a-c)/4$—namely, $3(a-c)/8$—rather than its Stackelberg quantity $(a-c)/2$. Thus, firm 2 should not believe that firm 1 has chosen its Stackelberg quantity. Rather, the unique Nash equilibrium of this

"Stackelberg equilibrium" often mean that the game is sequential- rather than simultaneous-move. As noted in the previous section, however, sequential-move games sometimes have multiple Nash equilibria, only one of which is associated with the backwards-induction outcome of the game. Thus, "Stackelberg equilibrium" can refer both to the sequential-move nature of the game and to the use of a stronger solution concept than simply Nash equilibrium.

modified sequential-move game is for both firms to choose the quantity $(a-c)/3$—precisely the Nash equilibrium of the Cournot game, where the firms move simultaneously.[5] Thus, having firm 1 know that firm 2 knows q_1 hurts firm 2.

2.1.C Wages and Employment in a Unionized Firm

In Leontief's (1946) model of the relationship between a firm and a monopoly union (i.e., a union that is the monopoly seller of labor to the firm), the union has exclusive control over wages, but the firm has exclusive control over employment. (Similar qualitative conclusions emerge in a more realistic model in which the firm and the union bargain over wages but the firm retains exclusive control over employment.) The union's utility function is $U(w, L)$, where w is the wage the union demands from the firm and L is employment. Assume that $U(w, L)$ increases in both w and L. The firm's profit function is $\pi(w, L) = R(L) - wL$, where $R(L)$ is the revenue the firm can earn if it employs L workers (and makes the associated production and product-market decisions optimally). Assume that $R(L)$ is increasing and concave.

Suppose the timing of the game is: (1) the union makes a wage demand, w; (2) the firm observes (and accepts) w and then chooses employment, L; (3) payoffs are $U(w, L)$ and $\pi(w, L)$. We can say a great deal about the backwards-induction outcome of this game even though we have not assumed specific functional forms for $U(w, L)$ and $R(L)$ and so are not able to solve for this outcome explicitly.

First, we can characterize the firm's best response in stage (2), $L^*(w)$, to an arbitrary wage demand by the union in stage (1), w. Given w, the firm chooses $L^*(w)$ to solve

$$\max_{L \geq 0} \pi(w, L) = \max_{L \geq 0} R(L) - wL,$$

the first-order condition for which is

$$R'(L) - w = 0.$$

[5]This is an example of a claim we made in Section 1.1.A: in a normal-form game the players choose their strategies simultaneously, but this does not imply that the parties necessarily *act* simultaneously; it suffices that each choose his or her action without knowledge of the others' choices. For further discussion of this point, see Section 2.4.A.

will be informal (but can be made formal). Suppose that there is a backwards-induction outcome of the game as a whole in which players 1 and 2 receive the payoffs s and $1 - s$, respectively. We can use these payoffs in the game beginning in the third period, should it be reached, and then work backwards to the first period (as in the three-period model) to compute a new backwards-induction outcome for the game as a whole. In this new backwards-induction outcome, player 1 will offer the settlement $(f(s), 1 - f(s))$ in the first period and player 2 will accept, where $f(s) = 1 - \delta(1 - \delta s)$ is the share taken by player 1 in the first period of the three-period model above when the settlement $(s, 1 - s)$ is exogenously imposed in the third period.

Let s_H be the highest payoff player 1 can achieve in any backwards-induction outcome of the game as a whole. Imagine using s_H as the third-period payoff to player 1, as previously described: this will produce a new backwards-induction outcome in which player 1's first-period payoff is $f(s_H)$. Since $f(s) = 1 - \delta + \delta^2 s$ is increasing in s, $f(s_H)$ is the highest possible first-period payoff because s_H is the highest possible third-period payoff. But s_H is also the highest possible first-period payoff, so $f(s_H) = s_H$. Parallel arguments show that $f(s_L) = s_L$, where s_L is the lowest payoff player 1 can achieve in any backwards-induction outcome of the game as a whole. The only value of s that satisfies $f(s) = s$ is $1/(1+\delta)$, which we will denote by s^*. Thus, $s_H = s_L = s^*$, so there is a unique backwards-induction outcome in the game as a whole: in the first period, player 1 offers the settlement $(s^* = 1/(1 + \delta), 1 - s^* = \delta/(1 + \delta))$ to player 2, who accepts.

2.2 Two-Stage Games of Complete but Imperfect Information

2.2.A Theory: Subgame Perfection

We now enrich the class of games analyzed in the previous section. As in dynamic games of complete and perfect information, we continue to assume that play proceeds in a sequence of stages, with the moves in all previous stages observed before the next stage begins. Unlike in the games analyzed in the previous section,

however, we now allow there to be simultaneous moves within each stage. As will be explained in Section 2.4, this simultaneity of moves within stages means that the games analyzed in this section have imperfect information. Nonetheless, these games share important features with the perfect-information games considered in the previous section.

We will analyze the following simple game, which we (uninspiredly!) call a two-stage game of complete but imperfect information:

1. Players 1 and 2 simultaneously choose actions a_1 and a_2 from feasible sets A_1 and A_2, respectively.

2. Players 3 and 4 observe the outcome of the first stage, (a_1, a_2), and then simultaneously choose actions a_3 and a_4 from feasible sets A_3 and A_4, respectively.

3. Payoffs are $u_i(a_1, a_2, a_3, a_4)$ for $i = 1, 2, 3, 4$.

Many economic problems fit this description.[8] Three examples (later discussed in detail) are bank runs, tariffs and imperfect international competition, and tournaments (e.g., competition among several vice presidents in a firm to be the next president). Other economic problems can be modeled by allowing for a longer sequence of stages, either by adding players or by allowing players to move in more than one stage. There could also be fewer players: in some applications, players 3 and 4 are players 1 and 2; in others, either player 2 or player 4 is missing.

We solve a game from this class by using an approach in the spirit of backwards induction, but this time the first step in working backwards from the end of the game involves solving a real game (the simultaneous-move game between players 3 and 4 in stage two, given the outcome from stage one) rather than solving a single-person optimization problem as in the previous section. To keep things simple, in this section we will assume that for each feasible outcome of the first-stage game, (a_1, a_2), the second-stage game that remains between players 3 and 4 has a unique Nash equilibrium, denoted by $(a_3^*(a_1, a_2), a_4^*(a_1, a_2))$. In Section 2.3.A (on

[8] As in the previous section, the feasible action sets of players 3 and 4 in the second stage, A_3 and A_4, could be allowed to depend on the outcome of the first stage, (a_1, a_2). In particular, there may be values of (a_1, a_2) that end the game.

repeated games) we consider the implications of relaxing this assumption.

If players 1 and 2 anticipate that the second-stage behavior of players 3 and 4 will be given by $(a_3^*(a_1,a_2), a_4^*(a_1,a_2))$, then the first-stage interaction between players 1 and 2 amounts to the following simultaneous-move game:

1. Players 1 and 2 simultaneously choose actions a_1 and a_2 from feasible sets A_1 and A_2, respectively.

2. Payoffs are $u_i(a_1,a_2,a_3^*(a_1,a_2),a_4^*(a_1,a_2))$ for $i = 1,2$.

Suppose (a_1^*,a_2^*) is the unique Nash equilibrium of this simultaneous-move game. We will call $(a_1^*,a_2^*,a_3^*(a_1^*,a_2^*).a_4^*(a_1^*,a_2^*))$ the *subgame-perfect outcome* of this two-stage game. This outcome is the natural analog of the backwards-induction outcome in games of complete and perfect information, and the analogy applies to both the attractive and the unattractive features of the latter. Players 1 and 2 should not believe a threat by players 3 and 4 that the latter will respond with actions that are not a Nash equilibrium in the remaining second-stage game, because when play actually reaches the second stage at least one of players 3 and 4 will not want to carry out such a threat (exactly because it is not a Nash equilibrium of the game that remains at that stage). On the other hand, suppose that player 1 is also player 3, and that player 1 does not play a_1^* in the first stage: player 4 may then want to reconsider the assumption that player 3 (i.e., player 1) will play $a_3^*(a_1,a_2)$ in the second stage.

2.2.B Bank Runs

Two investors have each deposited D with a bank. The bank has invested these deposits in a long-term project. If the bank is forced to liquidate its investment before the project matures, a total of $2r$ can be recovered, where $D > r > D/2$. If the bank allows the investment to reach maturity, however, the project will pay out a total of $2R$, where $R > D$.

There are two dates at which the investors can make withdrawals from the bank: date 1 is before the bank's investment matures; date 2 is after. For simplicity, assume that there is no discounting. If both investors make withdrawals at date 1 then each receives r and the game ends. If only one investor makes

of tournaments in which losers in one round do not proceed to the next.

Section 2.3: Benoit and Krishna (1985) analyze finitely repeated games. On renegotiation, see Benoit and Krishna (1989) for finitely repeated games and Farrell and Maskin (1989) for infinitely repeated games and a review of the literature. Tirole (1988, Chapter 6) surveys dynamic oligopoly models. Akerlof and Yellen's (1986) book collects many of the important papers on efficiency wages and provides an integrative introduction. On monetary policy, see Ball (1990) for a summary of the stylized facts, a review of existing models, and a model that explains the time-path of inflation.

Section 2.4: See Kreps and Wilson (1982) for a formal treatment of extensive-form games, and Kreps (1990, Chapter 11) for a more discursive account.

2.6 Problems

Section 2.1

2.1. Suppose a parent and child play the following game, first analyzed by Becker (1974). First, the child takes an action, A, that produces income for the child, $I_C(A)$, and income for the parent, $I_P(A)$. (Think of $I_C(A)$ as the child's income net of any costs of the action A.) Second, the parent observes the incomes I_C and I_P and then chooses a bequest, B, to leave to the child. The child's payoff is $U(I_C + B)$; the parent's is $V(I_P - B) + kU(I_C + B)$, where $k > 0$ reflects the parent's concern for the child's well-being. Assume that: the action is a nonnegative number, $A \geq 0$; the income functions $I_C(A)$ and $I_P(A)$ are strictly concave and are maximized at $A_C > 0$ and $A_P > 0$, respectively; the bequest B can be positive or negative; and the utility functions U and V are increasing and strictly concave. Prove the "Rotten Kid" Theorem: in the backwards-induction outcome, the child chooses the action that maximizes the family's aggregate income, $I_C(A) + I_P(A)$, even though only the parent's payoff exhibits altruism.

2.2. Now suppose the parent and child play a different game, first analyzed by Buchanan (1975). Let the incomes I_C and I_P be fixed exogenously. First, the child decides how much of the

Section 2.2

2.6. Three oligopolists operate in a market with inverse demand given by $P(Q) = a - Q$, where $Q = q_1 + q_2 + q_3$ and q_i is the quantity produced by firm i. Each firm has a constant marginal cost of production, c, and no fixed cost. The firms choose their quantities as follows: (1) firm 1 chooses $q_1 \geq 0$; (2) firms 2 and 3 observe q_1 and then simultaneously choose q_2 and q_3, respectively. What is the subgame-perfect outcome?

2.7. Suppose a union is the sole supplier of labor to all the firms in an oligopoly, such as the United Auto Workers is to General Motors, Ford, Chrysler, and so on. Let the timing of moves be analogous to the model in Section 2.1.C: (1) the union makes a single wage demand, w, that applies to *all* the firms; (2) the firms observe (and accept) w and then simultaneously choose employment levels, L_i for firm i; (3) payoffs are $(w - w_a)L$ for the union, where w_a is the wage that union members can earn in alternative employment and $L = L_1 + \cdots + L_n$ is total employment in the unionized firms, and profit $\pi(w, L_i)$ for firm i, where the determinants of firm i's profit are described next.

All firms have the following production function: output equals labor; $q_i = L_i$. The market-clearing price is $P(Q) = a - Q$ when the aggregate quantity on the market is $Q = q_1 + \cdots + q_n$. To keep things simple, suppose that firms have no costs other than wages. What is the subgame-perfect outcome of this game? How (and why) does the number of firms affect the union's utility in the subgame-perfect outcome?

2.8. Modify the tournament model in Section 2.2.D so that worker i's output is $y_i = e_i - (1/2)s_j + \varepsilon_i$, where $s_j \geq 0$ represents sabotage by worker j, and worker i's disutility of (productive and destructive) effort is $g(e_i) + g(s_i)$, as in Lazear (1989). Show that the optimal prize $w_H - w_L$ is smaller than when there is no possibility of sabotage (as in the text).

2.9. Consider two countries. At date 1, the countries both have such high tariffs that there is no trade. Within each country, wages and employment are determined as in the monopoly-union model in Section 2.1.C. At date 2, all tariffs disappear. Now each union sets the wage in its country but each firm produces for both markets.

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Andreu Mas-Colell, Michael D. Whinston, Jerry R. Green - Mic

- Dominant and dominated strategies (block 1): §8.B (p. 236–241)
- Mixed-strategy Nash equilibrium (block 2): §8.D (p. 246–252)

equilibrium. In Section 8.F, we explore the implications of players entertaining the possibility that, with some small but positive probability, their opponents might make a mistake in choosing their strategies. We define the notion of a (*normal form*) *trembling-hand perfect Nash equilibrium*, an extension of the Nash equilibrium concept that requires that equilibria be robust to the possibility of small mistakes.

Throughout the chapter, we study simultaneous-move games using their normal form representations (see Section 7.D). Thus, we use $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ when we consider only pure (nonrandom) strategy choices and $\Gamma_N = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$ when we allow for the possibility of randomized choices by the players (see Section 7.E for a discussion of randomized choices). We often denote a profile of pure strategies for player i's opponents by $s_{-i} = (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_I)$, with a similar meaning applying to the profile of mixed strategies σ_{-i}. We then write $s = (s_i, s_{-i})$ and $\sigma = (\sigma_i, \sigma_{-i})$. We also let $S = S_1 \times \cdots \times S_I$ and $S_{-i} = S_1 \times \cdots \times S_{i-1} \times S_{i+1} \times \cdots \times S_I$.

8.B Dominant and Dominated Strategies

We begin our study of simultaneous-move games by considering the predictions that can be made based on a relatively simple means of comparing a player's possible strategies: that of *dominance*.

To keep matters as simple as possible, we initially ignore the possibility that players might randomize in their strategy choices. Hence, our focus is on games $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ whose strategy sets allow for only pure strategies.

Consider the game depicted in Figure 8.B.1, the famous *Prisoner's Dilemma*. The story behind this game is as follows: Two individuals are arrested for allegedly engaging in a serious crime and are held in separate cells. The district attorney (the DA) tries to extract a confession from each prisoner. Each is privately told that if he is the only one to confess, then he will be rewarded with a light sentence of 1 year while the recalcitrant prisoner will go to jail for 10 years. However, if he is the only one not to confess, then it is he who will serve the 10-year sentence. If both confess, they will both be shown some mercy: they will each get 5 years. Finally, if neither confesses, it will still be possible to convict both of a lesser crime that carries a sentence of 2 years. Each player wishes to minimize the time he spends in jail (or maximize the negative of this, the payoffs that are depicted in Figure 8.B.1).

What will the outcome of this game be? There is only one plausible answer: (confess, confess). To see why, note that playing "confess" is each player's best strategy *regardless of what the other player does*. This type of strategy is known as a *strictly dominant strategy*.

		Prisoner 2: Don't Confess	Prisoner 2: Confess
Prisoner 1	Don't Confess	−2, −2	−10, −1
	Confess	−1, −10	−5, −5

Figure 8.B.1
The Prisoner's Dilemma.

Definition 8.B.1: A strategy $s_i \in S_i$ is a *strictly dominant strategy* for player i in game $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ if for all $s_i' \neq s_i$, we have

$$u_i(s_i, s_{-i}) > u_i(s_i', s_{-i})$$

for all $s_{-i} \in S_{-i}$.

In words, a strategy s_i is a strictly dominant strategy for player i if it maximizes uniquely player i's payoff for any strategy that player i's rivals might play. (The reason for the modifier *strictly* in Definition 8.B.1 will be made clear in Definition 8.B.3.) If a player has a strictly dominant strategy, as in the Prisoner's Dilemma, we should expect him to play it.

The striking aspect of the (confess, confess) outcome in the Prisoner's Dilemma is that although it is the one we expect to arise, it is not the best outcome for the players *jointly*; both players would prefer that neither of them confess. For this reason, the Prisoner's Dilemma is the paradigmatic example of self-interested, rational behavior *not* leading to a socially optimal result.

One way of viewing the outcome of the Prisoner's Dilemma is that, in seeking to maximize his own payoff, each prisoner has a negative effect on his partner; by moving away from the (don't confess, don't confess) outcome, a player reduces his jail time by 1 year but increases that of his partner by 8 (in Chapter 11, we shall see this as an example of an *externality*).

Dominated Strategies

Although it is compelling that players should play strictly dominant strategies if they have them, it is rare for such strategies to exist. Often, one strategy of player i's may be best when his rivals play s_{-i} and another when they play some other strategies s_{-i}' (think of the standard Matching Pennies game in Chapter 7). Even so, we might still be able to use the idea of dominance to eliminate some strategies as possible choices. In particular, we should expect that player i will not play *dominated* strategies, those for which there is some alternative strategy that yields him a greater payoff regardless of what the other players do.

Definition 8.B.2: A strategy $s_i \in S_i$ is *strictly dominated* for player i in game $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ if there exists another strategy $s_i' \in S_i$ such that for all $s_{-i} \in S_{-i}$,

$$u_i(s_i', s_{-i}) > u_i(s_i, s_{-i}).$$

In this case, we say that strategy s_i' *strictly dominates* strategy s_i.

With this definition, we can restate our definition of a strictly dominant strategy (Definition 8.B.1) as follows: Strategy s_i is a strictly dominant strategy for player i in game $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ if it strictly dominates every other strategy in S_i.

Example 8.B.1: Consider the game shown in Figure 8.B.2. There is no strictly dominant strategy, but strategy D for player 1 is strictly dominated by strategy M (and also by strategy U). ■

Definition 8.D.3 presents a related, weaker notion of a dominated strategy that is of some importance.

		Player 2 L	Player 2 R
Player 1	U	1, −1	−1, 1
	M	−1, 1	1, −1
	D	−2, 5	−3, 2

Figure 8.B.2 (left)
Strategy D is strictly dominated.

		Player 2 L	Player 2 R
Player 1	U	5, 1	4, 0
	M	6, 0	3, 1
	D	6, 4	4, 4

Figure 8.B.3 (right)
Strategies U and M are weakly dominated.

Definition 8.B.3: A strategy $s_i \in S_i$ is *weakly dominated* in game $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ if there exists another strategy $s_i' \in S_i$ such that for all $s_{-i} \in S_{-i}$,

$$u_i(s_i', s_{-i}) \geq u_i(s_i, s_{-i}),$$

with strict inequality for *some* s_{-i}. In this case, we say that strategy s_i' *weakly dominates* strategy s_i. A strategy is a *weakly dominant strategy* for player i in game $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ if it weakly dominates every other strategy in S_i.

Thus, a strategy is weakly dominated if another strategy does at least as well for all s_{-i} and strictly better for some s_{-i}.

Example 8.B.2: Figure 8.B.3 depicts a game in which player 1 has two weakly dominated strategies, U and M. Both are weakly dominated by strategy D. ■

Unlike a strictly dominated strategy, a strategy that is only weakly dominated cannot be ruled out based solely on principles of rationality. For any alternative strategy that player i might pick, there is at least one profile of strategies for his rivals for which the weakly dominated strategy does as well. In Figure 8.B.3, for example, player 1 could rationally pick M if he was *absolutely sure* that player 2 would play L. Yet, if the probability of player 2 choosing strategy R was perceived by player 1 as positive (no matter how small), then M would not be a rational choice for player 1. *Caution* might therefore rule out M. More generally, weakly dominated strategies could be dismissed if players always believed that there was at least some positive probability that any strategies of their rivals could be chosen. We do not pursue this idea here, although we return to it in Section 8.F. For now, we continue to allow a player to entertain any conjecture about what an opponent might play, even a perfectly certain one.

Iterated Deletion of Strictly Dominated Strategies

As we have noted, it is unusual for elimination of strictly dominated strategies to lead to a unique prediction for a game (e.g., recall the game in Figure 8.B.2). However, the logic of eliminating strictly dominated strategies can be pushed further, as demonstrated in Example 8.B.3.

Example 8.B.3: In Figure 8.B.4, we depict a modification of the Prisoner's Dilemma, which we call the *DA's Brother.*

The story (a somewhat far-fetched one!) is now as follows: One of the prisoners, prisoner 1, is the DA's brother. The DA has some discretion in the fervor with which

		Prisoner 2 Don't Confess	Prisoner 2 Confess
Prisoner 1	Don't Confess	0, −2	−10, −1
	Confess	−1, −10	−5, −5

Figure 8.B.4
The DA's Brother.

he prosecutes and, in particular, can allow prisoner 1 to go free if neither of the prisoners confesses. With this change, if prisoner 2 confesses, then prisoner 1 should also confess; but "don't confess" has become prisoner 1's best strategy if prisoner 2 plays "don't confess." Thus, we are unable to rule out either of prisoner 1's strategies as being dominated, and so elimination of strictly dominated (or, for that matter, weakly dominated) strategies does not lead to a unique prediction.

However, we can still derive a unique prediction in this game if we push the logic of eliminating strictly dominated strategies further. Note that "don't confess" is still strictly dominated for prisoner 2. Furthermore, once prisoner 1 eliminates "don't confess" as a possible action by prisoner 2, "confess" is prisoner 1's unambiguously optimal action; that is, it is his strictly dominant strategy once the strictly dominated strategy of prisoner 2 has been deleted. Thus, the unique predicted outcome in the DA's Brother game should still be (confess, confess). ■

Note the way players' common knowledge of each other's payoffs and rationality is used to solve the game in Example 8.B.3. Elimination of strictly dominated strategies requires only that each player be rational. What we have just done, however, requires not only that prisoner 2 be rational but also that prisoner 1 *know* that prisoner 2 is rational. Put somewhat differently, a player need not know anything about his opponents' payoffs or be sure of their rationality to eliminate a strictly dominated strategy from consideration as his own strategy choice; but for the player to eliminate one of his strategies from consideration because it is dominated if his opponents never play *their* dominated strategies *does* require this knowledge.

As a general matter, if we are willing to assume that all players are rational *and* that this fact and the players' payoffs are common knowledge (so everybody knows that everybody knows that . . . everybody is rational), then we do not need to stop after only two iterations. We can eliminate not only strictly dominated strategies and strategies that are strictly dominated after the first deletion of strategies but also strategies that are strictly dominated after this *next* deletion of strategies, and so on. Note that with each elimination of strategies, it becomes possible for additional strategies to become dominated because the fewer strategies that a player's opponents might play, the more likely that a particular strategy of his is dominated. However, each additional iteration requires that players' knowledge of each others' rationality be one level deeper. A player must now know not only that his rivals are rational but also that they know that he is, and so on.

One feature of the process of iteratively eliminating strictly dominated strategies is that the order of deletion does not affect the set of strategies that remain in the end (see Exercise 8.B.4), That is, if at any given point several strategies (of one or

several players) are strictly dominated, then we can eliminate them all at once or in any sequence without changing the set of strategies that we ultimately end up with. This is fortunate, since we would worry if our prediction depended on the arbitrarily chosen order of deletion.

Exercise 8.B.5 presents an interesting example of a game for which the iterated removal of strictly dominated strategies yields a unique prediction: the *Cournot duopoly game* (which we will discuss in detail in Chapter 12).

The iterated deletion of *weakly* dominated strategies is harder to justify. As we have already indicated, the argument for deletion of a weakly dominated strategy for player i is that he contemplates the possibility that every strategy combination of his rivals occurs with positive probability. However, this hypothesis clashes with the logic of iterated deletion, which assumes, precisely, that eliminated strategies are not expected to occur. This inconsistency leads the iterative elimination of weakly dominated strategies to have the undesirable feature that it *can* depend on the order of deletion. The game in Figure 8.B.3 provides an example. If we first eliminate strategy U, we next eliminate strategy L, and we can then eliminate strategy M; (D, R) is therefore our prediction. If, instead, we eliminate strategy M first, we next eliminate strategy R, and we can then eliminate strategy U; now (D, L) is our prediction.

Allowing for Mixed Strategies

When we recognize that players may randomize over their pure strategies, the basic definitions of strictly dominated and dominant strategies can be generalized in a straightforward way.

Definition 8.B.4: A strategy $\sigma_i \in \Delta(S_i)$ is *strictly dominated* for player i in game $\Gamma_N = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$ if there exists another strategy $\sigma_i' \in \Delta(S_i)$ such that for all $\sigma_{-i} \in \prod_{j \neq i} \Delta(S_j)$,

$$u_i(\sigma_i', \sigma_{-i}) > u_i(\sigma_i, \sigma_{-i}).$$

In this case, we say that strategy σ_i' *strictly dominates* strategy σ_i. A strategy σ_i is a *strictly dominant strategy* for player i in game $\Gamma_N = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$ if it strictly dominates every other strategy in $\Delta(S_i)$.

Using this definition and the structure of mixed strategies, we can say a bit more about the set of strictly dominated strategies in game $\Gamma_N = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$.

Note first that when we test whether a strategy σ_i is strictly dominated by strategy σ_i' for player i, we need only consider these two strategies' payoffs against the *pure* strategies of i's opponents. That is,

$$u_i(\sigma_i', \sigma_{-i}) > u_i(\sigma_i, \sigma_{-i}) \quad \text{for all } \sigma_{-i}$$

if and only if

$$u_i(\sigma_i', s_{-i}) > u_i(\sigma_i, s_{-i}) \quad \text{for all } s_{-i}.$$

This follows because we can write

$$u_i(\sigma_i', \sigma_{-i}) - u_i(\sigma_i, \sigma_{-i}) = \sum_{s_{-i} \in S_{-i}} \left[\prod_{k \neq i} \sigma_k(s_k) \right] [u_i(\sigma_i', s_{-i}) - u_i(\sigma_i, s_{-i})].$$

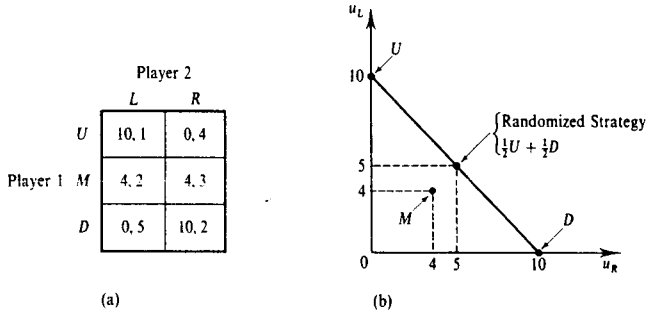


Figure 8.B.5
Domination of a pure strategy by a randomized strategy.

This expression is positive for all σ_{-i} if and only if $[u_i(\sigma_i', s_{-i}) - u_i(\sigma_i, s_{-i})]$ is positive for all s_{-i}. One implication of this point is presented in Proposition 8.B.1.

Proposition 8.B.1: Player i's pure strategy $s_i \in S_i$ is strictly dominated in game $\Gamma_N = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$ if and only if there exists another strategy $\sigma_i' \in \Delta(S_i)$ such that

$$u_i(\sigma_i', s_{-i}) > u_i(s_i, s_{-i})$$

for all $s_{-i} \in S_{-i}$.

Proposition 8.B.1 tells us that to test whether a pure strategy s_i is dominated when randomized play is possible, the test given in Definition 8.B.2 need only be augmented by checking whether any of player i's mixed strategies does better than s_i against every possible profile of pure strategies by i's rivals.

In fact, this extra requirement can eliminate additional pure strategies because a pure strategy s_i may be dominated only by a randomized combination of other pure strategies; that is, to dominate a strategy, even a pure one, it may be necessary to consider alternative strategies that involve randomization. To see this, consider the two-player game depicted in Figure 8.B.5(a). Player 1 has three strategies: U, M, and D. We can see that U is an excellent strategy when player 2 plays L but a poor one against R and that D is excellent against R and poor against L. Strategy M, on the other hand, is a good but not great strategy against both L and R. None of these three pure strategies is strictly dominated by any of the others. But if we allow player 1 to randomize, then playing U and D each with probability $\frac{1}{2}$ yields player 1 an expected payoff of 5 regardless of player 2's strategy, strictly dominating M (remember, payoffs are levels of von Neumann–Morgenstern utilities). This is shown in Figure 8.B.5(b), where player 1's expected payoffs from playing U, D, M, and the randomized strategy $\frac{1}{2}U + \frac{1}{2}D$ are plotted as points in $\mathbb{R}^2$ (the two dimensions correspond to a strategy's expected payoff for player 1 when player 2 plays R, denoted by u_R, and L, denoted by u_L). In the figure, the payoff vectors achievable by randomizing over U and D, and that from the randomized strategy $\frac{1}{2}U + \frac{1}{2}D$ in particular, lie on the line connecting points (0, 10) and (10, 0). As can be seen, the payoffs from $\frac{1}{2}U + \frac{1}{2}D$ strictly dominate those from strategy M.

Once we have determined the set of undominated pure strategies for player i, we need to consider which mixed strategies are undominated. We can immediately eliminate any mixed strategy that uses a dominated pure strategy; if pure strategy s_i is strictly dominated for player i, then so is every mixed strategy that assigns a positive probability to this strategy.

Exercise 8.B.6: Prove that if pure strategy s_i is a strictly dominated strategy in game $\Gamma_N = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$, then so is any strategy that plays s_i with positive probability.

8.D Nash Equilibrium

In this section, we present and discuss the most widely used solution concept in applications of game theory to economics, that of *Nash equilibrium* [due to Nash (1951)]. Throughout the rest of the book, we rely on it extensively.

For ease of exposition, we initially ignore the possibility that players might randomize over their pure strategies, restricting our attention to game $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$. Mixed strategies are introduced later in the section.

We begin with Definition 8.D.1.

Definition 8.D.1: A strategy profile $s = (s_1, \ldots, s_I)$ constitutes a *Nash equilibrium* of game $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ if for every $i = 1, \ldots, I$,

$$u_i(s_i, s_{-i}) \geq u_i(s_i', s_{-i})$$

for all $s_i' \in S_i$.

In a Nash equilibrium, each player's strategy choice is a best response (see Definition 8.C.1) to the strategies *actually played* by his rivals. The italicized words distinguish the concept of Nash equilibrium from the concept of rationalizability studied in Section 8.C. Rationalizability, which captures the implications of the players' common knowledge of each others' rationality and the structure of the game, requires only that a player's strategy be a best response to some reasonable conjecture about what his rivals will be playing, where *reasonable* means that the conjectured play of his rivals can also be so justified. Nash equilibrium adds to this the requirement that players be *correct* in their conjectures.

Examples 8.D.1 and 8.D.2 illustrate the use of the concept.

Example 8.D.1: Consider the two-player simultaneous-move game shown in Figure 8.D.1. We can see that (M, m) is a Nash equilibrium. If player 1 chooses M, then the best response of player 2 is to choose m; the reverse is true for player 2. Moreover, (M, m) is the only combination of (pure) strategies that is a Nash equilibrium. For example, strategy profile (U, r) cannot be a Nash equilibrium because player 1 would prefer to deviate to strategy D given that player 2 is playing r. (Check the other possibilities for yourself.) ■

Example 8.D.2: *Nash Equilibrium in the Game of Figure 8.C.1.* In this game, the unique Nash equilibrium profile of (pure) strategies is (a_2, b_2). Player 1's best response to b_2 is a_2, and player 2's best response to a_2 is b_2, so (a_2, b_2) is a Nash equilibrium.

		Player 2		
		ℓ	m	r
Player 1	U	5, 3	0, 4	3, 5
	M	4, 0	(5, 5)	4, 0
	D	3, 5	0, 4	5, 3

Figure 8.D.1
A Nash equilibrium.

		Mr. Schelling	
		Empire State	Grand Central
Mr. Thomas	Empire State	(100, 100)	0, 0
	Grand Central	0, 0	(100, 100)

Figure 8.D.2
Nash equilibria in the Meeting in New York game.

At any other strategy profile, one of the players has an incentive to deviate. [In fact, (a_2, b_2) is the unique Nash equilibrium even when randomization is permitted; see Exercise 8.D.1.]

This example illustrates a general relationship between the concept of Nash equilibrium and that of rationalizable strategies: *Every strategy that is part of a Nash equilibrium profile is rationalizable* because each player's strategy in a Nash equilibrium can be justified by the Nash equilibrium strategies of the other players. Thus, as a general matter, the Nash equilibrium concept offers at least as sharp a prediction as does the rationalizability concept. In fact, it often offers a *much* sharper prediction. In the game of Figure 8.C.1, for example, the rationalizable strategies a_1, a_3, b_1, and b_3 are eliminated as predictions because they cannot be sustained when players' beliefs about each other's play are required to be correct. ■

In the previous two examples, the Nash equilibrium concept yields a unique prediction. However, this is not always the case. Consider the Meeting in New York game.

Example 8.D.3: *Nash Equilibria in the Meeting in New York Game.* Figure 8.D.2 depicts a simple version of the Meeting in New York game. Mr. Thomas and Mr. Schelling each have two choices: They can meet either at noon at the top of the Empire State Building or at noon at the clock in Grand Central Station. There are two Nash equilibria (ignoring the possibility of randomization): (Empire State, Empire State) and (Grand Central, Grand Central). ■

Example 8.D.3 emphasizes how strongly the Nash equilibrium concept uses the assumption of mutually correct expectations. The theory of Nash equilibrium is silent on *which* equilibrium we should expect to see when there are many. Yet, the players are assumed to correctly forecast which one it will be.

A compact restatement of the definition of a Nash equilibrium can be obtained through the introduction of the concept of a player's *best-response correspondence*. Formally, we say that player i's best-response correspondence $b_i: S_{-i} \to S_i$ in the game $\Gamma_N = [I, \{S_i\}\ \{u_i(\cdot)\}]$, is the correspondence that assigns to each $s_{-i} \in S_{-i}$ the set

$$b_i(s_{-i}) = \{s_i \in S_i: u_i(s_i, s_{-i}) \geq u_i(s_i', s_{-i}) \text{ for all } s_i' \in S_i\}.$$

With this notion, we can restate the definition of a Nash equilibrium as follows: The strategy profile $(s_1, \ldots, s_I)$ is a Nash equilibrium of game $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ if and only if $s_i \in b_i(s_{-i})$ for $i = 1, \ldots, I$.

Discussion of the Concept of Nash Equilibrium

Why might it be reasonable to expect players' conjectures about each other's play to be correct? Or, in sharper terms, why should we concern ourselves with the concept of Nash equilibrium?

A number of arguments have been put forward for the Nash equilibrium concept and you will undoubtedly react to them with varying degrees of satisfaction. Moreover, one argument might seem compelling in one application but not at all convincing in another. Until very recently, all these arguments have been informal, as will be our discussion. The issue is one of the more important open questions in game theory, particularly given the Nash equilibrium concept's widespread use in applied problems, and it is currently getting some formal attention.

(i) *Nash equilibrium as a consequence of rational inference.* It is sometimes argued that because each player can think through the strategic considerations faced by his opponents, rationality alone implies that players must be able to correctly forecast what their rivals will play. Although this argument may seem appealing, it is faulty. As we saw in Section 8.C, the implication of common knowledge of the players' rationality (and of the game's structure) is precisely that each player must play a rationalizable strategy. Rationality need not lead players' forecasts to be correct.

(ii) *Nash equilibrium as a necessary condition if there is a unique predicted outcome to a game.* A more satisfying version of the previous idea argues that if there is a unique predicted outcome for a game, then rational players will understand this. Therefore, for no player to wish to deviate, this predicted outcome must be a Nash equilibrium. Put somewhat differently [as in Kreps (1990)], if players think and share the belief that there is an *obvious* (in particular, a unique) way to play a game, then it must be a Nash equilibrium.

Of course, this argument is only relevant if there is a unique prediction for how players will play a game. The discussion of rationalizability in Section 8.C, however, shows that common knowledge of rationality alone does not imply this. Therefore, this argument is really useful only in conjunction with some reason why a particular profile of strategies might be the obvious way to play a particular game. The other arguments for Nash equilibrium that we discuss can be viewed as combining this argument with a reason why there might be an "obvious" way to play a game.

(iii) *Focal points.* It sometimes happens that certain outcomes are what Schelling (1960) calls *focal*. For example, take the Meeting in New York game depicted in Figure 8.D.2, and suppose that restaurants in the Grand Central area are so much better then those around the Empire State Building that the payoffs to meeting at Grand Central are (1000, 1000) rather than (100, 100). Suddenly, going to Grand Central seems like the obvious thing to do. Focal outcomes could also be culturally determined. As Schelling pointed out in his original discussion, two people who do not live in New York will tend to find meeting at the top of the Empire State building (a famous tourist site) to be focal, whereas two native New Yorkers will find Grand Central Station (the central railroad station) a more compelling choice. In both examples, one of the outcomes has a natural appeal. The implication of argument (ii) is that this kind of appeal can lead an outcome to be the clear prediction in a game only if the outcome is a Nash equilibrium.

(iv) *Nash equilibrium as a self-enforcing agreement.* Another argument for Nash equilibrium comes from imagining that the players can engage in nonbinding communication prior to playing the game. If players agree to an outcome to be played, this naturally becomes the obvious candidate for play. However, because players cannot bind themselves to their agreed-upon strategies, any agreement that the players reach must be self-enforcing if it is to be meaningful. Hence, any meaningful agreement must involve the play of a Nash equilibrium strategy profile. Of course, even though players have reached an agreement to play a Nash equilibrium, they could still deviate from it if they expect others to do so. In essence, this justification assumes that once the players have agreed to a choice of strategies, this agreement becomes focal.

(v) *Nash equilibrium as a stable social convention.* A particular way to play a game might arise over time if the game is played repeatedly and some stable social convention emerges. If it does, it may be "obvious" to all players that the convention will be maintained. The convention, so to speak, becomes focal.

A good example is the game played by New Yorkers every day: Walking in Downtown Manhattan. Every day, people who walk to work need to decide which side of the sidewalk they will walk on. Over time, the stable social convention is that everyone walks on the right side, a convention that is enforced by the fact that any individual who unilaterally deviates from it is sure to be severely trampled. Of course, on any given day, it is *possible* that an individual might decide to walk on the left by conjecturing that everyone else suddenly expects the convention to change. Nevertheless, the prediction that we will remain at the Nash equilibrium "everyone walks on the right" seems reasonable in this case. Note that if an outcome is to become a stable social convention, it must be a Nash equilibrium. If it were not, then individuals would deviate from it as soon as it began to emerge.

The notion of an equilibrium as a rest point for some dynamic adjustment process underlies the use and the traditional appeal of equilibrium notions in economics. In this sense, the stable social convention justification of Nash equilibrium is closest to the tradition of economic theory.

To formally model the emergence of stable social conventions is not easy. One difficulty is that the repeated one-day game may itself be viewed as a larger dynamic game. Thus, when we consider rational players choosing their strategies in this overall game, we are merely led back to our original conundrum: Why should we expect a Nash equilibrium in this larger game? One response to this difficulty currently getting some formal attention imagines that players follow simple rules of thumb concerning their opponents' likely play in situations where play is repeated (note that this implies a certain withdrawal from the assumption of complete rationality). For example, a player could conjecture that whatever his opponents did yesterday will be repeated today. If so, then each day players will play a best response to yesterday's play. If a combination of strategies arises that is a stationary point of this process (i.e., the

		Player 2: Heads	Player 2: Tails
Player 1	Heads	−1, +1	+1, −1
	Tails	+1, −1	−1, +1

Figure 8.D.3
Matching Pennies.

play today is the same as it was yesterday), it must be a Nash equilibrium. However, it is less clear that from any initial position, the process will converge to a stationary outcome; convergence turns out to depend on the game.[4]

Mixed Strategy Nash Equilibria

It is straightforward to extend the definition of Nash equilibrium to games in which we allow the players to randomize over their pure strategies.

Definition 8.D.2: A mixed strategy profile $\sigma = (\sigma_1, \ldots, \sigma_I)$ constitutes a *Nash equilibrium* of game $\Gamma_N = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$ if for every $i = 1, \ldots, I$,

$$u_i(\sigma_i, \sigma_{-i}) \geq u_i(\sigma_i', \sigma_{-i})$$

for all $\sigma_i' \in \Delta(S_i)$.

Example 8.D.4: As a very simple example, consider the standard version of Matching Pennies depicted in Figure 8.D.3. This is a game with no pure strategy equilibrium. On the other hand, it is fairly intuitive that there is a mixed strategy equilibrium in which each player chooses H or T with equal probability. When a player randomizes in this way, it makes his rival indifferent between playing heads or tails, and so his rival is also willing to randomize between heads and tails with equal probability. ■

It is not an accident that a player who is randomizing in a Nash equilibrium of Matching Pennies is indifferent between playing heads and tails. As Proposition 8.D.1 confirms, this indifference among strategies played with positive probability is a general feature of mixed strategy equilibria.

Proposition 8.D.1: Let $S_i^+ \subset S_i$ denote the set of pure strategies that player i plays with positive probability in mixed strategy profile $\sigma = (\sigma_1, \ldots, \sigma_I)$. Strategy profile σ is a Nash equilibrium in game $\Gamma_N = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$ if and only if for all $i = 1, \ldots, I$,

(i) $u_i(s_i, \sigma_{-i}) = u_i(s_i', \sigma_{-i})$ for all $s_i, s_i' \in S_i^+$;
(ii) $u_i(s_i, \sigma_{-i}) \geq u_i(s_i', \sigma_{-i})$ for all $s_i \in S_i^+$ and all $s_i' \notin S_i^+$.

Proof: For necessity, note that if either of conditions (i) or (ii) does not hold for some player i, then there are strategies $s_i \in S_i^+$ and $s_i' \in S_i$ such that $u_i(s_i', \sigma_{-i}) > u_i(s_i, \sigma_{-i})$. If so, player i can strictly increase his payoff by playing strategy s_i' whenever he would have played strategy s_i.

4. This approach actually dates to Cournot's (1838) myopic adjustment procedure. A recent example can be found in Milgrom and Roberts (1990). Interestingly, this work explains the "ultrarational" Nash outcome by *relaxing* the assumption of rationality. It also can be used to try to identify the likelihood of various Nash equilibria arising when multiple Nash equilibria exist.

For sufficiency, suppose that conditions (i) and (ii) hold but that σ is not a Nash equilibrium. Then there is some player i who has a strategy σ_i' with $u_i(\sigma_i', \sigma_{-i}) > u_i(\sigma_i, \sigma_{-i})$. But if so, then there must be some pure strategy s_i' that is played with positive probability under σ_i' for which $u_i(s_i', \sigma_{-i}) > u_i(\sigma_i, \sigma_{-i})$. Since $u_i(\sigma_i, \sigma_{-i}) = u_i(s_i, \sigma_{-i})$ for all $s_i \in S_i^+$, this contradicts conditions (i) and (ii) being satisfied. ■

Hence, a necessary and sufficient condition for mixed strategy profile σ to be a Nash equilibrium of game $\Gamma_N = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$ is that each player, given the distribution of strategies played by his opponents, is indifferent among all the pure strategies that he plays with positive probability and that these pure strategies are at least as good as any pure strategy he plays with zero probability.

An implication of Proposition 8.D.1 is that to test whether a strategy profile σ is a Nash equilibrium it suffices to consider only pure strategy deviations (i.e., changes in a player's strategy σ_i to some pure strategy s_i'). As long as no player can improve his payoff by switching to any pure strategy, σ is a Nash equilibrium. We therefore get the comforting result given in Corollary 8.D.1.

Corollary 8.D.1: Pure strategy profile $s = (s_1, \ldots, s_I)$ is a Nash equilibrium of game $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ if and only if it is a (degenerate) mixed strategy Nash equilibrium of game $\Gamma_N' = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$.

Corollary 8.D.1 tells us that to identify the pure strategy equilibria of game $\Gamma_N' = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$, it suffices to restrict attention to the game $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ in which randomization is not permitted.

Proposition 8.D.1 can also be of great help in the computation of mixed strategy equilibria as Example 8.D.5 illustrates.

Example 8.D.5: *Mixed Strategy Equilibria in the Meeting in New York Game.* Let us try to find a mixed strategy equilibrium in the variation of the Meeting in New York game where the payoffs of meeting at Grand Central are (1000, 1000). By Proposition 8.D.1, if Mr. Thomas is going to randomize between Empire State and Grand Central, he must be indifferent between them. Suppose that Mr. Schelling plays Grand Central with probability σ_s. Then Mr. Thomas' expected payoff from playing Grand Central is $1000\sigma_s + 0(1 - \sigma_s)$, and his expected payoff from playing Empire State is $100(1 - \sigma_s) + 0\sigma_s$. These two expected payoffs are equal only when $\sigma_s = 1/11$. Now, for Mr. Schelling to set $\sigma_s = 1/11$, he must also be indifferent between his two pure strategies. By a similar argument, we find that Mr. Thomas' probability of playing Grand Central must also be 1/11. We conclude that each player going to Grand Central with a probability of 1/11 is a Nash equilibrium. ■

Note that in accordance with Proposition 8.D.1, the players in Example 8.D.5 have no real preference over the probabilities that they assign to the pure strategies they play with positive probability. What determines the probabilities that each player uses is an equilibrium consideration: the need to make the *other* player indifferent over *his* strategies.

This fact has led some economists and game theorists to question the usefulness of mixed strategy Nash equilibria as predictions of play. They raise two concerns: First, if players always have a pure strategy that gives them the same expected payoff as their equilibrium mixed strategy, it is not clear why they will bother to randomize.

One answer to this objection is that players may not actually randomize. Rather, they may make definite choices that are affected by seemingly inconsequential variables ("signals") that only they observe. For example, consider how a pitcher for a major league baseball team "mixes his pitches" to keep batters guessing. He may have a completely deterministic plan for what he will do, but it may depend on which side of the bed he woke up on that day or on the number of red traffic lights he came to on his drive to the stadium. As a result, batters view the behavior of the pitcher as random even though it is not. We touched briefly on this interpretation of mixed strategies as behavior contingent on realizations of a signal in Section 7.E, and we will examine it in more detail in Section 8.E.

The second concern is that the stability of mixed strategy equilibria seems tenuous. Players must randomize with exactly the correct probabilities, but they have no positive incentive to do so. One's reaction to this problem may depend on why one expects a Nash equilibrium to arise in the first place. For example, the use of the correct probabilities may be unlikely to arise as a stable social convention, but may seem more plausible when the equilibrium arises as a self-enforcing agreement.

Up to this point, we have assumed that players' randomizations are independent. In the Meeting in New York game in Example 8.D.5, for instance, we could describe a mixed strategy equilibrium as follows: Nature provides *private and independently distributed* signals $(\theta_1, \theta_2) \in [0, 1] \times [0, 1]$ to the two players, and each player i assigns decisions to the various possible realizations of his signal θ_i.

However, suppose that there are also *public* signals available that both players observe. Let $\theta \in [0, 1]$ be such a signal. Then many new possibilities arise. For example, the two players could both decide to go to Grand Central if $\theta < \frac{1}{2}$ and to Empire State if $\theta \geq \frac{1}{2}$. Each player's strategy choice is still random, but the coordination of their actions is now perfect and they always meet. More importantly, the decisions have an equilibrium character. If one player decides to follow this decision rule, then it is also optimal for the other player to do so. This is an example of a *correlated equilibrium* [due to Aumann (1974)]. More generally, we could allow for correlated equilibria in which nature's signals are partly private and partly public.

Allowing for such correlation may be important because economic agents observe many public signals. Formally, a correlated equilibrium is a special case of a Bayesian Nash equilibrium, a concept that we introduce in Section 8.E; hence, we defer further discussion to the end of that section.

Existence of Nash Equilibria

Does a Nash equilibrium necessarily exist in a game? Fortunately, the answer turns out to be "yes" under fairly broad circumstances. Here we describe two of the more important existence results; their proofs, based on mathematical fixed point theorems, are given in Appendix A of this chapter. (Proposition 9.B.1 of Section 9.B provides another existence result.)

Proposition 8.D.2: Every game $\Gamma_N = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$ in which the sets $S_1, \ldots, S_I$ have a finite number of elements has a mixed strategy Nash equilibrium.

Thus, for the class of games we have been considering, a Nash equilibrium always exists as long as we are willing to accept equilibria in which players randomize. (If you want to be convinced without going through the proof, try Exercise 8.D.6.) Allowing for randomization is essential for this result. We have already seen in (standard) Matching Pennies, for example, that a pure strategy equilibrium may not exist in a game with a finite number of pure strategies.

Up to this point, we have focused on games with finite strategy sets. However, in economic applications, we frequently encounter games in which players have strategies naturally modeled as continuous variables. This can be helpful for the existence of a pure strategy equilibrium. In particular, we have the result given in Proposition 8.D.3.

Proposition 8.D.3: A Nash equilibrium exists in game $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ if for all $i = 1, \ldots, I$,

(i) S_i is a nonempty, convex, and compact subset of some Euclidean space $\mathbb{R}^M$.

(ii) $u_i(s_1, \ldots, s_I)$ is continuous in $(s_1, \ldots, s_I)$ and quasiconcave in s_i.

Proposition 8.D.3 provides a significant result whose requirements are satisfied in a wide range of economic applications. The convexity of strategy sets and the nature of the payoff functions help to smooth out the structure of the model, allowing us to achieve a pure strategy equilibrium.[5]

Further existence results can also be established. In situations where quasiconcavity of the payoff functions $u_i(\cdot)$ fails but they are still continuous, existence of a mixed strategy equilibrium can still be demonstrated. In fact, even if continuity of the payoff functions fails to hold, a mixed strategy equilibrium can be shown to exist in a variety of cases [see Dasgupta and Maskin (1986)].

Of course, these results do not mean that we *cannot* have an equilibrium if the conditions of these existence results do not hold. Rather, we just cannot be *assured* that there is one.

8.E Games of Incomplete Information: Bayesian Nash Equilibrium

Up to this point, we have assumed that players know all relevant information about each other, including the payoffs that each receives from the various outcomes of the game. Such games are known as games of *complete information*. A moment of thought, however, should convince you that this is a very strong assumption. Do two firms in an industry necessarily know each other's costs? Does a firm bargaining with a union necessarily know the disutility that union members will feel if they go out on strike for a month? Clearly, the answer is "no." Rather, in many circumstances, players have what is known as *incomplete information*.

The presence of incomplete information raises the possibility that we may need to consider a player's beliefs about other players' preferences, his beliefs about their beliefs about his preferences, and so on, much in the spirit of rationalizability.[6]

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5. Note that a finite strategy set S_i cannot be convex. In fact, the use of mixed strategies in Proposition 8.D.2 helps us to obtain existence of equilibrium in much the same way that Proposition 8.D.3's assumptions assure existence of a pure strategy Nash equilibrium: It convexifies players' strategy sets and yields well-behaved payoff functions. (See Appendix A for details.)

6. For more on this problem, see Mertens and Zamir (1985).

EXERCISES

8.B.1[A] There are I firms in an industry. Each can try to convince Congress to give the industry a subsidy. Let h_i denote the number of hours of effort put in by firm i, and let $c_i(h_i) = w_i(h_i)^2$, where w_i is a positive constant, be the cost of this effort to firm i. When the effort levels of the firms are $(h_1, \ldots, h_I)$, the value of the subsidy that gets approved is $\alpha\sum_i h_i + \beta(\prod_i h_i)$, where α and β are constants.

Consider a game in which the firms decide simultaneously and independently how many hours they will each devote to this effort. Show that each firm has a strictly dominant strategy if and only if $\beta = 0$. What is firm i's strictly dominant strategy when this is so?

8.B.2[B] **(a)** Argue that if a player has two weakly dominant strategies, then for every strategy choice by his opponents, the two strategies yield him equal payoffs.

(b) Provide an example of a two-player game in which a player has two weakly dominant pure strategies but his opponent prefers that he play one of them rather than the other.

8.B.3[B] Consider the following auction (known as a *second-price*, or *Vickrey*, auction). An object is auctioned off to I bidders. Bidder i's valuation of the object (in monetary terms) is v_i. The auction rules are that each player submit a bid (a nonnegative number) in a sealed envelope. The envelopes are then opened, and the bidder who has submitted the highest bid gets the object but pays the auctioneer the amount of the *second-highest* bid. If more than one bidder submits the highest bid, each gets the object with equal probability. Show that submitting a bid of v_i with certainty is a weakly dominant strategy for bidder i. Also argue that this is bidder i's unique weakly dominant strategy.

8.B.4[C] Show that the order of deletion does not matter for the set of strategies surviving a process of iterated deletion of strictly dominated strategies.

8.B.5[C] Consider the Cournot duopoly model (discussed extensively in Chapter 12) in which two firms, 1 and 2, simultaneously choose the quantities they will sell on the market, q_1 and q_2. The price each receives for each unit given these quantities is $P(q_1, q_2) = a - b(q_1 + q_2)$. Their costs are c per unit sold.

(a) Argue that successive elimination of strictly dominated strategies yields a unique prediction in this game.

(b) Would this be true if there were three firms instead of two?

8.B.6[B] In text.

8.B.7[B] Show that any strictly dominant strategy in game $[I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$ must be a pure strategy.

8.C.1[A] Argue that if elimination of strictly dominated strategies yields a unique prediction in a game, this prediction also results from eliminating strategies that are never a best response.

8.C.2[C] Prove that the order of removal does not matter for the set of strategies that survives a process of iterated deletion of strategies that are never a best response.

8.C.3[C] Prove that in a two-player game (with finite strategy sets), if a pure strategy s_i for player i is never a best response for any mixed strategy by i's opponent, then s_i is strictly dominated by some mixed strategy $\sigma_i \in \Delta(S_i)$. [*Hint:* Try using the supporting hyperplane theorem presented in Section M.G of the Mathematical Appendix.]

8.C.4[B] Consider a game Γ_N with players 1, 2, and 3 in which $S_1 = \{L, M, R\}$, $S_2 = \{U, D\}$, and $S_3 = \{\ell, r\}$. Player 1's payoffs from each of his three strategies conditional on the strategy choices of players 2 and 3 are depicted as (u_L, u_M, u_R) in each of the four boxes shown below, where $(\pi, \varepsilon, \eta) \gg 0$. Assume that $\eta < 4\varepsilon$.

Player 2's Strategy \ Player 3's Strategy	ℓ	r
U	$\pi + 4\varepsilon, \pi - \eta, \pi - 4\varepsilon$	$\pi - 4\varepsilon, \pi + \frac{\eta}{2}, \pi + 4\varepsilon$
D	$\eta + 4\varepsilon, \pi + \frac{\eta}{2}, \pi - 4\varepsilon$	$\pi - 4\varepsilon, \pi - \eta, \pi + 4\varepsilon$

(a) Argue that (pure) strategy M is never a best response for player 1 to any independent randomizations by players 2 and 3.

(b) Show that (pure) strategy M is not strictly dominated.

(c) Show that (pure) strategy M can be a best response if player 2's and player 3's randomizations are allowed to be correlated.

8.D.1[B] Show that (a_2, b_2) being played with certainty is the unique mixed strategy Nash equilibrium in the game depicted in Figure 8.C.1.

8.D.2[B] Show that if there is a unique profile of strategies that survives iterated removal of strictly dominated strategies, this profile is a Nash equilibrium.

8.D.3[B] Consider a first-price sealed-bid auction of an object with two bidders. Each bidder i's valuation of the object is v_i, which is known to both bidders. The auction rules are that each player submits a bid in a sealed envelope. The envelopes are then opened, and the bidder who has submitted the highest bid gets the object and pays the auctioneer the amount of his bid. If the bidders submit the same bid, each gets the object with probability $\frac{1}{2}$. Bids must be in dollar multiples (assume that valuations are also).

(a) Are any strategies strictly dominated?

(b) Are any strategies weakly dominated?

(c) Is there a Nash equilibrium? What is it? Is it unique?

8.D.4[B] Consider a bargaining situation in which two individuals are considering undertaking a business venture that will earn them 100 dollars in profit, but they must agree on how to split the 100 dollars. Bargaining works as follows: The two individuals each make a demand simultaneously. If their demands sum to more than 100 dollars, then they fail to agree, and each gets nothing. If their demands sum to less than 100 dollars, they do the project, each gets his demand, and the rest goes to charity.

(a) What are each player's strictly dominated strategies?

(b) What are each player's weakly dominated strategies?

(c) What are the pure strategy Nash equilibria of this game?

8.D.5[B] Consumers are uniformly distributed along a boardwalk that is 1 mile long. Ice-cream prices are regulated, so consumers go to the nearest vendor because they dislike walking (assume that at the regulated prices all consumers will purchase an ice cream even if they

have to walk a full mile). If more than one vendor is at the same location, they split the business evenly.

(a) Consider a game in which two ice-cream vendors pick their locations simultaneously. Show that there exists a unique pure strategy Nash equilibrium and that it involves both vendors locating at the midpoint of the boardwalk.

(b) Show that with three vendors, no pure strategy Nash equilibrium exists.

8.D.6[B] Consider any two-player game of the following form (where letters indicate arbitrary payoffs):

Player 1 \ Player 2	b_1	b_2
a_1	u, v	ℓ, m
a_2	w, x	y, z

Show that a mixed strategy Nash equilibrium always exists in this game. [*Hint:* Define player 1's strategy to be his probability of choosing action a_1 and player 2's to be his probability of choosing b_1; then examine the best-response correspondences of the two players.]

8.D.7[C] (*The Minimax Theorem*) A two-player game with finite strategy sets $\Gamma_N = [I, \{S_1, S_2\}, \{u_1(\cdot), u_2(\cdot)\}]$ is a *zero-sum* game if $u_2(s_1, s_2) = -u_1(s_1, s_2)$ for all $(s_1, s_2) \in S_1 \times S_2$.

Define i's *maximin* expected utility level $\underline{w}_i$ to be the level he can guarantee himself in game $[I, \{\Delta(S_1), \Delta(S_2)\}, \{u_1(\cdot), u_2(\cdot)\}]$:

$$\underline{w}_i = \operatorname*{Max}_{\sigma_i} \left[\operatorname*{Min}_{\sigma_{-i}} u_i(\sigma_i, \sigma_{-i}) \right].$$

Define player i's *minimax* utility level $\underline{v}_i$ to be the worst expected utility level he can be forced to receive if he gets to respond to his rival's actions:

$$\underline{v}_i = \operatorname*{Min}_{\sigma_{-i}} \left[\operatorname*{Max}_{\sigma_i} u_i(\sigma_i, \sigma_{-i}) \right].$$

(a) Show that $\underline{v}_i \geq \underline{w}_i$ in any game.

(b) Prove that in any mixed strategy Nash equilibrium of the zero-sum game $\Gamma_N = [I, \{\Delta(S_1), \Delta(S_2)\}, \{u_1(\cdot), u_2(\cdot)\}]$, player i's expected utility u_i^* satisfies $u_i^* = \underline{v}_i = \underline{w}_i$. [*Hint:* Such an equilibrium must exist by Proposition 8.D.2.]

(c) Show that if (σ_1', σ_2') and (σ_1'', σ_2'') are both Nash equilibria of the zero-sum game $\Gamma_N = [I, \{\Delta(S_1), \Delta(S_2)\}, \{u_1(\cdot), u_2(\cdot)\}]$, then so are (σ_1', σ_2'') and (σ_1'', σ_2').

8.D.8[C] Consider a simultaneous-move game with normal form $[I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$. Suppose that, for all i, S_i is a convex set and $u_i(\cdot)$ is strictly quasiconvex. Argue that any mixed strategy Nash equilibrium of this game must be degenerate, with each player playing a single pure strategy with probability 1.

8.D.9[B] Consider the following game [based on an example from Kreps (1990)]:

Player 1 \ Player 2	LL	L	M	R
U	100, 2	−100, 1	0, 0	−100, −100
D	−100, −100	100, −49	1, 0	100, 2

(a) If you were player 2 in this game and you were playing it once without the ability to engage in preplay communication with player 1, what strategy would you choose?

(b) What are all the Nash equilibria (pure and mixed) of this game?

(c) Is your strategy choice in **(a)** a component of any Nash equilibrium strategy profile? Is it a rationalizable strategy?

(d) Suppose now that preplay communication were possible. Would you expect to play something different from your choice in **(a)**?

8.E.1[B] Consider the following strategic situation. Two opposed armies are poised to seize an island. Each army's general can choose either "attack" or "not attack." In addition, each army is either "strong" or "weak" with equal probability (the draws for each army are independent), and an army's type is known only to its general. Payoffs are as follows: The island is worth M if captured. An army can capture the island either by attacking when its opponent does not or by attacking when its rival does if it is strong and its rival is weak. If two armies of equal strength both attack, neither captures the island. An army also has a "cost" of fighting, which is s if it is strong and w if it is weak, where $s < w$. There is no cost of attacking if its rival does not.

Identify all pure strategy Bayesian Nash equilibria of this game.

8.E.2[C] Consider the first-price sealed-bid auction of Exercise 8.D.3, but now suppose that each bidder i observes only his own valuation v_i. This valuation is distributed uniformly and independently on $[0, \bar{v}]$ for each bidder.

(a) Derive a symmetric (pure strategy) Bayesian Nash equilibrium of this auction. (You should now suppose that bids can be any real number.) [*Hint:* Look for an equilibrium in which bidder i's bid is a linear function of his valuation.]

(b) What if there are I bidders? What happens to each bidder's equilibrium bid function $s(v_i)$ as I increases?

8.E.3[B] Consider the linear Cournot model described in Exercise 8.B.5. Now, however, suppose that each firm has probability μ of having unit costs of c_L and $(1 - \mu)$ of having unit costs of c_H, where $c_H > c_L$. Solve for the Bayesian Nash equilibrium.

8.F.1[C] Prove Proposition 8.F.1.

8.F.2[B] Consider the following three-player game [taken from van Damme (1983)], in which player 1 chooses rows ($S_1 = \{U, D\}$), player 2 chooses columns ($S_2 = \{L, R\}$), and player 3 chooses boxes ($S_3 = \{B_1, B_2\}$):

B_1

	L	R
U	(1, 1, 1)	(1, 0, 1)
D	(1, 1, 1)	(0, 0, 1)

B_2

	L	R
U	(1, 1, 0)	(0, 0, 0)
D	(0, 1, 0)	(1, 0, 0)

Each cell describes the payoffs to the three players (u_1, u_2, u_3) from that strategy combination. Both (D, L, B_1) and (U, L, B_1) are pure strategy Nash equilibria. Show that (D, L, B_1) is not (normal form) trembling-hand perfect even though none of these three strategies is weakly dominated.

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Exercise 12—IDSDS and Rationality[A]

Consider the following 3 × 3 matrix (Fig. 1.30):

a. Does any player have a strictly dominant strategy?
b. Does any player have a strictly dominated strategy?
c. Use your results from part (b) to delete players' strictly dominated strategies assuming rationality, but not assuming common knowledge of rationality. [Hint: This means that every player can delete the strategies he finds to be strictly dominated but cannot put himself in the shoes of his opponent to delete the strategies his opponent finds to be strictly dominated.]
d. Use your results from part (b) to delete players' strictly dominated strategies assuming common knowledge of rationality. [Hint: This means that players can put themselves in the shoes of their opponents to delete strategies each other finds to be strictly dominated, as in the standard approach of IDSDS.]

Answer

Part (a). Let us start by identifying the strategies of player 1 (the row player) that are strictly dominating the other strategies. When player 1 chooses *F* in the first row, his payoff is 0 when player 2 chooses *F* or 2 when player 2 chooses *C* or *B*. However, when player 1 chooses *C* in the second row, his payoff is 2 when player 2 chooses *F*, 0 if player 2 chooses *C*, and 3 if player 2 chooses *B*. Furthermore, when player 1 chooses *B* in the third row, his payoff is 5 when player 2 chooses *F*, 2 if player 2 chooses *C*, and 2 if player 2 chooses *B*. Hence, there is no strictly dominant strategy for player 1, as there is no strategy that yields an unambiguously higher payoff for player 1 regardless of the strategy that player 2 selects.

Next, we move on to identify the strategies of player 2 (the column player) that are strictly dominating the other strategies. When player 2 chooses *F* in the first column, his payoff is 5 when player 1 chooses *F*, 3 when player 1 chooses *C*, and 0 when player 1 chooses *B*. However, when player 2 chooses *C* in the second column, his payoff is 4 when player 1 chooses *F*, 5 if player 1 chooses *C*, and 4 if player 1 chooses *B*. Furthermore, when player 2 chooses B in the third column, his payoff is

		Player 2		
		F	*C*	*B*
Player 1	*F*	0, 5	2, 4	2, 3
	C	2, 3	0, 5	3, 2
	B	5, 0	2, 4	2, 3

Fig. 1.30 Normal-form representation of a three-players game

3 when player 1 chooses F, 2 if player 1 chooses C, and 3 if player 1 chooses B. Hence, there is no strictly dominant strategy for player 2, as there is no strategy that yields an unambiguously higher payoff for player 2 regardless of the strategy that player 1 selects.

In summary, we could not find any strictly dominant strategies for either player, and then we could not delete any strategies in our application of strict dominance. Mathematically, this is equivalent to saying that we cannot find a strategy $s_i' \in S_i$ for player i, where $i = \{1, 2\}$ that satisfies

$$u_i\left(s_i', s_{-i}\right) > u_i(s_i, s_{-i})$$

for any two strategies $s_i \neq s_i'$, and for all strategies of his opponents $s_{-i} \in S_{-i}$.

Part (b). Starting with the player 1 (the row player), we cannot find a strictly dominated strategy for him by applying the above logic. Specifically, his payoff is higher from B that gives him a payoff of 5 than other strategies when player 2 plays F (see left-hand column of the matrix), but would be indifferent between F and B if player 2 plays C (see middle column) that gives him a payoff of 2, and is higher from C than other strategies when player 2 plays B (see right-hand column of the matrix) that gives him a payoff of 3. In short, we cannot identify a strategy that yields strictly lower payoffs for player 1, regardless of what player 2 does. There does not exist any such strictly dominated strategies that we could delete from player 1's strategy space, i.e., graphically there are no rows that we could delete as never being played by player 1.

A similar argument applies to player 2 (column player). Her payoff is higher from playing F than either C or B when player 1 plays F (see top row of the matrix) that gives her a payoff of 5, and from playing C than either F or B when player 1 plays C or B (see middle and bottom rows of the matrix) that also gives her a payoff of 5. In both cases, player 2 will not play B, such that there exists a strategy for player 2 (that is, column B) that we can identify as strictly dominated, i.e., strategies yielding a strictly lower payoff than another strategy regardless of what player 1 chooses.

In summary, we could find strictly dominated strategies for player 2 but not player 1 for which we can delete. Mathematically, this is equivalent to saying that we can find a strategy $s_i' \in S_i$ for player i, where $i = \{1, 2\}$ that satisfies

$$u_i\left(s_i', s_{-i}\right) < u_i(s_i, s_{-i})$$

for any two strategies $s_i \neq s_i'$, and for all strategy profiles of his opponents, $s_{-i} \in S_{-i}$.

Part (c). Following part (b), after player 2 deletes strategy B, the reduced form matrix is as follows (Fig. 1.31):

Fig. 1.31 Reduced normal-form game

		Player 2	
		F	*C*
Player 1	*F*	0, 5	2, 4
	C	2, 3	0, 5
	B	5, 0	2, 4

That yields the following strategy space assuming rationality for the players without common knowledge:

$$\textit{Strategies surviving rationality} = \{(F,F),(F,C),(C,F),(C,C),(B,F),(B,C)\}$$

Part (d). Now, player 1 can delete strategy *C* because his payoff is higher from *B* than other strategies when player 2 plays *F* (see left-hand column of the matrix), but would be indifferent between *F* and *B* if player 2 plays *C* (see middle column in the original matrix, which now became the right-hand column). In short, we can identify strategy *C* that yields strictly lower payoffs for player 1, regardless of what player 2 does. There is a strictly dominated strategy that we could delete from player 1's strategy space by IDSDS, i.e., graphically there is the middle row that we could delete as never being played by player 1, rendering a reduced-form matrix as follows (Fig. 1.32):

Here, starting with the player 1 (the row player), we cannot find any more strictly dominated strategy for him. Indeed, his payoff is higher from *B* than from *F* when player 2 plays *F* (see left-hand column of the matrix) that gives him a payoff of 5 instead of 0, but would be indifferent if player 2 plays *C* (see right-hand column) that gives him a payoff of 2. In short, we cannot identify additional strategies that yields strictly lower payoffs for player 1, regardless of what player 2 does. In this context, there aren't any strictly dominated strategies we could delete from player 1's strategy space, i.e., graphically there are no rows that we could delete as never being played by player 1.

A similar argument applies to player 2 (column player). Her payoff is higher from playing *F* than *C* when player 1 plays *F* (see top row of the matrix) that gives her a payoff of 5 instead of 4, but lower if player 1 plays *B* (bottom row of the matrix) that gives her a payoff of 0 instead of 4. Therefore, there are no strategies for player 2 (that is, no columns) that we identify as strictly dominated, i.e., strategies yielding a strictly lower payoff than another strategy regardless of what player 1 chooses.

In summary, we could find strictly dominated strategies for player 1 but not for player 2, and then we could delete no more strategies in our application of IDSDS. As a result, our most precise equilibrium prediction after applying IDSDS is the

msNE. Consider a strategy profile $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_N)$ where σ_i denotes a mixed strategy for player i. We then say that strategy profile σ is a msNE if and only if

$$u_i(\sigma_i, \sigma_{-i}) \geq u_i(s_i, \sigma_{-i}) \text{ for all } s_i \in S_i \text{ and for every player } i.$$

Intuitively, mixed strategy σ_i is a best response of player i to the strategy profile σ_{-i} selected by other players.[1]

The chapter starts with games of two players who choose among two available strategies. A natural examples is the Battle of the Sexes game, where husband and wife simultaneously and independently choose whether to attend to a football game or the opera. We then extend our analysis to settings with more than two players, and to contexts in which players are allowed to choose among more than two strategies. We illustrate how to find the convex hull of Nash equilibrium payoffs (where players use pure or mixed strategies), and afterwards explain how to find correlated equilibria when players use a publicly observable random variable. For completeness, we also examine an exercise which illustrates the difference between the equilibrium payoffs that players can reach when playing the mixed strategy Nash equilibrium of a game and when playing the correlated equilibrium, showing that they do not necessarily coincide. At the end of the chapter, we focus on strictly competitive games: first, describing how to systematically check whether a game is strictly competitive; and second, explaining how to find "maxmin strategies" (also referred to as "security strategies") using a two-step graphical procedure.

Exercise 1—Game of Chicken[A]

Consider the game of Chicken (depicted in Fig. 3.1), in which two players driving their cars against each other must decide whether or not to swerve.

Part (a) Is there any strictly dominated strategy for Player 1? And for Player 2?
Part (b) What are the best responses for Player 1? And for Player 2?
Part (c) Can you find any pure strategy Nash Equilibrium (psNE) in this game?
Part (d) Find the mixed strategy Nash Equilibrium (msNE) of the game. *Hint:* denote by p the probability that Player 1 chooses Straight and by $(1-p)$ the

[1]Note that we compare player i's expected payoff from mixed strategy σ_i, $u_i(\sigma_i, \sigma_{-i})$, against his expected payoff from selecting pure strategy s_i', $u_i(s_i', \sigma_{-i})$. We could, instead, compare $u_i(\sigma_i, \sigma_{-i})$ against $u_i(\sigma_i', \sigma_{-i})$, where $\sigma_i' \neq \sigma_i$. However, for player i to play mixed strategy σ_i', he must be indifferent between at least two pure strategies, e.g., s_i' and s_i''. Otherwise, player i would not be mixing, but choosing a pure strategy. Hence, his indifference between pure strategies s_i' and s_i'' entails that $u_i(s_i', \sigma_{-i}) = u_i(s_i'', \sigma_{-i})$, implying that it suffices to check if mixed strategy σ_i yields a higher expected payoff than all pure strategies that player i could use in his randomization. (This is a convenient result, as we will not need to compare the expected payoff of mixed strategy σ_i against all possible mixed strategies $\sigma_i' \neq \sigma_i$; which would entail studying the expected payoff from all feasible randomizations.)

		Player 2	
		Straight	Swerve
Player 1	Straight	0,0	3,1
	Swerve	1,3	2,2

Fig. 3.1 Game of Chicken

probability that he chooses to Swerve. Similarly, let q denote the probability that Player 2 chooses Straight and $(1-q)$ the probability that she chooses to Swerve.
Part (e) Show your result of part (d) by graphically representing every player i's best response function $BRF_i(s_j)$, where $s_j = \{Swerve, Straight\}$ is the strategy selected by player $j \neq i$.

Answer

Part (a) No, there are no strictly dominated strategies for any player. In particular, Player 1 prefers to play Straight when Player 2 plays Swerve getting a payoff of 3 instead of 2. However, he prefers to Swerve when Player 2 plays Straight, getting a payoff of 1 by swerving instead of a payoff of 0 if he chooses to go straight. Hence, there is no strictly dominated strategy for player 1, i.e., a strategy that Player 1 would never use regardless of what strategy his opponent selects. Since payoffs are symmetric, a similar argument applies to Player 2.
Part (b) *Player 1*. If player 2 chooses straight (fixing our attention in the left-hand column), player 1's best response is to Swerve, since his payoff from doing so, 1, is larger than that from choosing Straight. Hence,

$$BR_1(Straight) = Swerve, \text{ obtaining a payoff of } 1$$

If player 2 instead chooses to Swerve (in the right-hand column), player 1's best response is Straight, since his payoff from Straight, 3, exceeds that from Swerve, 2. That is,

$$BR_1(Swerve) = Straight, \text{ obtaining a payoff of } 3$$

Player 2. A similar argument applies to player 2's best responses; since players are symmetric in their payoffs. Hence,

$$BR_2(Straight) = Swerve, \text{ obtaining a payoff of } 1$$
$$BR_2(Swerve) = Straight, \text{ obtaining a payoff of } 3$$

These best responses illustrate that the Game of Chicken is an Anti-Coordination game, where each player responds by choosing exactly the opposite strategy of his/her competitor.

Part (c) Note that in strategy profiles (Swerve, Straight) and (Straight, Swerve), there is a mutual best response by both players. To see this more graphically, the matrix in Fig. 3.2 underlines the payoffs corresponding to each player's selection of his/her best response. For instance, $BR_1(Straight) = Swerve$ implies that, when Player 2 chooses Straight (in the left-hand column), Player 1 responds Swerving (in the bottom left-hand cell), obtaining a payoff of 1; while when Player 2 chooses Swerve (in the right-hand column) $BR_1(Swerve) = Straight$, and thus Player 1 obtains a payoff of 3 (top right-hand side cell). A similar argument applies to Player 2. There are, hence, two cells where the payoffs of all players have been underlined, i.e., mutual best responses (Swerve, Straight) and (Straight, Swerve). These are the two pure strategy Nash equilibria of the Chicken game.

Part (d) *Player 1*. Let q denote the probability that Player 2 chooses Straight, and $(1 - q)$ the probability that she chooses Swerve as depicted in Fig. 3.3.

In this context, the expected value that Player 1 obtains from playing Straight (in the top row) is:

$$EU_1(Straight) = 0q + 3(1 - q)$$

since, fixing our attention in the top row, Player 1 gets a payoff of zero when Player 2 chooses Straight (which occurs with probability q) or a payoff of 3 when Player 2

		Player 2	
		Straight	Swerve
Player 1	Straight	0,0	3,1
	Swerve	1,3	2,2

Fig. 3.2 Nash equilibria in the Game of Chicken

			Player 2	
			Straight	Swerve
			q	1 - q
Player 1	Straight	p	0,0	3,1
	Swerve	1 - p	1,3	2,2

Fig. 3.3 Searching for a mixed strategy equilibrium in the Game of Chicken

selects Swerve (which happens with the remaining probability $1 - q$). If, instead, Player 1 chooses to Swerve (in the bottom row), his expected utility becomes

$$EU_1(Swerve) = 1q + 2(1 - q)$$

since Player 1 obtains a payoff of 1 when Player 2 selects Straight (which occurs with the probability q) or a payoff of 2 when Player 2 chooses Swerve (which happens with probability $1 - q$). Player 1 must be indifferent between choosing Straight or Swerve. Otherwise, he would not be randomizing, since one pure strategy would generate a larger expected payoff than the other, leading Player 1 to select such a pure strategy. We therefore need that

$$\begin{aligned} EU_1(Straight) &= EU_1(Swerve) \\ 0q + 3(1 - q) &= 1q + 2(1 - q) \end{aligned}$$

rearranging and solving for probability q yields

$$3 + 3q = q + 2 - 2q \Rightarrow q = 1/2$$

That is, Player 2 must be choosing Straight with 50% probability, i.e., $q = \frac{1}{2}$, since otherwise Player 1 would not be indifferent between Straight and Swerve.

Player 2. When he chooses Straight (in the left-hand column), he obtains an expected utility of

$$EU_2(Straight) = 0p + 3(1 - p)$$

since he can get a payoff of zero when player 1 chooses Straight (which occurs with probability p, as depicted in Fig. 3.3), or a payoff of 3 when Player 1 selects to Swerve (which happens with probability $1 - p$). When, instead, Player 2 chooses Swerve (directing our attention to the right-hand column), he obtains an expected utility of

$$EU_2(Swerve) = 1p + 2(1 - p)$$

since his payoff is 1 if player 1 chooses Straight (which occurs with probability p) or a payoff of 2 if Player 1 selects to Swerve (which happens with probability $1 - p$). Therefore, Player 2 is indifferent between choosing Straight or Swerve when

$$\begin{aligned} EU_2(Straight) &= EU_2(Swerve) \\ 0p + 3(1 - p) &= 1p + 2(1 - p) \end{aligned}$$

Solving for probability p in the above indifference condition, we obtain

$$1 = 2p \Rightarrow p = 1/2$$

Hence, the mixed strategy Nash equilibrium of the Chicken game prescribes that every player randomizes between driving Straight and Swerve half of the time. More formally, the mixed strategy Nash equilibrium (msNE) is given by

$$msNE = \left\{ \left(\frac{1}{2} Straight,\ \frac{1}{2} Swerve \right),\ \left(\frac{1}{2} Straight,\ \frac{1}{2} Swerve \right) \right\}$$

where the first parenthesis indicates the probability distribution over Straight and Swerve for Player 1, and the second parenthesis represents the analogous probability distribution for Player 2.

Part (e) In Fig. 3.4, we first draw the thresholds which specify the mixed strategy Nash equilibrium (msNE): $p = 1/2$ and $q = 1/2$. Then, we draw the two pure strategy Nash equilibria found in question (c):

- One psNE in which Player 1 plays Straight and Player 2 plays Swerve (which in probability terms means $p = 1$ and $q = 0$), graphically depicted at the lower right-hand corner of Fig. 3.4 (in the southeast); and
- Another psNE in which Player 1 plays Swerve and Player 2 plays Straight (which in probability terms means $p = 0$ and $q = 1$), graphically depicted at the upper left-hand corner of Fig. 3.4 (in the northwest).

Player 1's best response function. Once we have drawn these pure strategy equilibria, notice that from Player 1's best response function, $BR_1(q)$, we know that, if $q < \frac{1}{2}$, then $EU_1(Straight) > EU_1(Swerve)$, thus implying that Player 1 plays Straight using pure strategies, i.e., $p = 1$. Intuitively, when Player 1 knows that Player 2 is rarely selecting Straight, i.e., $q < \frac{1}{2}$, his best response is to play Straight, i.e., $p = 1$ for all $q < \frac{1}{2}$, as illustrated in the vertical segment of $BR_1(q)$ in the

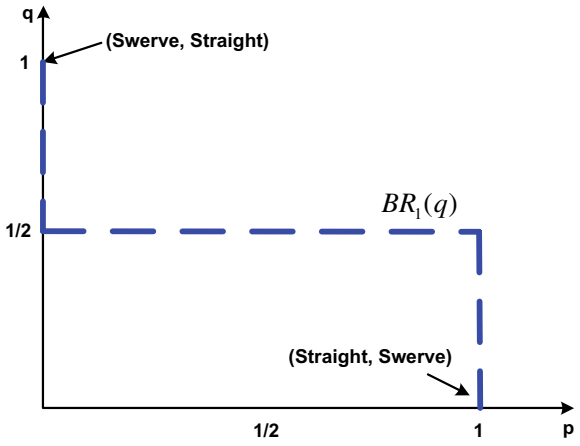


Fig. 3.4 Game of Chicken—best response function of Player 1

right-hand side of Fig. 3.4. In contrast, $q > \frac{1}{2}$ entails $EU_1(Straight) < EU_1(Swerve)$, and therefore $p = 0$. In this case, Player 2 is likely playing Straight, leading Player 1 to respond with Swerve, i.e., $p = 0$, as depicted in the vertical segment of $BR_1(q)$ that overlaps the vertical axis in the left-hand side of Fig. 3.4.

Player 2's best response function. A similar analysis applies to Player 2's best response function, $BR_2(p)$, depicted in Fig. 3.5: (1) when $p < \frac{1}{2}$, Player 2's expected utility comparison satisfies $EU_2(Straight) > EU_2(Swerve)$, thus implying $q = 1$, as depicted in the horizontal segment of $BR_2(p)$ at the top left-hand side of Fig. 3.5; and (2) when $p > \frac{1}{2}$, $EU_2(Straight) < EU_2(Swerve)$, ultimately leading to $q = 0$, as illustrated by the horizontal segment of $BR_2(p)$ that overlaps the horizontal axis in the right-hand side of the figure.

Superimposing both players' best response functions, we obtain Fig. 3.6, which depicts the two psNE of this game (southeast and northwest corners), as well as the msNE (strictly interior point, where players are randomizing their strategies).

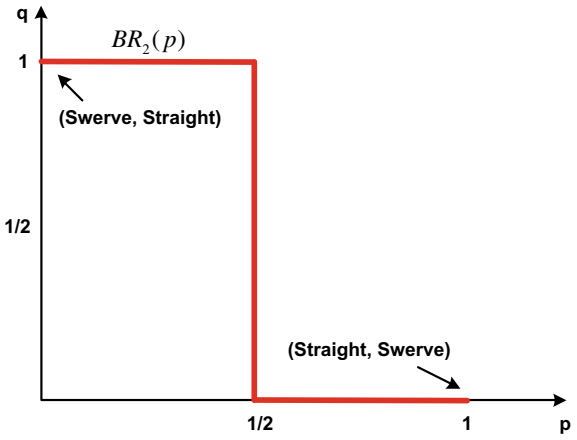


Fig. 3.5 Game of Chicken—Best response function of player 2

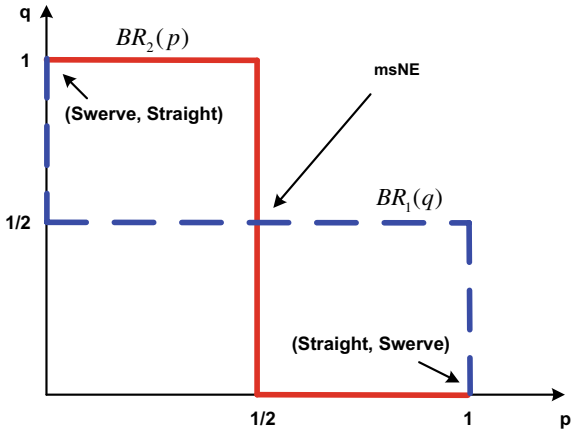


Fig. 3.6 Game of Chicken—Best response functions, psNE and msNE

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Exercise 3—Sequential Version of the Prisoner's Dilemma Game[A]

Consider the game tree in Fig. 4.9, representing the sequential-move version of the Prisoner's dilemma game, where player 1 acts first, and observing its move player 2 responds.

(a) Find the best responses of the second mover, for each of the first mover's actions.
(b) Find the first mover's equilibrium action and describe the subgame perfect Nash equilibrium of the game.
(c) Does the equilibrium behavior in the sequential-move game differ from its simultaneous-move version?

Answer

Part (a). We first circle the subgames in this extensive-form game, where Subgame 1 is the whole game, and Subgames 2 and 3 are initiated after player 1 chooses *Confess* and *Not Confess*, respectively (Fig. 4.10).

Using blue arrows, we depict player 2's best responses after observing player 1's move:

- *Subgame 2.* After player 1 chooses *Confess*, player 2 responds with *Confess* since his payoff from doing so (−5, that is, 5 years in jail) is higher than from *Not Confess* (−15, implying 15 years in jail). Intuitively, player 2 observes that player 1 has already confessed, providing evidence to the police against player 2, ultimately inducing player 2 to confess as well, which saves him 10 additional years in jail.
- *Subgame 3.* After player 1 chooses *Not Confess*, player 2 responds with *Confess* since his payoff from doing so (0, that is, no time in jail) is higher than from *Not Confess* (−1, which means spending one year in jail). Intuitively, player 2

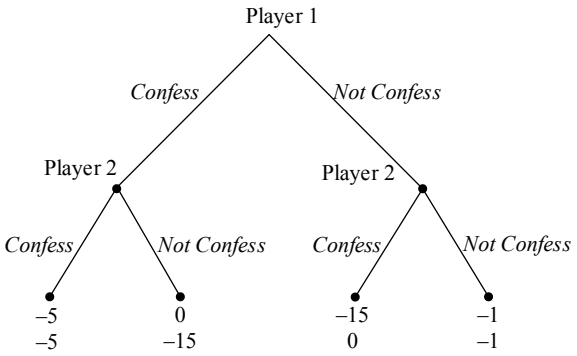


Fig. 4.9 Prisoner's dilemma sequential-move game

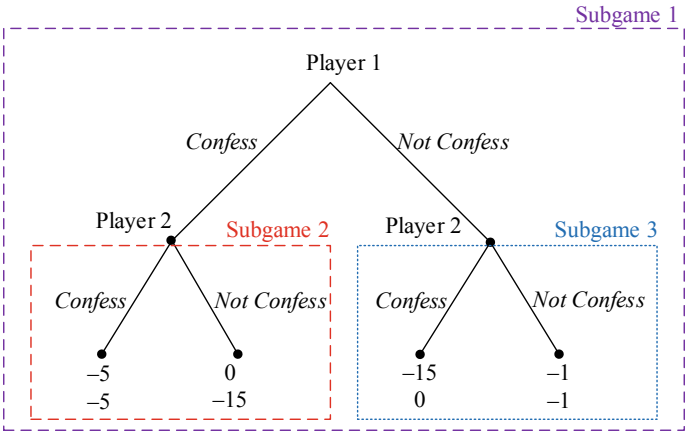


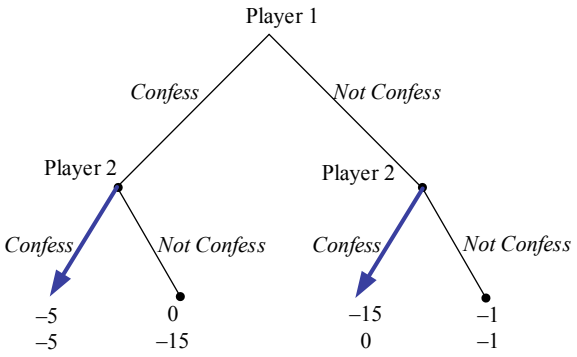
Fig. 4.10 Prisoner's dilemma sequential-move game with subgames

observes that player 1 didn't confessed. In this setting, player 2 is better off being the only prisoner who confesses—and thus leaving the police station immediately—than remaining silent as player 1 did, since by doing so he would serve a year in jail.

We can identify that, in summary, player 2 responds confessing in both Subgames 2 and 3, implying that *Not Confess* is a strictly dominated strategy for player 2 in the sequential-move game (as it is in its simultaneous-move version) (Fig. 4.11).

Part (b). By backwards induction, player 1 anticipates that player 2 will respond with *Confess* regardless of his choice at the first stage of the game, i.e., player 2 chooses *Confess* in both Subgames 2 and 3. Therefore, the game tree can be depicted as the reduced-form version in Fig. 4.12, where the payoff pairs after player 1 chooses *Confess* are (−5, −5) since he anticipates player 2 responding with

Fig. 4.11 Prisoner's dilemma sequential-move game with best responses



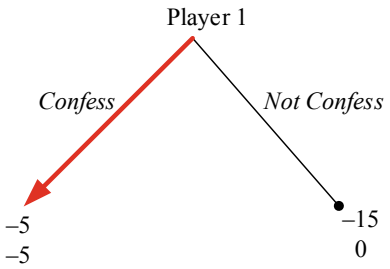


Fig. 4.12 Reduced prisoner's dilemma sequential-move game with best responses

Confess in Subgame 2. Similarly, payoff pairs after player 1 selects *Not Confess* are $(-15, 0)$ since he expects player 2 responding with *Confess* in Subgame 3.

Therefore, player 1 chooses *Confess* since his payoff from doing so, -5, is higher than from not confessing, -15. In words, since he anticipates that player 2 will respond with *Confess* regardless what action player 1 chooses in the first stage, player 1 is better off choosing *Confess* too.

As a result, the strategy profile

$$\text{SPNE} = \{Confess, (Confess, Confess)\}$$

is the Subgame Perfect Nash Equilibrium (SPNE) of this sequential-move Prisoner's dilemma game, where the first component of the triplet indicates player 1's action in equilibrium, the second element represents player 2's response to player 1 choosing *Confess* (in Subgame 2), and the third element describes player 2's response to player 1 selecting *Confess* (in Subgame 3). Note that the equilibrium path is different from the SPNE of the game, as it would only say {*Confess*, *Confess*} which indicates that player 1 chooses to *Confess*, and player 2 responds to player 1's confession (in equilibrium) confessing as well. In the SPNE, however, we list not only the action that player 2 chooses in equilibrium (that is, in Subgame 2) but also off-the-equilibrium (in Subgame 3, which should never be reached).

Part (c). Equilibrium behavior in the simultaneous-move version of the Prisoner's Dilemma game (where the Nash equilibrium predicts that both players choose *Confess*) coincides with that in its sequential-move version (where, in equilibrium, both players choose to *Confess*). This coincidence occurs because players find *Confess* to be strictly dominant both in the simultaneous and sequential version of the game, leading both players to choose *Confess* in all contingencies.

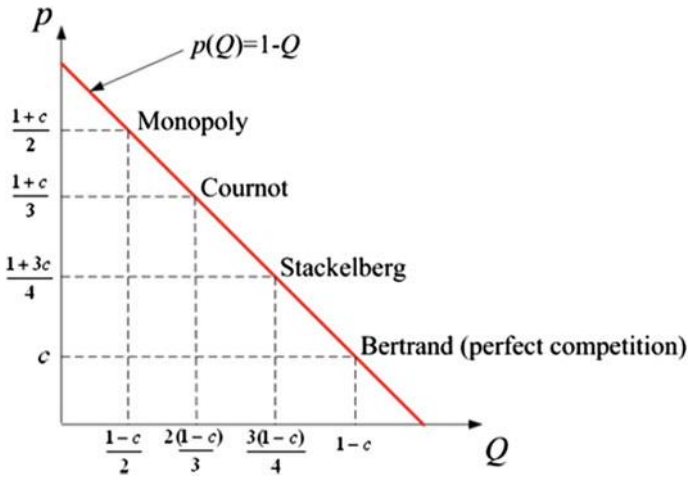


Fig. 4.34 Aggregate output and prices across different market structures

Taking first order conditions with respect to Q yields

$$1 - 2Q - c = 0$$

and, solving for Q, we obtain a monopoly output of

$$Q^{Monop} = \frac{1-c}{2}$$

which implies a monopoly price of

$$P\left(Q^{Monop}\right) = 1 - \frac{1-c}{2} = \frac{1+c}{2}.$$

Comparison. Figure 4.34 depicts the inverse linear demand $P(Q) = 1 - Q$, and locates the aggregate output levels found in the four market structures described above in the horizontal axis, afterwards mapping their corresponding prices in the vertical axis. Aggregate output is the lowest at a monopolistic market structure, larger under Cournot competition (when firms simultaneously and independently select output levels), larger under Stackelberg (when firms' competition is in quantities, but sequentially), and yet larger in a perfect competitive industry. The opposite ranking applies for equilibrium prices.

Exercise 12—Stackelberg Competition with 2, or 3 Firms[B]

Consider a Stackelberg model of output competition where firm 1 is the industry leader and firm 2 is the follower. Firms face a linear inverse demand curve $p(Q) = 10 - Q$, where $Q = q_1 + q_2$ denotes the aggregate output sold by both

firms. In addition, consider that firms are cost symmetric, facing a common marginal cost of $MC_i = 2$ for every firm $i = \{1, 2\}$ and no fixed costs.

(a) Find the equilibrium in this Stackelberg game.
(b) Assume now that a new firm (firm 3) joins the industry, acting last (after firm 2 chooses its output). Find the equilibrium in this Stackelberg game. Compare your results with those in part (a). How is the output of firm 1 and 2 affected by firm 3's entry?

Answer

Part (a). *Firm 2 (follower).* Firm 2 chooses output q_2 that solves the following profit maximization problem:

$$\begin{aligned}\max_{q_2 \geq 0} \pi_2(q_2) &= p(Q)q_2 - TC_2(q_2) \\ &= (8 - q_1 - q_2)q_2\end{aligned}$$

Taking the first order condition with respect to q_2, and assuming an interior solution for q_2,

$$8 - q_1 - 2q_2 = 0 \text{ if } q_1 > 0$$

After rearranging, the best response function of firm 2 in response to firm 1 is given by

$$q_2(q_1) = 4 - \frac{q_1}{2}$$

Firm 1 (leader). Firm 1, anticipating that firm 2 will respond with best response function $q_2(q_1) = 4 - \frac{q_1}{2}$, chooses its own output level, q_1, that solves the following profit maximization problem:

$$\begin{aligned}\max_{q_1 \geq 0} \pi_1(q_1) &= p(Q)q_1 - TC_1(q_1) \\ &= (10 - q_1 - q_2(q_1))q_1 - 2q_1 \\ &= \left(8 - q_1 - \left(4 - \frac{q_1}{2}\right)\right)q_1 \\ &= \left(4 - \frac{q_1}{2}\right)q_1\end{aligned}$$

Taking the first order condition with respect to q_1, and assuming an interior solution for q_1,

$$4 - q_1 = 0 \quad \text{if } q_1 > 0$$

such that the optimal output of firm 1 becomes

$$q_1^* = 4$$

Finding equilibrium output for the follower. Substituting the optimal output of firm 1 into firm 2's best response function, the optimal output of firm 2 becomes

$$q_2^* = 4 - \frac{4}{2} = 2$$

Therefore, when both firms are symmetric in costs, the leading firm 1 produces 4 units of output while the following firm 2 produces 2 units of output.

Part (b). *Firm 3 (last mover).* Firm 3 chooses output q_3 that solves the following profit maximization problem:

$$\begin{aligned}\max_{q_3 \geq 0} \pi_3(q_3) &= p(Q)q_3 - TC_3(q_3) \\ &= (8 - q_1 - q_2 - q_3)q_3\end{aligned}$$

Taking the first order condition with respect to q_3, and assuming an interior solution for q_3,

$$8 - q_1 - q_2 - 2q_3 = 0 \quad \text{if } q_3 > 0$$

After rearranging, the best response function of firm 3 in response to firms 1 and 2 is

$$q_3(q_1, q_2) = 4 - \frac{q_1 + q_2}{2}$$

Firm 2 (second mover). Firm 2, anticipating that firm 3 will respond with its best response function $q_3(q_1, q_2) = 4 - \frac{q_1+q_2}{2}$, chooses its own output level, q_2, that solves the following profit maximization problem:

$$\begin{aligned}\max_{q_2 \geq 0} \pi_2(q_2) &= p(Q)q_2 - TC_2(q_2) \\ &= (10 - q_1 - q_2 - q_3(q_1, q_2))q_2 - 2q_2 \\ &= \left(8 - q_1 - q_2 - \left(4 - \frac{q_1 + q_2}{2}\right)\right)q_2 \\ &= \left(4 - \frac{q_1 + q_2}{2}\right)q_2\end{aligned}$$

Taking the first order condition with respect to q_2, and assuming an interior solution for q_2,

$$4 - \frac{q_1}{2} - q_2 = 0 \text{ if } q_1 > 0$$

After rearranging, the best response function of firm 2 in response to firm 1 is

$$q_2(q_1) = 4 - \frac{q_1}{2}$$

such that the best response function of firm 3 can be rewritten as

$$q_3(q_1) = 4 - \frac{q_1}{2} - \frac{1}{2}\left(4 - \frac{q_1}{2}\right) = 2 - \frac{q_1}{4}$$

Firm 1 (first mover). Firm 1, anticipating that firms 2 and 3 will respond with their best response functions $q_2(q_1) = 4 - \frac{q_1}{2}$ and $q_3(q_1) = 2 - \frac{q_1}{4}$ respectively, chooses its own output level, q_1, that solves the following profit maximization problem:

$$\begin{aligned}\max_{q_1 \geq 0} \pi_1(q_1) &= p(Q)q_1 - TC_1(q_1) \\ &= (10 - q_1 - q_2(q_1) - q_3(q_1))q_1 - 2q_1 \\ &= \left(8 - q_1 - \left(4 - \frac{q_1}{2}\right) - \left(2 - \frac{q_1}{4}\right)\right)q_1 \\ &= \left(2 - \frac{q_1}{4}\right)q_1\end{aligned}$$

Taking the first order condition with respect to q_1, and assuming an interior solution for q_1,

$$2 - \frac{q_1}{2} = 0 \text{ if } q_1 > 0$$

such that the optimal output of firm 1 becomes

$$q_1^* = 4$$

Finding equilibrium output for the second and third movers. Substituting the optimal output of firm 1 into firm 2's best response function, the optimal output of firm 2 becomes

$$q_2^* = 4 - \frac{4}{2} = 2$$

Substituting the optimal output of firm 1 into firm 3's best response function, the optimal output of firm 3 becomes

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Player 2. Taking as given that player 1 sticks to the punishment scheme:

- If player 2 plays (B, R) in the first stage, then she obtains an overall payoff of $u_2(B,R) = 4+1 = 5$. As we can see, player 2 doesn't have incentives to deviate, because his best response function is in fact $BR_2(B) = R$ when we take as given that player 1 is playing B.

Therefore, no player has incentives to deviate from (B, R) in the first stage of the game. As a consequence, the efficient payoff (4, 4) can be sustained in the first stage of the game in a pure strategy SPNE strategy profile.

Exercise 8—Infinite-Horizon Bargaining Game Between Three Players[C]

Consider a three-player variant of Rubinstein's infinite-horizon bargaining game, where players divide a surplus (pie) of size 1, and the set of possible agreements is

$$X = \{(x_1, x_2, x_3) | x_i \geq 0 \quad \text{for} \quad i = 1, 2, 3 \quad \text{and} \quad x_1 + x_2 + x_3 = 1\}.$$

Players take turns to offer agreements, starting with player 1 in round $t = 0$, and the game ends when *both* responders accept an offer. If agreement x is reached in round t, player i's utility is given by $\delta^t x_i$, where $\frac{1}{2} < \delta < 1$. Show that, if each player is restricted to using stationary strategies (in which she makes the same proposal every time she is the proposer, and uses the same acceptance rule whenever she is a responder), there is a unique subgame perfect equilibrium outcome. How much does each player receive in equilibrium?

Answer

First, note that every proposal is voted using the unanimity rule. For instance, if player 1 offers x_2 to player 2 and x_3 to player 3, then players 2 and 3 independently and simultaneously decide if they accept or reject player 1's proposal. If they both accept, players get $x = (x_1, x_2, x_3)$, while if either player rejects, player 1's offer is rejected (because of unanimity rule) and player 2 becomes the proposer in the following period, making an offer to players 1 and 3, that they choose whether to accept or reject.

Consider any player i, and let x^{prop} denote the offer the proposer at this time period makes to himself, x^{next} the offer that he makes to the player who will become the proposer in the next period, and x^{two} the offer he makes to the player who will become the proposer two periods from now. In addition, we know that the sum of all offers must be equal to the size of the pie, $x^{\text{prop}} + x^{\text{next}} + x^{\text{two}} = 1$.

We know that the offer the proposer makes must satisfy two main conditions:

1. First, his offer to the player who will become the proposer in the next period, x^{next}, must be weakly higher than the discounted value of the offer that such player would make to himself during the next period (when he becomes the proposer), δx^{prop}. That is, $x^{\text{next}} \geq \delta x^{\text{prop}}$. Moreover, since the proposer seeks to minimize the offers he makes, he wants to reduce x^{next} as much as possible, but still guarantee that his offer is accepted, i.e., $x^{\text{next}} = \delta x^{\text{prop}}$.
2. Second, the offer he makes to the player who will be making proposals two periods from now, x^{two}, must be weakly higher than the discounted value of the offer that such a player will make to himself two periods from now as a proposer, $\delta^2 x^{\text{prop}}$. That is, $x^{\text{two}} \geq \delta^2 x^{\text{prop}}$, and since the proposer seeks to minimize the offer x^{two}, he will choose a proposal that satisfies $x^{\text{two}} = \delta^2 x^{\text{prop}}$.

Using the above three equations, $x^{\text{prop}} + x^{\text{next}} + x^{\text{two}} = 1$, $x^{\text{next}} = \delta x^{\text{prop}}$, and $x^{\text{two}} = \delta^2 x^{\text{prop}}$, we can simultaneously solve for x^{prop}, x^{next} and x^{two}. In particular, plugging $x^{\text{next}} = \delta x^{\text{prop}}$ and $x^{\text{two}} = \delta^2 x^{\text{prop}}$ into the constraint $x^{\text{prop}} + x^{\text{next}} + x^{\text{two}} = 1$, we obtain

$$x^{\text{prop}} + \delta x^{\text{prop}} + \delta^2 x^{\text{prop}} = 1,$$

and solving for x^{prop} yields

$$x^{prop} = \frac{1}{1+\delta+\delta^2}$$

Using this result into the other two equations, we obtain

$$x^{next} = \delta x^{prop} = \frac{\delta}{1+\delta+\delta^2} \quad \text{and} \quad x^{two} = \delta^2 x^{prop} = \frac{\delta^2}{1+\delta+\delta^2},$$

which constitute the equilibrium payoffs in this game. Intuitively, this implies that player 1 offers x^{prop} to himself in the first round of play, i.e., in period $t = 0$, x^{next} to player 2, and x^{two} to player 3, his offers are then accepted by players 2 and 3, and the game ends.

Comparative statics: Let us next check how these payoffs are affected by the common discount factor, δ, as depicted in Fig. 6.10. When players are very impatient, i.e., δ is close to zero, the player who makes the first proposal fares better than do others, and retains most of the pie, i.e., x^{prop} is close to one, while x^{next} and x^{two} are close to zero. However, when players become more patient, the proposer is forced to make more equal divisions of the pie, since otherwise his proposal would be rejected. In the extreme case of perfectly patient players, when $\delta = 1$, all players obtain a third of the pie.